The Photonic Holonet

A Single Self-Entangled Photon as Universal Computer, Universal Network, and Clock on the $W(3,3)$ Substrate

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Abstract

We present a complete, machine-verified architecture in which one self-entangled photon, traversing one finite geometry, is simultaneously the hardware, the software, and the network of a universal quantum computer. The geometry is the symplectic generalized quadrangle $W(3,3)$: forty points, forty lines, automorphism group with faithful projective subgroup $\mathrm{PSp}(4,\mathbb{F}_3)$ of order 25920 and full Weyl extension $W(E_6) \cong \mathrm{PSp}(4,3){:}2$ of order 51840. The symplectic double cover $\mathrm{Sp}(4,3)$ also has order 51840, but its central involution is invisible on projective objects; the two extensions must not be conflated. The photon carries $W(3,3)$ twice over — once as the forty Witting rays of its path–polarization space $\mathbb{C}^4$ (states), once as the forty Pauli displacement classes of its past–future time-bin space $\mathbb{C}^9$ (operators) — and the two carriers realize the same incidence geometry, making the machine's states and gates one set of objects. Every architectural layer is a theorem with an executable witness: the preparation circuit (two Clifford gates of tabletop optics), the routing fabric (an atlas of 540 hypercube charts glued along the 1620 apartments of the Tits building), the protected memory (the Steinberg module of dimension 81), the dual-torsor compiler (three regular copies of both the flat $\mathbb{F}_3^3$ program shell and the noncommutative Heisenberg state shell, with canonical 27+27+27 triality selected by the Heisenberg center), the oriented control plane (top homology $H_2 = 5_\omega \oplus 5_{\omega^2} \oplus 30$, whose conjugate 5-sectors fuse to one Weyl-visible 10-channel), the mirror middleware (a 2160-slot D_{12} transport bus whose full Clifford lift factors as $24 \cdot 45 \cdot 48 = 51840$), the fuel (the matter shell is exactly the magic sector, with exact Kochen–Specker budget 36/40 and contextual fraction 1/10), the clock (an internal $\mathbb{Z}_{12}$ gauge clock, two external references $\mathbb{Z}_7, \mathbb{Z}_{13}$, and an irrational Boerdijk–Coxeter drive realizing a discrete time quasicrystal), and universality (the photon's own tritter, phase plate, and modulator generate the complete Clifford group; the matter shell supplies the magic). We also give the recursive engineering law: a level-n holonet has 40^n photonic leaves, $(40^n - 1)/39$ $W(3,3)$ instances, and reversible routing diameter bounded by $8n = 8\log_{40} N$, while durable commits use a separate Császár/tomotope clock $T(n) = 4(7^n - 1)$. The runtime contains an exact parabolic packet decoder: the Bell-line stabilizer splits the 1296 compass incidences into 648+324+162+162, giving the complete binary prefix code $\{0, 10, 110, 111\}$ for mirror, schedule, and two chiral caches, while the dual point stabilizer supplies two persistent 648-slot address sheets. The two cache branches are now resolved internally: each is the same homogeneous space H/C_4, each is a two-sheet cover of one canonical 81-route base H/D_8, and their 324-slot union has full equivariant commutant D_8 acting in 81 four-state fibers. This split address bus is provably distinct from the older non-split ternary $81 \to 162 \to 81$ transport memory. At the communication edge, the Witting delayed-query desk is now a frame compiler: 40 tetrads times 4 Alice slots times 4 Bob slots give a 640-entry admission ROM, with 480 off-diagonal data handshakes and 160 diagonal contextual witness apertures. The corresponding analyzer bank is sparse physical optics rather than a generic multiport: the 40 Witting bases split as $1 + 12 + 27$ analyzers with at most 12 nonzero matrix entries and only cube-root phase weights over $\sqrt{3}$. The newest storage layer turns the tomotope cover pathology into an engineering

feature: durable storage is cover-indexed by a $\mathbb{Z}_k^3$ fiber over the stable 48-block packet ABI, sentinel-aware compilation can trade commit-time slack for lower $g = 15$ fault energy, the first exact guard band is the arithmetically forced desynchronization at $n = 5$ with remainder $24 = f$, and the Grünbaum–Coxeter Wythoff operations of the tomotope are exactly the packet-growth ladder $4, 12, 24, 48, 96$.

Beyond the architecture, the same substrate is a candidate *physics*. The choice $q = 3$ is triply forced — by the geometric master equation $q! = 2q$, by the minimal magic (contextuality) dimension, and by the holographic central charge $c = 24 = f$ — and from it unfolds an exact exceptional skeleton: the Eisenstein tower $q = 3 \to \{3, 7\}$ Hurwitz surfaces $\to$ the Witting polytope $\to \mathrm{E}_6/\mathrm{E}_7/\mathrm{E}_8 \to$ the complex Leech $\to$ the Monster at $c = 24$, threaded by $\mathrm{G}_2 = \mathrm{Aut}(\mathbb{O})$ and welded by a single order-3 Eisenstein element. The Standard Model descends from E_6 by trinification ($\dim \mathrm{SU}(3)^3 = 24 = f$, dim SM $= 12 = k$), with the three generations, gauge unification ($\sin^2\theta_W = 3/8$), and tri-bimaximal PMNS mixing all one S_3 triality. The falsifiable layer is on a live test sheet: the cosmological predictions ($n_s = 29/30$, $r = 1/300$, $f_{NL} = 1/72$) are consistent, the neutrino sum is resolved by the seesaw ($\sum m_\nu \approx 0.06\,\mathrm{eV}$), and proton decay ($\tau_p \sim 10^{35\text{–}36}$ yr, Hyper-K) together with two demonstrator readouts (contextual fraction $1/10$, pump Chern 2) are sharp near-term tests. No fitted parameters appear anywhere: every constant is substrate arithmetic at $q = 3$.

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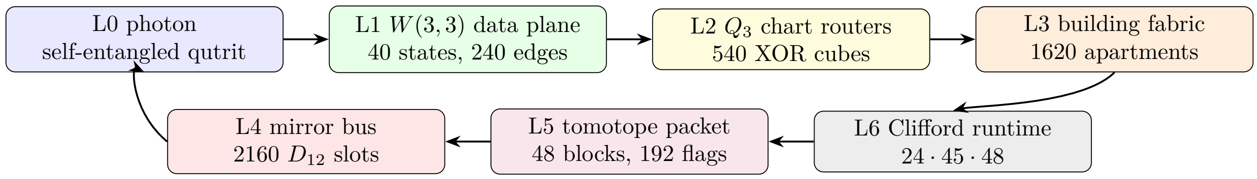
1 Overture: the machine is the geometry is the carrier

Classical architecture separates three things: the processor that acts, the program that directs, and the network that connects. The thesis of this paper is that at the level of a single photon on a finite symplectic geometry the three are one object, and that this is not a metaphor but a chain of theorems.

Principle 1.1 (Identity of the three layers)**.** *Transporting the photon through the mesh* is *applying a gate* is *routing a packet. One physical process; three descriptions.*

The identification rests on seven exact pillars, each proved by explicit computation in the project repository (every theorem below carries its witness script):

1. **Operator–operand duality** (§2.1): the same $W(3,3)$ is drawn by the photon's *states* (orthogonality of the 40 Witting rays in $\mathbb{C}^4$) and by its *operators* (commutation of the 40 two-qutrit Pauli classes on $\mathbb{C}^9$).
2. **Routing–gate identity** (§4): the mesh is an atlas of 540 three-dimensional hypercube charts whose native XOR routing moves are exactly the register operations, glued along the 1620 apartments of the Tits building of $\mathrm{Sp}(4,\mathbb{F}_3)$.
3. **Parabolic address–route duality** (§2.3): the point parabolic supplies two 648-slot address sheets, whereas the Bell-line parabolic decodes the compass fabric by the complete prefix code $0, 10, 110, 111$; its cache quarter is an exact D_8-over-81 four-state address fiber. Address and route are the two nodes of the same C_2 building, while the separate ternary transport extension carries temporal state.
4. **Dual-torsor state–program compilation**: the Steinberg memory restricts to both canonical order-27 shell groups as three regular copies. Bell-shell programs use commuting $\mathbb{F}_3^3$ coordinates; point-shell states use Heisenberg 3^{1+2} coordinates; both compile into the same 81 logical qutrits.
5. **Mirror middleware** (§5): the cyclic selector clock and the mirror transport bus are different 2160 worlds. The photon is clocked by the C_{12} side but routed by the D_{12} side, whose full Clifford lift is the exact product $24 \cdot 45 \cdot 48$.
6. **Universality from optics** (§6.5): the photon's own beam-splitting elements generate the complete Clifford group (symplectic closure = 51840, exact), and its matter shell is precisely its magic supply.
7. **Fractal holonet scaling** (§7): recursively replacing every site by a fresh $W(3,3)$ core gives a universal computer/network with exact 40^n leaves and logarithmic reversible routing.



One loop: carrier → data plane → router → building → runtime → tomotope packet → mirror bus → carrier.

Figure 1: The photonic holonet as an engineering stack. The same photon is data, instruction, and packet; the same finite geometry supplies the network, middleware, and runtime.

2 The substrate

Definition 2.1 ($W(3,3)$)**.** *Let $V = \mathbb{F}_3^4$ with the alternating form $\langle u, v\rangle = u_1v_3 - u_3v_1 + u_2v_4 - u_4v_2$. The points of $W(3,3)$ are the 40 projective points of V; the lines are the 40 totally isotropic projective lines. Each point lies on 4 lines and each line carries 4 points; collinearity defines the strongly regular graph* $\mathrm{SRG}(40, 12, 2, 4)$.

Three related symmetry groups occur, and their roles are different:

$$\underbrace{\mathrm{Sp}(4,3)}_{2\cdot\mathrm{PSp}(4,3),\ \text{Clifford double cover}}, \qquad \underbrace{\mathrm{PSp}(4,3)}_{25920,\ \text{faithful projective action}}, \qquad \underbrace{W(E_6) \cong \mathrm{PSp}(4,3){:}2}_{51840,\ \text{outer Weyl extension}}.$$

The first and third groups both have order 51840 but are not the same extension: the center of $\mathrm{Sp}(4,3)$ acts trivially on projective points and lines, while the outer involution in $W(E_6)$ acts nontrivially. The projective group acts with permutation rank 3. The ATLAS index-40 subgroups are the two building parabolics $3^{1+2}{:}2A_4$ and $3^3{:}S_4$ [21]. The substrate primitives are

$$q = 3, \quad \lambda = 2, \quad \mu = 4, \quad k = 12, \quad f = 24, \quad g = 15, \quad \Phi_6 = 7, \quad \Phi_4 = 10, \quad \Phi_3 = 13,$$

and every numerical claim in this paper reduces to arithmetic in these.

Formal grounding (the master formalism). In the master manuscript [1] the whole structure is forced by one Diophantine seed: the *master equation* $q! = 2q$ has the unique positive-integer solution $q = 3$, and the SRG quadratic $x^2 - q!\,x + 2^q = (x-2)(x-4)$ then delivers $\lambda = 2$, $\mu = 4$ and the full parameter closure ($k = 2^q + q + 1$, $v = 40$, multiplicities $f = 24$, $g = 15$, cyclotomic ladder $\Phi_3 = 13$, $\Phi_6 = 7$, $\Phi_{12} = q^4 - q^2 + 1 = 73$, $\Theta = q^2 + 1 = 10$). Two of those closure constants reappear in this paper as *spectral field discriminants*: the chart-web's irrational eigenvalue pairs $1 \pm \sqrt{10}$ and $(-1 \pm \sqrt{73})/2$ (§4) live precisely in $\mathbb{Q}(\sqrt{\Theta})$ and $\mathbb{Q}(\sqrt{\Phi_{12}})$ — the machine's transport spectrum speaks the same cyclotomic dialect as the physics tier ladder.

2.1 The two carriers and the operator–operand duality

Theorem 2.2 (Two realizations, one geometry)**.** (i) *The 40 Witting rays in $\mathbb{C}^4$ (the photon's path$\otimes$polarization space) have orthogonality graph isomorphic to the collinearity graph of $W(3,3)$.* (ii) *The 40 projective classes of two-qutrit Pauli displacements $D(v) = X^aZ^b \otimes X^cZ^d$ on $\mathbb{C}^9$ (the photon's past$\otimes$future time-bin space) have commutation graph equal to the same $W(3,3)$, via $D(u)D(v) = \omega^{\langle u,v\rangle} D(v)D(u)$.* Witnesses: `bt817_self_entangled_photon_atlas.py`, `bt821_operator_operand_duality.py`.

Thus a point of the substrate is simultaneously a pure state the photon can occupy and a unitary the photon can enact; a line is simultaneously a measurement context (orthonormal tetrad) and a stabilizer group (maximal commuting Pauli set with its 9-state joint eigenbasis). The $360 = 40 \times 9 = f \cdot g$ two-qutrit stabilizer states are the joint eigenbases of the forty lines.

2.2 The five vacua and the Schläfli geography

Every maximal subgroup class of $\mathrm{PSp}(4,3)$ decomposes the substrate in its own canonical way; the maximal indices are the classical inventory of the 27 lines on a cubic surface.

index	subgroup	points	lines	vacuum	cubic surface
27	$2^4{:}A_5$	$[40]$	$[40]$	homogeneous (icosahedral)	27 lines
36	S_6	$[40]$	$[10, 30]$	spread fibration	36 double-sixes
40	parabolic	$[1, 12, 27]$	$[4, 36]$	holonet point vacuum	trihedra I
40	parabolic	$[4, 36]$	$[1, 12, 27]$	dual line vacuum	trihedra II
45	$(2T{\times}2T){:}2$	$[8, 32]$	$[16, 24]$	binary polar	45 tritangents

The physical decomposition $40 = 1 + 12 + 27$ (self + gauge shell + matter shell) is the *choice of the point-parabolic vacuum* among five inequivalent readings; the icosahedral vacuum sees a featureless substrate. The platonic ladder threads the table: the binary tetrahedral group $2T$ (the 24-cell) lives in every polar-pair stabilizer, the real octahedral group O_h ($= S_4 \times \mathbb{Z}_2$, order 48) is the chart symmetry of §4, and the binary icosahedral group $2I = \mathrm{SL}(2,5)$ (the 600-cell) is the core of every spread stabilizer, with point orbits $[20, 20]$ and line orbits $[10, 30]$ echoing the Boerdijk–Coxeter decomposition $600 = 20 \times 30$. *Witnesses:* `bt808-bt813`.

2.3 The parabolic router: address and route are dual

The two index-40 parabolics are not duplicate descriptions of one local neighborhood. They execute different halves of a packet protocol. This becomes visible only after the compass census: there are two 216-element classes of icosahedral A_5 cores, and exactly $1296 = 216 \cdot 6$ cross-class pairs meet in D_{10}.

Theorem 2.3 (Bell–compass parabolic router)**.** *Fix an isotropic Bell line. Its projective Siegel parabolic* $3^3{:}S_4$ *has four orbits on the* 1296 *incident compass pairs:*

$$162_L + 162_R + 324 + 648.$$

They are distinguished objectwise by the Bell line lying in the pentad core's 5_L*,* 5_R*,* 10*, or* 20 *line orbit. The normalized orbit sizes satisfy the Kraft equality*

$$2^{-3} + 2^{-3} + 2^{-2} + 2^{-1} = 1,$$

so they admit the complete prefix decoder

$$110 \mapsto \textit{left cache}, \quad 111 \mapsto \textit{right cache}, \quad 10 \mapsto \textit{schedule}, \quad 0 \mapsto \textit{dark mirror}.$$

The outer Weyl involution fuses exactly the two 162 *cache orbits, leaving* $324_{\text{cache}} + 324_{\text{schedule}} + 648_{\text{mirror}}$*. By contrast, the dual point parabolic has two regular projective orbits of size* 648*; the outer extension preserves rather than fuses these address sheets.* Witness: `bt859_bell_compass_parabolic_router.py`.

The equality case of Kraft–McMillan means the routing words fill a binary decision tree with no unused branch [22]. Here the code lengths were not optimized from assumed traffic probabilities: they were read from exact group orbits. A packet first asks whether its Bell line is in the dark mirror half; if not, whether it is in the common schedule quarter; only the remaining cache quarter needs the third, chiral bit. The decoder is self-delimiting and constant-depth, with expected word length

$$\frac{1}{2}(1) + \frac{1}{4}(2) + 2 \cdot \frac{1}{8}(3) = \frac{7}{4}$$

under the invariant uniform measure on the 1296 compass pairs.

orbit	Bell-line position	prefix	engineering action
648	dark 20-orbit	0	enter mirror bus
324	common schedule 10-orbit	10	timetable-local route
162	absolute pentad 5_L	110	left cache
162	absolute pentad 5_R	111	right cache

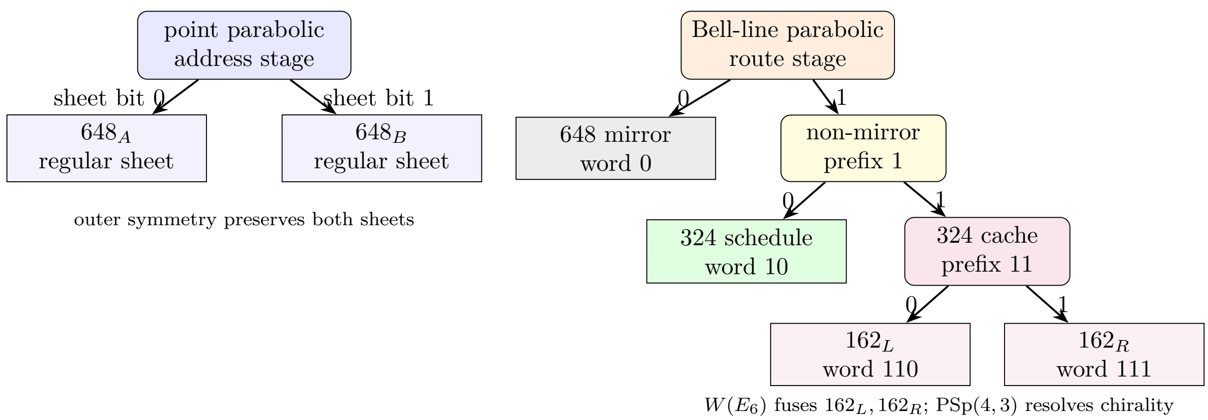


Figure 2: The two C_2 parabolics implement a dual protocol. Point localization selects one of two address torsors; Bell-line localization decodes a complete variable-length route word.

This theorem also closes an honesty boundary. The Bell stabilizer and the compass incidence set both have cardinality 1296, but they are not a regular G-set identification. Projectively, the Bell stabilizer has order 648 and the orbit stabilizers are $1, 2, 4$. The useful result is stronger than the count: a hierarchical decoder and a persistent address bit are forced by the two parabolic actions.

3 The carrier: how to self-entangle a photon

Entanglement is a relation between tensor factors; nothing requires the factors to be distinct particles. Self-entanglement is entanglement between one photon's own registers.

3.1 Stage A: spatial (one element)

A diagonal photon meets a polarizing beam splitter: $\frac{1}{\sqrt{2}}(|H\rangle + |V\rangle) \mapsto \frac{1}{\sqrt{2}}(|H, a\rangle + |V, b\rangle)$. Path and polarization are now maximally entangled *within* the photon; this two-qubit $\mathbb{C}^4$ is the Witting carrier of Theorem 2.2(i).

3.2 Stage B: temporal (two Clifford gates)

A tritter (symmetric 3-port coupler) *is* the qutrit Fourier gate F_3; a three-bin delay ladder $(0, \tau, 2\tau)$ defines past and future registers; one electro-optic modulator applies $\mathrm{CX}_{p\to f}$ conditioned on bin index. The temporal Bell qutrit

$$|\Omega\rangle \;=\; \mathrm{CX}_{p\to f}\,(F_3 \otimes I)|0\rangle_p|0\rangle_f \;=\; \frac{1}{\sqrt{3}}\sum_{j=0}^{2}|j\rangle_p|j\rangle_f$$

is verified exactly; the photon is entangled with its own future. Inserting any U in the future arm, the recombination visibility is the *trace–Choi witness*

$$V(U) \;=\; \frac{|\mathrm{Tr}\,U|}{3}, \qquad V(F_3) = \tfrac{1}{3}, \quad V(X) = V(Z) = 0,$$

so the photon implements the Choi–Jamiołkowski isomorphism on itself: it measures channels against its own past. *Witness:* `bt820_self_entanglement_protocol.py`.

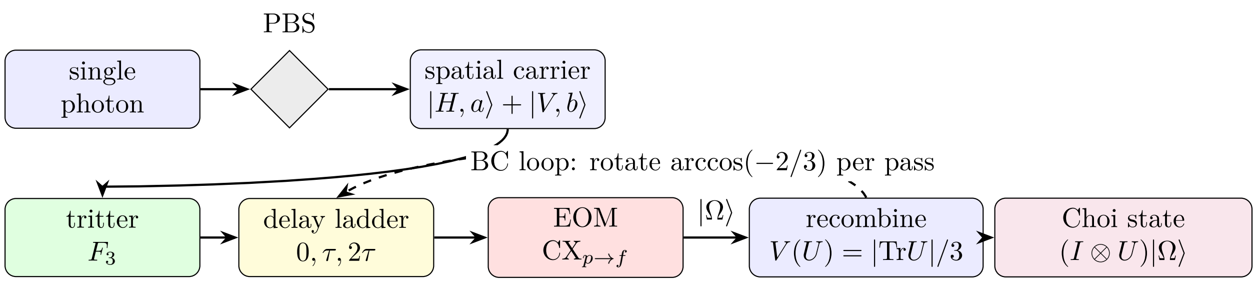


Figure 3: Preparation schematic. One photon, four registers. The PBS entangles path with polarization (Stage A); the tritter–delay–EOM chain prepares the temporal Bell qutrit $|\Omega\rangle$ (Stage B); the dashed loop is the irrational Boerdijk–Coxeter drive of §11.

4 The network: atlas, building, and routing

Theorem 4.1 (The hypercube atlas). *$W(3,3)$ contains exactly 540 skew line pairs. For each pair the non-collinearity graph on its 8 points is the cube graph Q_3, with full symmetry group O_h of order 48 (these are the only order-48 stabilizers; the binary $2O$ does not occur). Each chart admits an exact $\mathbb{F}_2^3$ Gray-code addressing in which chart edges are unit XOR moves and the antipode of a vertex is its unique collinear partner. Every $W(3,3)$ nonedge is a chart edge in exactly $12 = k$ charts; every edge is an antipode pair in exactly $9 = q^2$.* Witnesses: `bt773`, `bt777`, `bt778`.

4.1 Hypercube networking, not hypercube decoration

The cube is not only a picture for one chart. It is the local interconnect primitive. A chart gives eight simultaneously valid addresses

$$000,\; 001,\; 010,\; 011,\; 100,\; 101,\; 110,\; 111 \in \mathbb{F}_2^3,$$

and the elementary network instruction is the dimension-order hypercube hop

$$a \longmapsto a \oplus e_i, \qquad i \in \{1,2,3\}.$$

This is the standard e-cube rule of hypercube networks: compare source and target, flip one differing coordinate at a time, and finish in at most three local hops [13]. In the holonet that same XOR flip is also a register operation, because the chart address is a finite "which-register" word on the photon's self-entangled carrier. The local routing primitive is therefore simultaneously

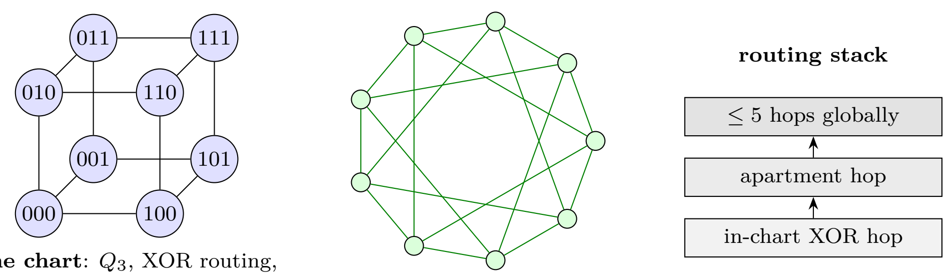
$$\text{bit flip} \quad = \quad \text{qutrit-frame update} \quad = \quad \text{optical path choice.}$$

A packet in the photonic holonet is not merely a payload. The natural packet header is

$$\mathcal{P} = (\text{payload},\ \text{Pauli frame},\ \text{chart word},\ \text{apartment hop},\ \text{mirror slot},\ \text{clock phase}).$$

The payload is the quantum state or teleported gate; the Pauli frame is the two-trit feedforward record; the chart word is the $\mathbb{F}_2^3$ route inside the current cube; the apartment hop selects the next chart; the mirror slot chooses the D_{12} transport bus of §5; and the clock phase is the cyclic C_{12} interrupt that keeps the selector coherent. This is why the same object can be called a computer architecture and a network architecture. A node never sends data to a processor: the act of transport is the processor's gate action.

Theorem 4.2 (The fabric is the building)**.** *The chart-adjacency web (*540 *nodes, two charts adjacent when one's axis is a disjoint transversal pair of the other) is* 6*-regular with exactly* 1620 *edges, and these edges are in canonical bijection with the* 1620 *apartments of the Tits building of* $\mathrm{Sp}(4, \mathbb{F}_3)$*. The web has diameter* 5*; its adjacency module contains the Steinberg representation twice and presses against the Ramanujan bound only on a* 15*-dimensional sentinel eigenspace.* Witnesses: `bt777`, `bt779`, `bt744`.



one chart: Q_3, XOR routing, antipode = collinear partner

the web (sketch): 540 charts, 6-regular, 1620 links = apartments, diameter 5

Figure 4: Network schematics. Left: a single chart with its $\mathbb{F}_2^3$ addressing (XOR routing native). Center: the 540-chart web whose links are the apartments of the Tits building (sketched as a circulant fragment). Right: the two-level routing stack.

Routing a packet means choosing XOR moves inside charts and apartment hops between them; since chart moves *are* register operations (§6), routing is computation. At scale the recursion is the holonet ladder: 40^n cores at level n, the same $\mathrm{SRG}(40, 12, 2, 4)$ at every level, with the level-0 chart's $1 + 12 + 27$ degree split realized as self pole, gauge shell, and matter shell of each node.

Corollary 4.3 (Two-plane routing bound)**.** *Inside a chart the worst route length is* 3*, the diameter of* Q_3*. Between charts the worst route length is* 5*, the diameter of the apartment web. One recursive digit of a holonet address therefore costs at most*

$$3 + 5 = 8$$

reversible transport moves. BT827 globalizes this to $8n = 8 \log_{40} N$ *for* $N = 40^n$ *photonic leaves.*

5 The middleware: tomotope mirrors and the runtime bus

Definition 5.1 (Tomotope packet)**.** *In this paper the tomotope is used in the only way the finite calculation currently justifies: as a local middle-layer incidence packet. A tomotope packet has four vertex carriers, twelve local edge axes, sixteen triangular face labels, and forty-eight edge–face incidence blocks. The doubled base/shadow fiber gives* 192 *flags. Thus the tomotope is the durable packet format sitting between the fast cube router and the global mirror bus; it is not introduced as an ambient continuous torus, and it is not identified with the rectangle clock.*

The atlas contains two different 2160-element boundary worlds. They must not be collapsed. The rectangle side is cyclic: its stabilizer is C_{12} and it supplies the selector's phase clock. The chart-transversal side is dihedral: its stabilizer is D_{12} and it supplies the mirror transport bus. Equal cardinality is the red herring; different stabilizer structure is the theorem.

Theorem 5.2 (The D_{12} mirror bus)**.** *The* 540 *cube charts carry exactly four common-transversal slots each, so the global chart-transversal space has* $540 \cdot 4 = 2160$ *slots. The map*

$$(\textit{chart},\ \textit{common transversal line}) \longmapsto (\textit{chart},\ \textit{antipode pair cut out by that transversal})$$

is an equivariant bijection onto the atlas antipode-slot space. Its projective stabilizer has order 12 *and GAP identifies it as* D_{12}*, not* C_{12}*.* Witness: `bt815_global_2160_transversal_gset.py`.

Locally, a skew-line chart has four common isotropic transversals. Read those four transversals as the four vertex carriers of the tomotope middle layer. Each transversal is a K_4, and each K_4 has three opposite-edge axes and four triangular faces. Thus the chart carries

$$4 \text{ vertices}, \qquad 4 \cdot 3 = 12 \text{ edge labels}, \qquad 4 \cdot 4 = 16 \text{ face labels},$$

and the middle incidence packet has

$$4 \cdot 3 \cdot 4 = 48$$

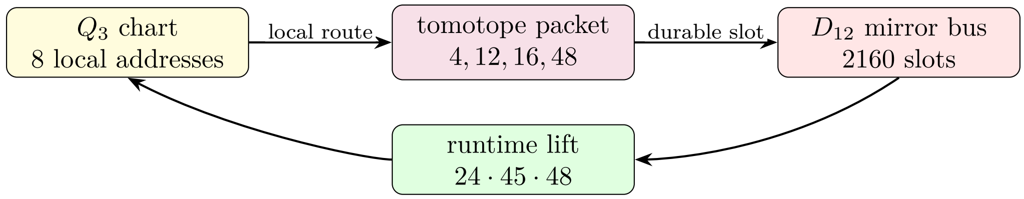
edge–face blocks, with the doubled base/shadow fiber restoring 192 flags. This is the local $\langle r_0, r_3 \rangle$ tomotope block layer, not an analogy to a toroidal polyhedron. *Witness: `bt814_tomotope_middle_layer_from_residual_tetrahedra.py`.*

Theorem 5.3 (Photonic mirror middleware)**.** *The full optical Clifford runtime factors as*

$$|\mathrm{Sp}(4, \mathbb{F}_3)| = 51840 = 24 \cdot 2160 = 24 \cdot 45 \cdot 48.$$

Here $24 = f$ *is the full* $\mathrm{Sp}(4,3)$ *lift of the projective* D_{12} *mirror-slot stabilizer,* 45 *is the hyperbolic polar-pair/tritangent geography, and* 48 *is the local tomotope edge–face middle layer (also the order of the chart group* O_h*). Equivalently, the photon is clocked by the cyclic selector side and routed by the dihedral mirror side; the complete Clifford group is the lift of that mirror bus through the tomotope middle carrier.* Witness: `bt826_photonic_mirror_middleware.py`.

This is the architectural closure missing from a bare universality claim. The tritter, phase plate, and controlled delay generate the whole Clifford group, but the group is *organized* as a middleware stack: full slot stabilizer, global mirror bus, polar vacuum geography, and tomotope local block. In software terms, C_{12} is the clock interrupt, D_{12} is the mirror router, 48 is the local packet format, and $24 \cdot 45 \cdot 48$ is the finite runtime.



fast XOR routing is committed through the tomotope middle layer and reflected by the mirror bus.

Figure 5: Middleware schematic. The cube is the reversible local router; the tomotope is the edge–face packet format; the D_{12} bus is the global mirror transport object.

5.1 Product-grid closure ABI v2

The 12 closure strands are modeled as

$$C_3 \times V_4, \qquad V_4 = C_2 \times C_2,$$

rather than as a bare C_{12}. The C_3 coordinate is the Szilassi/Fano channel and qutrit phase axis; the V_4 coordinate is the four-triangle E6 gauge-shell/D4 branch axis.

The E6 firewall square is now upstream in the claim DAG through the exact nodes

$$36 \to 72, \qquad 72 + 6 = 78, \qquad 72 + 9 = 81.$$

This supports the exact finite ABI/CSS branch while leaving blocked formula and physics claims downstream of the transcription gate.

The closure packet ABI is upgraded to v2. Each packet carries C_3 channel, V_4 branch, active/guard outputs, 72-row sector metadata, the nine-mode firewall gap, and claim-DAG dependencies.

5.2 ABI v2 consumer, E6 claim table v2, and D4/V4 branch audit

The closure ABI v2 is executable. It regenerates 12 packets, 48 events, and 72 rows while preserving the dual-axis metadata. The three C_3 channels each carry 24 rows, and the four V_4 triangles each carry 18 rows.

The E6-merged claim DAG now produces a table with explicit E6 provenance for the 72-sector, the 81 closure, and the $C_3 \times V_4$ grid. Blocked formula and physical claims remain downstream.

On branch classes, D4 shear fixes each V_4 triangle, while τ_4 acts as an order-two translation preserving the triangle partition as a set. The two operations still fail to commute on full branch/phase states, so retwining remains part of the exact decoder rule.

5.3 BT1486–BT1488 ABI v2 retwined CSS and splice manifest

BT1486 reruns the CSS join from the ABI v2 consumer output. The result is not just the old 72-row count: every row satisfies the retwined X- and Z-syndrome checks, while the C_3 channel profile remains $P_0 = P_1 = P_2 = 24$ rows and the V_4 triangle profile remains $T_0 = T_1 = T_2 = T_3 = 18$ rows.

BT1487 classifies the branch symmetries behind those four triangles. The full triangle-partition stabilizer is S_4 of order 24, giving the Fano point reading $7 \cdot 24 = 168$. The physically used square subgroup is D_4 of order 8, giving the Fano flag reading $21 \cdot 8 = 168$. Thus the active detector-bin count is Fano-native, not merely an optical count.

BT1488 updates the splice cascade: the E6 claim table v2 from BT1484 supersedes the BT1472 table as the preferred paper table, and the preferred exact finite insert packet is now BT1474, BT1480, BT1482, BT1484, BT1486, and BT1487. The Golden Quartic/Moebius-ball equation fill remains blocked pending rendered formula transcription; no public source-code repository is assumed.

5.4 BT1489–BT1491 row symmetry and Fano–E6 square

BT1489 lifts the S_4, D_4, and V_4 branch actions from triangle labels to the actual ABI v2 row layer. All 24 elements of S_4, all 8 elements of the square D_4 subgroup, and all 4 V_4 translations act as honest permutations of the 72 active/guard value rows. The lift preserves the C_3 channel, row kind, qutrit value, guard slot, and column formula, so the branch symmetry is now a decoder-row symmetry.

BT1490 is the new finite-geometry lock:

$$24 = 4 \cdot 6 = 3 \cdot 8, \qquad 72 = 3 \cdot 24, \qquad 168 = 7 \cdot 24 = 21 \cdot 8, \qquad 81 = 72 + 9.$$

The shared 24-state fiber is V_4 branch times six row-value slots on the ABI side, and three local Fano arms times the D_4 flag stabilizer on the Fano side. Thus the same finite fiber feeds the E6/CSS 72-sector, the 81-closure after adjoining the $q^2 = 9$ firewall gap, and the Fano-native 168 active-bin bus.

BT1491 makes the paper splice idempotent. The main holonet paper now imports the BT1480–BT1491 exact finite packet through one controlled block; rerunning the splicer replaces that block rather than duplicating it. The claim firewall remains intact: this is an exact finite ABI and incidence statement, not a detector calibration, optical-layout claim, or imported Golden Quartic/Mobius particle interpretation.

5.5 BT1492–BT1494 canonical Fano fiber and pulse release lock

BT1492 removes the last arbitrary index from the 24-state bridge. In the Fano plane, $\mathrm{GL}(3, 2)$ has order 168. Fixing one point leaves a point stabilizer of order 24, and this stabilizer acts as the full S_4 on the four Fano lines not through that point. Fixing one incident flag leaves an order-8 flag stabilizer with the D_4 order profile. Therefore the shared fiber is canonical:

$$24 = 3 \cdot 8,$$

namely three Fano lines through the anchor point times the D_4 flag stabilizer.

BT1493 compiles those row symmetries down to the existing physical holonet interfaces. A row action first selects a BT1411 detector slot, then hands that slot to the BT1374 rule mirror_slot mod 4, and finally occupies the matching BT1407 Hesse epilogue lane for its C_3 channel and row-value slot. All 24 S_4 actions compile across all 72 rows, while the order-8 D_4 square subgroup is the native square-pulse layer.

BT1494 repairs the broad photonic-QEC release lock. The legacy tests expected selected `PART_*` artifacts at the repository root, while the preserved copies lived under `manuscripts/parts`. The repair restores those root release-lock targets and regenerates the live scheduler, harmonic bus, fusion splice, and projective screen/bulk/QEC artifacts.

6 The software: braids, teleported gates, universality

6.1 Exact braid words

In the anyonic Fibonacci representation $\sigma_1 = \mathrm{diag}(\zeta^4, -\zeta^2)$ ($\zeta = e^{i\pi/5}$) one has *exactly* $\sigma_1^5 = Z$ and $\sigma_1^{10} = I$ — not merely projectively (verified in the cyclotomic ring $\mathbb{Z}[\zeta_{10}]$). In the dual ($\pm$) encoding of the local $K_{3,3}$ cycle code $\mathbb{F}_2^4$ this makes every register bit-flip an exact five-letter braid word, and the flat global register (the unique connected gluing with trivial holonomy across the atlas) carries the action coherently: a zero-Berry-phase memory. *Witnesses:* `bt740`*,* `bt741`*.*

6.2 The photon is its own gate

A gate U is stored as the self-entangled state $(I \otimes U)|\Omega\rangle$; Bell-projecting an input register against the photon's *past* register outputs U (times a known Pauli correction) on the *future* register — verified for all nine outcomes. The machine's gate library is a library of self-entangled states; executing software means interfering the photon with its own history. *Witness:* `bt821`*.*

6.3 Instruction set architecture

The practical instruction set is finite:

$$\mathcal{I}_{\mathrm{holo}} = \{F_p, F_f,\ S_p, S_f,\ \mathrm{CX}_{p\to f},\ \sigma^5 = Z,\ D_{12}\text{-mirror},\ M_{36}\text{-magic}\}.$$

F and S generate single-qutrit Clifford frame changes; CX couples past to future; $\sigma^5 = Z$ gives exact braid-register flips; the D_{12} mirror operation moves a packet through the atlas without confusing it with the cyclic selector clock; and M_{36} denotes injection from the thirty-six magic rays of §9. The Clifford part normalizes Pauli frames and is efficiently classically trackable; it is not universal by itself. Universality starts only when the Clifford runtime can consume the intrinsic magic sector. This is the engineering meaning of "matter equals magic": the same thirty-six rays that appear as the matter shell are the non-Clifford fuel for the processor.

6.4 The minimal classical shadow

The smallest classical headline compatible with the substrate is the pair

$$(\lambda, q) = (2, 3).$$

This is the same two-state, three-symbol signature made famous by the Wolfram–Smith weakly universal Turing machine. The present paper does not use that result as the proof of quantum universality; the proof is the Clifford closure plus magic/contextuality [17, 16]. The point is sharper: the classical shadow of the photonic holonet lands on the same minimality boundary. The quantum machine has forty rays and an $81 = q^4$ protected register, but its smallest visible finite-state control alphabet is already the substrate pair $2, 3$.

6.5 The Universality Theorem

Theorem 6.1 (Clifford completeness from optics)**.** *The symplectic images of the photon's physical elements — the tritter F_3 on either register, the quadratic phase plate S, and the delay-conditioned* CX *— generate all of* $\mathrm{Sp}(4, \mathbb{F}_3)$ *(closure of order* 51840*, exact). Hence tabletop optics realize the complete two-qutrit Clifford group modulo Paulis and phases.* Witness: `bt825_universality_theorem.py`.

Theorem 6.2 (Universality)**.** *Clifford completeness, together with the machine's intrinsic magic supply (§9) and the strict positivity of its contextual fraction (1/10, exact), implies universal quantum computation by magic-state injection; for qutrits, contextuality is necessary and sufficient for magic distillation (Howard–Wallman–Veitch–Emerson) [3]. One self-entangled photon on the $W(3,3)$ mesh is a universal quantum computer whose transport is its gate action.*

7 Fractal scaling: the computer is the network

The single-core theorem is not the end of the architecture. The substrate is self-similar: any node may be replaced by a fresh copy of the same forty-node holonet, with the parent line/point incidence becoming the routing grammar for the child layer.

Definition 7.1 (Recursive holonet)**.** *Let $\mathcal{H}_0$ be one self-entangled photonic qutrit. Define $\mathcal{H}_n$ recursively by replacing each point of one copy of $W(3,3)$ with a copy of $\mathcal{H}_{n-1}$. Thus $\mathcal{H}_1$ is the single-core machine of this paper and $\mathcal{H}_n$ is a 40-ary fractal computer/network.*

Theorem 7.2 (BT827 holonet scaling law)**.** *A level-n holonet has*

$$N_n = 40^n$$

photonic leaf cores and

$$I_n = 1 + 40 + \cdots + 40^{n-1} = \frac{40^n - 1}{39}$$

total $W(3,3)$ instances. Its total edge-qutrit slots are $240I_n$, its directed transport slots are $480I_n$, its chart routers are $540I_n$, its apartment links are $1620I_n$, its mirror slots are $2160I_n$, and its local Clifford runtime atoms are $51840I_n$. Worst-case reversible routing across the recursive address uses at most

$$8n = 8\log_{40} N_n$$

moves, where $8 = 3+5$ is one in-chart hypercube diameter plus one chart-web apartment diameter. Durable commits use the distinct Császár/tomotope clock

$$T(0) = 1, \qquad T(g) = 4(7^g - 1) \quad (g \geq 1).$$

Witness: `bt827_holonet_fractal_architecture.py`.

level n	leaves 40^n	$W(3,3)$ instances	mirror slots	runtime atoms	route bound
1	40	1	2160	51840	8
2	1600	41	88560	2125440	16
3	64000	1641	3544560	85069440	24
4	2560000	65641	141784560	3402829440	32

This resolves the computer-engineering question. A conventional machine grows by adding processors connected by a separate network. The holonet grows by adding copies of the same object; each copy contains its own processor, router, memory, clock, and packet format. There is no von Neumann bus to saturate. At every scale, the edge set is both communication bandwidth and entangling-gate availability, the chart atlas is both routing table and instruction decoder, and the Steinberg 81-sector is both protected memory and fault-tolerant logical space.

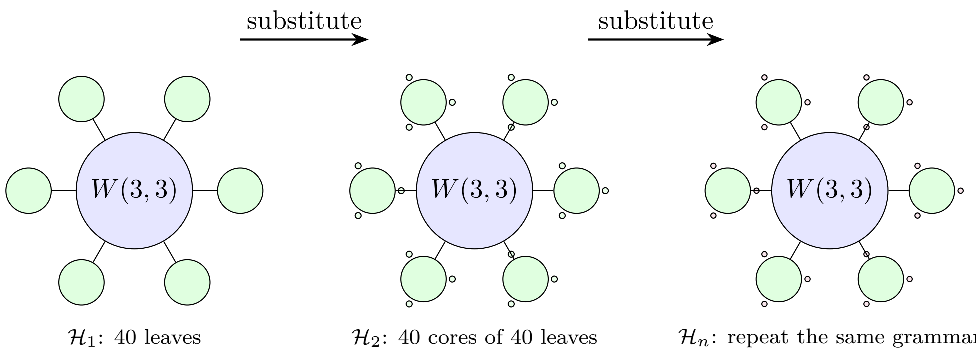


Figure 6: Fractal scaling schematic. The drawn fans show only a few children; the rule is exact substitution of forty child holonets at each substrate point. Routing grows logarithmically in the number of photonic leaves because each recursive digit costs at most eight reversible moves.

8 Runtime closure: compile, monitor, commit

The architecture now has an executable operating-system loop.

$$\text{address word} \longrightarrow \text{packet route} \longrightarrow \text{sentinel monitor} \longrightarrow \text{durable tomotope commit.}$$

BT828–BT831 make each arrow finite and testable.

Theorem 8.1 (BT828 packet compiler)**.** *A recursive address word with digits in* $\{0,\ldots,39\}$ *compiles to a sequence of digit packets*

$$(Q_3\text{-}XOR\ hops,\ apartment\ hops,\ D_{12}\ mirror\ slot,\ C_{12}\ phase,\ tomotope\ block).$$

Each digit uses at most three XOR hops and at most five apartment hops, hence at most eight reversible moves. The stress route

$$(0,7,14,21,28,35) \longrightarrow (39,32,25,18,11,4)$$

compiles to 47 *reversible moves, below the level-six bound* 48*.* Witness: `bt828_holonet_packet_compiler.py`.

Theorem 8.2 (BT829 sentinel monitor)**.** *The projector onto the* $g = 15$ *adjacency sector of* $W(3,3)$ *has exact entry profile*

$$P_{-4}(x,x) = \frac{3}{8}, \qquad P_{-4}(x,y) = \begin{cases} -\frac{1}{8}, & x \sim y, \\ \frac{1}{24}, & x \nsim y. \end{cases}$$

Consequently a full context line K_4 *is sentinel-invisible, while localized, mirror, and shell faults activate the sentinel with exact energies:*

fault	*energy*	*normalized fraction*
single point	3/8	5/13
adjacent edge	1/2	5/19
nonedge mirror pair	5/6	25/57
gauge shell $N(x)$	6	5/7
matter shell	27/8	5/13

The immune system is therefore not generic noise detection: it is blind to legal context-line traffic and tuned to off-context routing and shell congestion. Witness: `bt829_fault_sentinel_monitor.py`.

Theorem 8.3 (BT830 two-phase commit clock)**.** *The runtime has two clocks. The reversible prepare phase finishes in at most $8n$ moves. The durable commit phase waits for*

$$T(n) = 4(7^n - 1).$$

For $1 \le n \le 24$, $T(n)$ is always divisible by 24 and by 8, so it is aligned with both the local runtime stabilizer and the single-digit route window. But it is not always divisible by the full $8n$ route epoch: the first desynchronization is $n = 5$, with remainder $24 = f$. Thus the commit layer is deliberately slower and cover-like; it is not merely the route clock renamed. Witness: `bt830_two_phase_commit_clock.py`.

8.1 The infinite-cover warning

The tomotope contributes a harder boundary than the earlier packet picture showed. Monson, Pellicer, and Williams prove that the tomotope has infinitely many distinct minimal regular covers; for coprime odd $p, q > 1$ there are minimal covers R_p, R_q of the tomotope such that neither covers the other, and neither covers the special R_2 cover [20]. They also record the toroidal cover arithmetic

$$|\mathrm{Mon}(Q_k)| = 36864\, k^6, \qquad Q_k:\ (4, 24, 32, 8, 4)k^3$$

for vertices, edges, triangles, tetrahedra, and octahedra.

Architecturally this is a big correction. The 48-block tomotope middle layer is a stable local ABI, but the global regular cover is not unique. Durable storage must therefore carry a *cover index k* as an implementation gauge. Fast routes compile to the same invariant packet, while persistent storage may choose one minimal cover family without invalidating the other non-comparable families. *Witness:* `bt831_tomotope_minimal_cover_architecture.py`*.*

8.2 Cover-indexed storage, rerouting, and guard bands

The infinite-cover warning would be useless as engineering unless it changed the runtime. BT832–BT834 make it operational. The principle is that the local tomotope packet is the ABI and the chosen regular cover is the storage backend.

Theorem 8.4 (BT832 cover-indexed durable storage)**.** *For every supported cover index k, durable storage is the lifted packet space*

$$\{0, \dots, 47\} \times \mathbb{Z}_k^3, \qquad |slots| = 48k^3,$$

with projection back to the fast-route ABI by forgetting the $\mathbb{Z}_k^3$ coordinate. The kernel to the base tomotope packet has order

$$k^6 = (k^3)^2,$$

matching the Monson–Pellicer–Williams toroidal cover law $|\mathrm{Mon}(Q_k)| = 36864k^6$. Changing k changes durable capacity and kernel size, but not the packet program seen by the compiler. Tested cover gauges $k = 3, 5, 7, 11, 13$ all reduce exactly to the same 48-block ABI, and larger k reduces the load fraction of the same stress program. Witness: `bt832_cover_indexed_durable_storage.py`.

Theorem 8.5 (BT833 sentinel-aware packet rerouting). *The BT829 projector supplies a compiler-visible cost function. Before the durable commit, each digit route may insert up to two W33 waypoint points and minimize*

$$\textit{sentinel energy first, reversible move count second.}$$

The extra reversible moves are charged to the slower BT830 commit phase, not to the BT827 fast $8n$ *route bound. In the level-six stress program the direct route uses* 47 *moves and total sentinel energy* 4*; the sentinel-aware route uses* 58 *moves, still far below the* 470592*-tick commit window, and lowers the total sentinel energy to* 2*. Each tested digit route strictly lowers its own* $g = 15$ *energy; adjacent endpoints can be context-completed to zero, and nonedge mirror pairs drop below their direct* $5/6$ *activation.* Witness: `bt833_sentinel_aware_packet_rerouter.py`.

Theorem 8.6 (BT834 desynchronization guard-band arithmetic). *The two-phase clock has an exact number-theoretic boundary. Full route-epoch synchronization is*

$$8n \mid T(n) = 4(7^n - 1),$$

equivalently

$$2n \mid 7^n - 1.$$

For every odd prime power $p^a \mid n$ *with* $p \neq 7$*, this requires* $\mathrm{ord}_{p^a}(7) \mid n$*; if* $7 \mid n$*, synchronization is impossible because the base is not coprime. The first failure is* $n = 5$*:* $\mathrm{ord}_5(7) = 4 \nmid 5$*, and the remainder is*

$$T(5) \bmod 40 = 24 = f.$$

Thus the first cover-index where durable storage and the full route epoch separate is not empirical drift; it is the $f = 24$ *guard band of the tomotope-backed runtime.* Witness: `bt834_desync_guard_band_arithmetic.py`.

Theorem 8.7 (BT838 tomotope Wythoff runtime ladder). *The Grünbaum–Coxeter/tomotope operation tables give the local tomotope counts*

$$\textit{original},\ \textit{rectified},\ \textit{truncated},\ \textit{maximal expanded},\ \textit{omnitruncated} \quad = \quad 4,\ 12,\ 24,\ 48,\ 96.$$

These are exactly the runtime packet-growth layers:

$$4 = \mu \quad (\textit{vertex carriers}), \qquad 12 = k \quad (\textit{edge axes}), \qquad 24 = f \quad (D_{12}\ \textit{lift}),$$

$$48 = 2f \quad (\textit{BT814 packet ABI}), \qquad 96 = 4f \quad (\textit{half-flag boundary}).$$

The doubled omnitruncated boundary gives the verified full packet flags, $2 \cdot 96 = 192$*. After a cover lift this becomes*

$$48k^3 = \textit{BT832 lifted packet capacity}, \qquad 192k^3 = |W_k|,$$

the W_k *order already recorded in the tomotope cover architecture. Thus the Wythoff operations are not decorative variants; they are the compiler's packet expansion states:*

$$\textit{carrier} \to \textit{edge-axis} \to \textit{mirror lift} \to \textit{packet ABI} \to \textit{flag boundary}.$$

Witness: `bt838_tomotope_wythoff_runtime_ladder.py`.

9 The fuel: matter = magic

Theorem 9.1 (The triality). *In the photon's two-qubit reading, exactly the* 4 *basis rays are stabilizer states and all* 36 *ω-rays are magic (no stabilizer state has support* 3*). Consequently three classifications coincide exactly, on rays* $[4, 36]$ *and on contexts* $\{4{:}1,\ 1{:}12,\ 0{:}27\}$*: the holonet vacuum split, the Schmidt (self-entanglement) strata, and the magic strata.* The matter shell is the machine's non-classical resource sector. Witness: `bt822`.

Theorem 9.2 (Three grades and the cube inside). *Stabilizer fidelity splits the* 36 *magic rays into grades*

$$\underbrace{8}_{2^q}\ \textit{deep}\ \left(F = \tfrac{2+\sqrt{3}}{6}\right), \qquad \underbrace{24}_{f}\ \textit{mid}\ \left(F = \tfrac{5+2\sqrt{3}}{12}\right), \qquad \underbrace{4}_{\mu}\ \textit{shallow}\ \left(F = \tfrac{3}{4} = \tfrac{q}{\mu}\right),$$

the shallow grade sitting exactly at the Werner decoherence threshold. The 8 *deep rays are precisely the point set of a hyperbolic polar pair (the* 24*-cell vacuum axis), and the* 27 *fully-magic contexts stratify by grade signature as* $1+8+12+6$ *— the cell census of the cube.* Witnesses: `bt822`, `bt823`.

Theorem 9.3 (Exact contextuality budget). *The maximum number of contexts a classical* $\{0, 1\}$ *valuation can satisfy exactly-once is exactly* $36 = (q!)^2$ *(complete branch-and-bound over all* 2^{40} *markings): classicality saturates at the spread count, the deficit is exactly* $\mu = 4$*, and the contextual fraction is exactly* $1/10 = 1/\Phi_4$*. A perfect valuation would be an ovoid, and* $W(3, 3)$ *has none: the true independence number is* $\alpha = 7 = \Phi_6$*, attained only by the beacon heptads of §12, against the Lovász bound* $\vartheta = 10$ *— the* $\vartheta - \alpha$ *gap* $3 = q$ *is the combinatorial face of Kochen–Specker.* Witnesses: `bt818`, `bt823`.

10 Memory, protection, and the immune system

Theorem 10.1 (The code register is the Steinberg module). *The companion paper's CSS code* $[[240, 81, 4, 3]]_3$ *stores its* 81 *logical qutrits in* H_1 *of the* $W(3, 3)$ 2*-complex (*40 *vertices,* 240 *edges,* 160 *in-line triangles). Complete character computation over all* 25920 *group elements gives*

$$\langle \chi_{H_1}, \chi_{H_1} \rangle = 1.$$

Therefore $H_1 \otimes \mathbb{C}$ *is* irreducible *of dimension* 81*, hence equal to the unique* 81*-dimensional irreducible —* ***the Steinberg module*** *— with* $\chi_{H_1}(g) = \#\text{fixflags} - \#\text{fixpoints} - \#\text{fixlines} + 1$ *(the Solomon–Tits alternating sum) verified for every* g*. The QECC logical space and the Schur-protected register below are one object: any equivariant logical error is a scalar, so only symmetry-breaking noise can touch the code, and the sentinel/clock layers are exactly the symmetry monitors. The homological dictionary then completes:* H_0 *is trivial (the vacuum pole),* H_1 *is Steinberg (the register), and* H_2 *(dim* 40*, one tetrahedron-boundary* 2*-cycle per line) is the* sign-twisted *line module — a fixing symmetry acts on a line's* 2*-cycle by the parity of its induced* S_4 *action (verified for all* 25920 *elements;* not *the plain permutation module): the top homology is the* oriented *timetable carrier, feeding the chirality ledger.* Witnesses: `bt861_code_register_is_steinberg.py`, `bt862_h2_is_the_line_module.py`.

Corollary 10.2 (Three generations from Steinberg vanishing). $\chi_{\mathrm{St}}(g) = 0$ *iff* $3 \mid \mathrm{ord}(g)$ *(verified for all* 25920 *elements). Hence* any *order-*3 *symmetry of the substrate splits the matter register into eigenspaces of exactly* $27 + 27 + 27$ *— the three-generation decomposition is forced for every choice of triality element, not selected — and every order-*9 *element (*5760 *of them) refines it into nine* 9*-dimensional sub-generations. In the code's own characteristic the parameters survive:*

$\mathrm{rank}_3\partial_0 = 39$, $\mathrm{rank}_3\partial_1 = 120$, $\dim H_1(\mathbb{F}_3) = 81$, *so* $[[240, 81, 4, 3]]_3$ *is exact over* $\mathbb{F}_3$. *The number of fermion generations is the defining-characteristic degeneracy of the protected register.* Witness: `bt863_generations_from_steinberg_vanishing.py`.

Theorem 10.3 (The dual-torsor Steinberg compiler)**.** *Let* $H_{27} = 3_+^{1+2}$ *be the canonical Heisenberg* O_3 *of the point parabolic and let* $A_{27} = \mathbb{F}_3^3$ *be the canonical translation* O_3 *of the Bell-line parabolic. Their* 27*-point and* 27*-line shells are regular torsors (BT858). On the protected register,*

$$\mathrm{St}{\downarrow_{H_{27}}} = 3\,\mathbb{C}[H_{27}], \qquad \mathrm{St}{\downarrow_{A_{27}}} = 3\,\mathbb{C}[A_{27}],$$

because both restricted characters are exactly $\{81^1, 0^{26}\}$*. The statement survives in the code's own field and is constructive:*

$$H_1(W(3,3);\mathbb{F}_3){\downarrow_{H_{27}}} \cong \mathbb{F}_3[H_{27}]^{\oplus 3}, \qquad H_1(W(3,3);\mathbb{F}_3){\downarrow_{A_{27}}} \cong \mathbb{F}_3[A_{27}]^{\oplus 3}.$$

For each torsor the verifier finds three cycle seeds whose 27 *group translates add* $27 + 27 + 27$ *independent classes modulo the* 120*-dimensional boundary space. Thus the same logical memory has a flat program basis and a curved state basis:* A_{27} *is the commuting three-trit address compiler, while* H_{27} *is the noncommuting qutrit-phase compiler.*

The center $Z(H_{27}) = \langle z\rangle \cong C_3$ *selects a canonical triality. Over* $\mathbb{C}$ *its central-character blocks are*

$$H_1 = H_{1,1} \oplus H_{1,\omega} \oplus H_{1,\omega^2}, \qquad \dim H_{1,j} = 27.$$

The trivial block contains the nine linear characters with multiplicity 3*; the two nontrivial blocks contain the conjugate* 3*-dimensional Schrödinger representations with multiplicity* 9*. Over* $\mathbb{F}_3$*, where* $x^3 - 1 = (x-1)^3$*, the phases coalesce into a canonical unipotent flag. With* $N = z - I$*,*

$$(\mathrm{rank}\, N, \mathrm{rank}\, N^2, \mathrm{rank}\, N^3) = (54, 27, 0),$$

so

$$0 \subset \ker N \subset \ker N^2 \subset H_1, \qquad \dim = (0, 27, 54, 81),$$

has three successive quotients of dimension 27*. Equivalently,* z *acts by* 27 *Jordan blocks of length* 3*. Complex generations are phase sectors; native qutrit generations are the three layers of one depth-*3 *memory stack.*

The freeness is canonical, but the displayed orbit bases are not: H_{27} *and* A_{27} *are nonisomorphic, and a basis-to-basis matrix still requires a chosen point, line, shell origin, and cycle seeds. No canonical intertwiner is claimed.* Witness: `bt865_dual_torsor_steinberg_compiler.py`.

Theorem 10.4 (Steinberg memory)**.** *The Levi-graph cycle space and the chart-overlap* 81*-sector are both irreducible and equal to the Steinberg representation of* $\mathrm{PSp}(4,3)$ *— the Solomon–Tits homology of the building, whose canonical basis is the* 81 *apartments through any chamber. Schur's lemma then forces the uniqueness of the chart–cycle bridge. Restricted to a Sylow* 3*-subgroup the module is free of rank one: the protected memory is one free generator over the substrate's ternary symmetry.* Witnesses: `bt742`, `bt744`.

Corollary 10.5 (The oriented timetable spectrum)**.** *Let* $P = 3^3{:}S_4$ *be the index-*40 *line parabolic. The ordinary line permutation module and the oriented top-homology module are the two inductions of its two linear characters:*

$$\mathrm{Ind}_P^{\mathrm{PSp}(4,3)}(1) = 1 \oplus 15 \oplus 24, \qquad H_2 = \mathrm{Ind}_P^{\mathrm{PSp}(4,3)}(\mathrm{sign}_{S_4}) = 5_\omega \oplus 5_{\omega^2} \oplus 30.$$

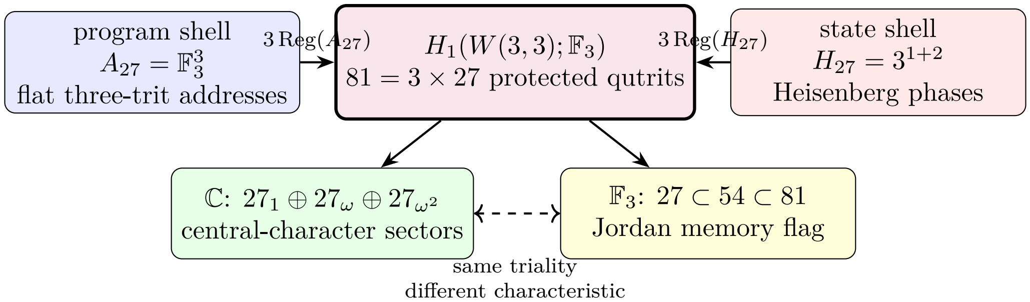


Figure 7: The dual-torsor compiler. Flat program translations and noncommutative state translations give two free rank-3 module coordinates on one Steinberg-protected register. The Heisenberg center reads as three phase sectors over $\mathbb{C}$ and as a three-step unipotent memory flag over $\mathbb{F}_3$.

The two 5-dimensional characters are complex conjugates over $\mathbb{Q}(\zeta_3)$; the 30 is rational. Under the outer extension $W(E_6) = U_4(2){:}2$, the conjugate pair is the restriction of one irreducible 10-dimensional representation, whereas the 30 has two inequivalent extensions. Thus the oriented timetable header reads as a Weyl-visible chiral channel 10 *plus a payload* $30^{\pm}$ *whose outer parity is not fixed by the projective module alone.*

The completed homology spectrum is

$$H_0 = 1, \qquad H_1 = \mathrm{St}_{81}, \qquad H_2 = 5_\omega \oplus 5_{\omega^2} \oplus 30,$$

and resolves the Euler charge representation by representation:

$$1 - 81 + (5 + 5 + 30) = -40 = -v.$$

BT867 resolves the tempting cache comparison negatively: neither 162*-cache branch is one of the two 5-dimensional sectors. The branches are isomorphic copies with identical character overlaps; the outer involution acts on their multiplicity label. The possible identification of the two 30-extensions with BT857's local gauge bit remains open.* Witness: `analysis/bt866_h2_oriented_irreducible_decomposition.py`.

Theorem 10.6 (The cache/transport extension boundary)**.** *Let $H = 3^3{:}S_4$ be the order-*648 *Bell-line parabolic. The two* 162*-slot BT859 cache branches have conjugate cyclic stabilizers and are therefore isomorphic transitive H-sets,*

$$\mathcal{C}_L \cong H/C_4 \cong \mathcal{C}_R.$$

Their permutation characters are equal. In particular, their scalar products with the restrictions of the three BT866 characters $(5_\omega, 5_{\omega^2}, 30)$ are both $(1, 1, 8)$: each cache sees both conjugate 5-sectors equally. "Left" and "right" are copy labels, not internal spectral chiralities.

The parabolic contains one conjugacy class of 81 *cyclic C_4 subgroups, and*

$$N_H(C_4) = D_8, \qquad H/C_4 \longrightarrow H/D_8, \qquad 162 \longrightarrow 81$$

is the canonical free two-sheet cache cover. Both copies cover this same stabilizer base. On their union GAP computes

$$C_{\mathrm{Sym}(324)}(H) \cong D_8, \qquad 324 = 81 \times 4 = 81 \times 2_{\mathrm{deck}} \times 2_{\mathrm{cache}},$$

with exactly 81 *commutant orbits of size four. Thus the four-state fiber carries the symmetry of a square, not merely a numerical pair of bits.*

This cache carrier is not the older ternary 162*-sector. On one cache fiber the deck swap and its two projectors are*

$$D = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad P_{\pm} = \frac{I \pm D}{2}, \qquad \mathbb{F}_3^2 = \operatorname{im} P_+ \oplus \operatorname{im} P_-.$$

Since 2 *is invertible in* $\mathbb{F}_3$*, the cache splits; indeed* $(D-I)^2 = D - I$*. By contrast, packet transport uses*

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad N^2 = 0, \qquad (I+N)^3 = I,$$

and tensors this indecomposable fiber with the 81*-register to obtain the non-split sequence* $0 \to 81 \to 162 \to 81 \to 0$*. No change of basis can identify the idempotent cache difference with the nilpotent transport shift. The architecture therefore separates a reversible* address/control plane *from a memory-bearing ternary* temporal data plane *by extension class, not by convention.* Witness: `analysis/bt867_cache_split_transport_nonsplit_boundary.py`.

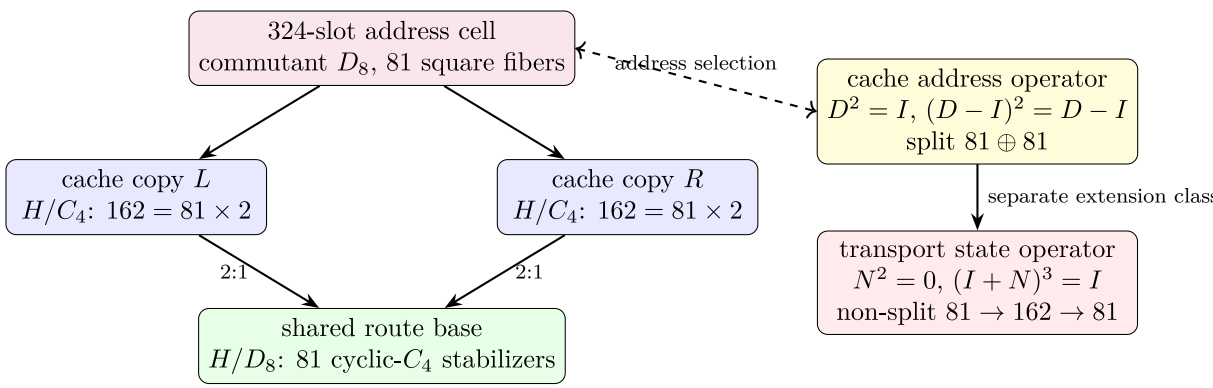


Figure 8: The local cache cell and the split/non-split boundary. The D_8 square fiber chooses a reversible address over one of 81 routes; the independent ternary nilpotent updates state while retaining temporal memory. Equal 162-dimensional counts do not imply equivalent modules.

The error layer is native: the $[[240, 81, 4, 3]]_3$ CSS code on the edge qutrits has all four parameters substrate-primitive; the dual Császár/Szilassi cross-check supplies parity-free error detection; and the web's unique non-Ramanujan eigenspace — dimension $15 = g$ — is the spectral sentinel where drift appears first.

11 Time: three clocks and a quasicrystal

Theorem 11.1 (Clock trichotomy)**.** *Of the program's three natural clocks, exactly one is internal:* $\mathbb{Z}_{12}$ *(the rectangle clock,* $12 \mid 51840$*) runs the selector and duo machinery, while* $\mathbb{Z}_7$ *(Császár) and* $\mathbb{Z}_{13}$ *(Singer) do not divide the group order and act as external references — both with decimal period* $6 = q!$*, so the digit expansion of* $1/7$ *is literally the internal clock's shadow modulo its duo bit.* Witnesses: `bt774`, `bt803`, `bt807`, `bt819`.

Theorem 11.2 (The Boerdijk–Coxeter time quasicrystal)**.** *Driving the loop by the BC twist* $\theta = \arccos(-2/3)$ *per round trip yields a stroboscopic orbit that provably never repeats (Niven), with* *the Steinhaus three-distance signature at all substrate step counts and — the substrate's signature moment — exactly* two *gap lengths at $n = 30 = h(E_8)$, the BC ring length, where closure happens in S^3 (the 600-cell) rather than the phase circle (deficit 0.0158). This is the photonic sibling of the Fibonacci-drive dynamical topological phase.* Witness: `bt820`.

Theorem 11.3 (The arrow of time)**.** *Without its two-trit feedforward record, the photon's self-configuration loop is the completely depolarizing channel with unique fixed point I/q — exactly the Bell marginal; with the record it is unitary. Decoherence is self-forgetting, priced at $\log_3 q^2 = 2$ trits per cycle.* Witness: `bt823`.

12 Synchronization: beacons and schedules

The maximum classically coherent constellations are the *beacon heptads*: seven states with all pairwise overlaps exactly $1/3 = 1/q$ — a complete K_7 interference mesh, the Császár toroidal node realized in light. There are exactly 2880 heptads in a single transitive family; each has a $\mathbb{Z}_3 \times \mathbb{Z}_3$ symmetry fixing a unique *master beacon* and cycling two triads ($7 = 1 + 3 + 3$), and single-beacon swaps connect the family into its own exchange network: the reference frames of the network form a network. The measurement timetable is rigid: there exist exactly 36 complete schedules (partitions of the 40 states into 10 contexts) — an exhaustive result that simultaneously proves every spread of $W(3,3)$ is regular — with each context appearing in exactly 9 schedules. *Witnesses: `bt817`, `bt819`.*

Theorem 12.1 (Schedule-overlap law)**.** *Two distinct schedules share exactly* 1 *or exactly* $4 = \mu$ *contexts: each schedule has* 15 near *partners (overlap* 4*) and* 20 far *partners (overlap* 1*), the two relation classes of the double-six diagonal* $[1, 15, 20]$*, with the exact counting identity* $1 \cdot 20 + 4 \cdot 15 = 80 = 10 \times (9 - 1)$*. Operationally: a cheap timetable switch preserves* 4 *of* 10 *calibrated contexts and a far switch only* 1*, so a controller hopping among near partners re-calibrates* 6 *contexts per hop — the timetable library is itself a metric space whose geometry is the double-six relation.* Witness: `bt835_schedule_overlap_theorem.py`.

The Grünbaum–Coxeter cells. Each schedule also has an internal shape. The full schedule stabilizer S_6 is 2-homogeneous on the 10 contexts, but its icosahedral core A_5 — the 600-cell level of the platonic ladder, and *exactly* the automorphism group of the hemi-dodecahedron $\{5,3\}_5$ — splits the $\binom{10}{2} = 45$ context pairs as $45 = 15 + 30$, and the 15-orbit graph on the 10 contexts is the **Petersen graph**: the edge skeleton of the hemi-dodecahedron, the cell of Coxeter's exceptional 57-cell. Dually, the hidden 6-set of the S_6 action realizes the hemi-icosahedron $\{3,5\}_5$ — the cell of Grünbaum's 11-cell — with incidence count $(6, 15, 10)$: six hidden points, fifteen duads, ten triple-splits = the contexts themselves. The exceptional polytopes' groups do not embed in $\mathrm{Sp}(4, \mathbb{F}_3)$, but they are substrate arithmetic ($|\mathrm{PSL}(2, 11)| = 660 = k \cdot N_{\mathrm{eff}}$, $|\mathrm{PSL}(2, 19)| = 3420 = k \cdot g \cdot 19$, Heawood genus $H(19) = 20$ = the BC ring count, and $11 = k - 1$ is the Ihara prime of the zeta critical circle $|u| = 1/\sqrt{11}$): the full 11-cell and 57-cell live elsewhere, yet their *cells* are carried by every measurement timetable of this machine, glued by the substrate's own group instead of $\mathrm{PSL}(2, p)$. *Witness: `bt836_gc_hemicells_in_spreads.py`.*

Theorem 12.2 (The library is a classical geometry)**.** *The timetable library carries its own rank-3 geometry, and the hemicell structure fibers over it with perfect uniformity: (i) every skew line pair lies in exactly* 3 *schedules, so the* (schedule $\supset$ pair) *flag count is* $36 \times 45 = 1620$ *— the apartment count of the Tits building — and* 540×3*; (ii) the near-partner graph on the* 36 *schedules is strongly* *regular* $\mathrm{SRG}(36,15,6,6)$*, the* other *classical rank-3 graph of* $U_4(2) \cong \mathrm{PSp}(4,\mathbb{F}_3)$*, so the control plane (timetables) and the data plane (*$W(3,3) = \mathrm{SRG}(40,12,2,4)$*) are the two rank-3 geometries of one group; (iii) each schedule stabilizer contains exactly* 6 *icosahedral* A_5 *cores (*$216 = 6^3$ *in all, each marking its unique schedule via its* $[10,30]$ *line orbits) giving* 6 *distinct Petersen splits; each internal pair is a Petersen edge under exactly* 2 *of the* 6 *cores, hence every skew pair of* $W(3,3)$ *has exactly* $6 = 3 \times 2$ *hemi-dodecahedral homes (*$3240 = 540 \times 6$ *Petersen flags).* Witness: `bt837_schedule_library_geometry.py`.

Theorem 12.3 (Two compasses, twelve Petersens, $4 \cdot K_{10}$)**.** $\mathrm{PSp}(4,\mathbb{F}_3)$ *has exactly two conjugacy classes of* A_5*, both with normalizer* S_5 *and* 216 *conjugates, unfused by the outer automorphism, and* both fiber over the same 36 schedules*: every schedule stabilizer contains* $12 = 6+6$ *icosahedral cores —* 6 duad *cores (line signature* $[10,30]$*, the Theorem-12.2(iii) cores) and* 6 pentad *cores (signature* $[5,5,10,20]$*: transitive on the schedule, plus two distinguished* pentads *of* 5 *pairwise-disjoint lines outside it). The two* 216*-sets are non-isomorphic G-sets (rank* 10 *suborbits* $[1,5,10,10,20,20,20,30,40,60]$ *vs* $[1,5,10,10,20,20,30,30,30,60]$*); the Petersen flags form the coset geometry* $\mathrm{PSp}(4,\mathbb{F}_3)/D_4$ *(*3240 *flags, dihedral stabilizer). There are* 432 *distinct pentads in* two chiral orbits *of* 216 *(stabilizer order* 120*); every core pairs one LEFT with one RIGHT pentad, and each pentad serves exactly one core. Pentads are* maximal partial spreads*: no schedule contains one. The two pentads of one core interlock as* $K_{5,5}$ *minus a perfect matching — mutual disjointness is* forbidden *by the overlap law (it would create a* 0*-overlap schedule) — and the deleted matching consists of* 5 skew *pairs, i.e. five hypercube charts; over the* 216 *cores these matchings cover the* 540*-chart atlas* ***exactly twice*** *(*$1080 = 540 \times 2$*): the routing fabric is readable off the pentad compasses, five charts per needle. Finally,* all twelve *cores split the* 45 *internal pairs as Petersen* 15*-orbits, and the twelve Petersen edge-sets cover every pair exactly* $\mu = 4$ *times:*

$$4 \cdot K_{10} \;=\; 12 \textit{ Petersens.}$$

Schwenk's classical obstruction (no edge-partition of K_{10} *into* 3 *Petersen graphs) dissolves at the substrate multiplicity* μ*. The numerical coincidence* $3240 = \#\{\textit{triangles of } Q\}$ *is refuted at the G-set level (orbits* $[360, 2880]$ *vs transitive).* Witnesses: `bt843_icosahedral_compass.py`, `bt844_double_five_and_six_petersens.py`, `bt845_pentad_exchange_and_chart_double_cover.py`.

Theorem 12.4 (The amalgamation ladder)**.** *The* 11*-cell and* 57*-cell are* universal *abstract polytopes: the unique ones with hemi-icosahedral facets and hemi-dodecahedral vertex figures (group* $L_2(11)$*, no proper quotients) and vice versa (*$L_2(19)$*). Their incidence is icosahedral-subgroup geometry: each of* $L_2(11)$*,* $L_2(19)$*,* $U_4(2)$ *has exactly two conjugacy classes of* A_5*, and in all three the distinguished incidence is* intersection in D_{10} *with valency* 6 *— the* 11*-cell's vertex–facet relation, the* 57*-cell's facet adjacency (the Perkel graph), and the substrate's compass incidence (*$216+216$*, bipartite* 6*-regular,* $216 \times 6 = 1296 = 6^4$ *incident pairs* $= 36$ *schedules* $\times\, 36$ *core pairs). The closures of one amalgam* $A_5 *_{D_{10}} A_5$ *(GAP):*

$$\langle A, B\rangle = \begin{cases} L_2(9) \cong A_6 \ (360) & \textit{in } U_4(2)\textit{: inside exactly one schedule stabilizer,}\\ L_2(11) \ (660) & \textit{the } 11\textit{-cell,}\\ L_2(19) \ (3420) & \textit{the } 57\textit{-cell.}\end{cases}$$

One local icosahedral amalgam, a $\mathrm{PSL}(2)$*-ladder of completions over* $9 = q^2$*,* $11 = k-1$ *(the Ihara prime),* 19 *(Heawood,* $H(19) = 20$*). Universality plus Lagrange (*$11, 19 \nmid 25920$*) shows the substrate carries the* complete local data *of both exceptional polytopes while their rank-*4 *amalgamation is arithmetically forbidden inside* $\mathrm{Sp}(4,\mathbb{F}_3)$*: the GC polytopes are the substrate's sibling completions,* *not subobjects. One step up the universal ladder (*$\{\{5,3\},\{3,5\}_5\}$*, 10006920 facets) the completion group is* $J_1 \times L_2(19)$ *— the first sporadic Janko group, whose involution centralizer* $2 \times A_5$ is *the compass datum. Finally, the compass Petersen* 15*-orbit carries exactly* 12 *pentagons splitting under the compass into two* 6*-orbits, each covering every edge exactly twice: two* chiral fully-faced *hemi-dodecahedra per needle, not merely skeletons. The dark sector closes the calculus: its* 190 *line pairs split* $[10,30,30,30,30,60]$ *under the core into a perfect matching of* 10 dark charts*, two chiral* $5\times K_4$ *partitions into skew tetrads, the two dark dodecahedra, and a* 6*-regular meeting frame; the dark matching is* schedule-shadow pairing *(each dark line meets exactly* 4 *schedule lines; matched lines share their shadow), and in* K_5 *terms the ten shadows are exactly the ten "edge + opposite triangle" configurations — a canonical bijection dark chart* $\leftrightarrow$ K_5 *edge* $\leftrightarrow$ *schedule line. The timetable is stored three ways (lit-chart transversals,* K_5 *edges, dark shadows): built-in erasure redundancy. Each chiral tetrad's four distinguished edges form a* K_5 star*, the five tetrads of each partition marking the five lit charts exactly once:* K_5 *vertices are stored three ways (lit charts, left/right tetrads), edges two ways (schedule lines, dark charts) — every* A_5*-orbit structure of the pentad core except the dodecahedra and the meeting frame is a* K_5 *element — and those two fold in as well: each chiral dark dodecahedron is the* antipodal double cover *of the Petersen (Kneser) graph on the* 10 *dark charts, with deck transformation the shadow matching (antipode = shadow partner, verified), and the meeting frame is the doubled Johnson graph* $T(5)$*. The* K_5*-relation census is constant on every orbit (matching = same edge, tetrads and meeting = adjacent, dodecahedra = disjoint): the dark sector is the canonical* 2:1 *cover of the* K_5 *edge-world, with the Petersen graph appearing three ways in one needle (schedule skeleton, dark Kneser quotient, doubled chiral covers).*

Theorem 12.5 (The dark charts are the mirror bus)**.** *The* (core, dark chart) *slot space —* 216 *pentad cores* $\times 10$ *dark charts* = 2160 *slots, with every skew pair occurring as a dark chart in exactly* 4 *cores (*$2160 = 540 \times 4$*, the repair-atlas factorization) — is* ***G-set isomorphic to the*** 2160***-slot*** D_{12} ***mirror bus*** *of the middleware: the slot stabilizer has order* 12 *with the* D_{12} *profile* $\{1{:}1, 2{:}7, 3{:}2, 6{:}2\}$ *and is* conjugate in $\mathrm{PSp}(4,\mathbb{F}_3)$ *to the antipode-slot stabilizer (direct search over all* 25920 *elements). The multiplicity-*4 *agreement is canonical: a dark chart's transversal tetrad is its schedule shadow, so the* 4 *host cores' schedules all contain the pair's* 4 *transversals. Hence the compass layer* generates *the middleware — timetables from its lit side (reconstruction), the transport bus from its dark side — and the mirror bus acquires a hardware identification: one slot = one needle's dark chart.* Witnesses: `bt856_dark_mirror_bus.py`; further: `bt853_dark_orbit_zoo.py`, `bt854_dark_chart_transversal_duality.py`, `bt855_dark_sector_k5_complete.py`.

Witnesses: `bt848_universal_amalgam_ladder.py`, `.tmp/gap_bt848*.g`; cf. Hartley's classification (17 universal rank-4 locally projective polytopes, 441 quotients) and Hartley–Leemans on $\{5,3,5\}$.

The barren rung and the maniplex workaround. Relaxing polytopes to *maniplexes* (flag graphs without the intersection property; the tomotope middleware of this machine is a uniform 4-polytope whose monodromy $18432 = 96 \times 192$ fails that property, giving it *infinitely many* distinct minimal regular covers — the canonical rank-4 cover pathology) does not lift the obstruction: exhaustive search shows $U_4(2)$ admits *no* rank-4 regular maniplex with a 5 in its type, and the amalgam closure $A_6 = L_2(9)$ admits no regular maniplex at any rank — it is the *barren rung* of the PSL(2) ladder, whose populated rungs are exactly the 11-cell (q=11), the 57-cell (q=19), and Leemans–Toledo's degenerate single-facet $\{5,5,5\}$ family (our control search rediscovers all of them, including both exceptional polytopes). This is why the Grünbaum–Coxeter content of the substrate is *delocalized* over the 36 schedules — Petersen splits, chiral pentads, chart covers, dark dodecahedra — rather than crystallized in one flag object, and why the machine's runtime uses the tomotope:

it is the flag structure that survives the obstruction. The tomotope's symmetry type is now exact: $|\mathrm{Aut}| = 96$ (independent centralizer computation on the 192 flags), two flag orbits $[96, 96]$, **class** $2_{\{0,1,2\}}$ — only the cell-color crosses orbits, and the orbits are precisely "flag in a tetrahedron" vs "flag in a hemioctahedron": the middleware *alternates spherical and projective cells*, the exact situation the classical "locally X" taxonomy cannot name, and the setting of Mochán's 2-orbit polytopality theory. Moreover the tomotope is a **Cayley maniplex over** $V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2$ (vertex action = full S_4 with a regular Klein subgroup; edge and face actions faithful and transitive): the middleware's vertex layer is a free $\mathbb{F}_2^2$-torsor, so register XOR-addressing extends from the chart atlas into the runtime with no added structure. Reading Hartley's full classification (seventeen universal locally projective rank-4 polytopes) against the substrate adds three facts: Aut(**tomotope**) **is itself a universal polytope group** — isomorphic to $\Gamma(\{\{4,3\}_3, \{3,4\}_3\})$, the quotient-free doubly-projective $\{4,3,4\}$ universal built from hemicubes and hemicrosses (the tomotope's own projective cell type), with the case-10 group $= C_2 \times \mathrm{Aut}$ of order $192 =$ the flag count; the mixed $\{3,5,3\}$ amalgam (icosahedral facets, hemi-dodecahedral vertex figures) *does not exist* — the construction closes into the 11-cell — while the dodecahedral mix is the J_1 giant; and the chart–compass universal $\{\{4,3\},\{3,5\}_5\} = 2^{\mathcal{I}}$ (cube facets $= Q_3$ charts, hemi-icosahedral vertex figures = compass cells, vertices $= \mathbb{F}_2^6$ on the hidden 6-set) has order $3840 \nmid 25920$: blocked like the rest of the ladder. *Witnesses:* `bt849_maniplex_barren_rung.py`, `bt850_tomotope_two_orbit_class.py`, `bt851_tomotope_cayley_test.py`, `bt852_seventeen_universals.py`, `.tmp/gap_bt849*.g`, `.tmp/gap_bt852*.g`.

Reconstruction, the chart K_5, and the dark dodecahedra. The pentad core's full signature $[5, 5, 10, 20]$ (lines), $[20, 20]$ (points) is rigid: both chiral pentads cover the *same* 20-point orbit (lit/dark halves of the substrate), the five matching charts' common-transversal bundles overlap to exactly the 10 schedule lines — **a pentad pair generates its timetable** — and the line $\to$ chart-pair map is a bijection onto the $\binom{5}{2} = 10$ edges of K_5: the schedule *is* a K_5 on its own charts (4 transversals per chart, 2 charts per line). The 20 dark lines complete the $\{5,3\}$ family: their 190 pairs split $[10, 30, 30, 30, 30, 60]$ under the core, and exactly two of the 30-orbits are **dodecahedron skeletons** (3-regular, girth 5, diameter 5) — a chiral pair of genuine dodecahedra beneath every compass needle, one level under the hemi-dodecahedral Petersen splits of the schedule itself. *Witnesses:* `bt846_pentad_core_anatomy.py`, `bt847_dark_dodecahedron.py`.

Theorem 12.6 (BT839 GC operation Euler/flag audit)**.** *Treating the full Grünbaum–Coxeter note as a data table gives a second invariant. The base tables for the* 11*-cell,* 57*-cell, tomotope partial a, and tomotope partial b are all Euler-neutral:*

$$V - E + F - C = 0.$$

But every Wythoff operation table has a fixed family charge:

$$\chi_{\mathrm{op}}(11\text{-}cell) = -11, \qquad \chi_{\mathrm{op}}(57\text{-}cell) = -57, \qquad \chi_{\mathrm{op}}(tomotope) = -4.$$

The omnitruncated rows are full-flag carriers:

$$660 = |\mathrm{PSL}(2,11)| = 11 \cdot A_5 = kN_{\mathrm{eff}},$$

$$3420 = |\mathrm{PSL}(2,19)| = 57 \cdot A_5 = k \cdot g \cdot 19,$$

and the tomotope has 96 *half-flags, whose double is the verified* 192*-flag BT814 packet. The* 57*-cell also locks to the W33 timetable library: BT837 gives* 3240 *Petersen homes, and*

$$3240 + k \cdot g = 3240 + 180 = 3420.$$

Thus one $k \cdot g$ sentinel sheet completes the W33 Petersen-home carrier to the 57-cell flag count.

The regular-polychoron alignment supported by the cell/symmetry data is

$$11\text{-}cell \to 600\text{-}cell, \qquad tomotope \to 24\text{-}cell, \qquad 57\text{-}cell \to 120\text{-}cell.$$

The $57 \to 120$ link is direct at the dodecahedral cell level; the $11 \to 600$ link is weaker but real, through hemi-icosahedral local structure and the quotient $2I \to A_5$ rather than through a direct tetrahedral-cell match. The swapped $11/57$ orientation is recorded as a lower-scoring dual/vertex-figure hook, not discarded. Witness: `bt839_gc_operation_euler_flag_audit.py`.

Theorem 12.7 (BT840–BT842 carrier closure)**.** *The three unresolved carriers in the Grünbaum–Coxeter paragraph are now explicit objects.*

(i) The* 180 *sentinel sheet. *The verified Clifford L/R boundary has 36 cells arranged as a 6×6 rook grid. Its zero-overlap relation is not a numerical afterthought: it is the union of the 6 row fibres and 6 column fibres, each fibre carrying the duads of a six-set. Hence*

$$12 \cdot \binom{6}{2} = 12 \cdot 15 = k \cdot g = 180.$$

This is exactly the sheet missing from BT837:

$$3240 + 180 = 3420 = |\mathrm{PSL}(2, 19)|.$$

Equivalently, the timetable library gives 216 Petersen cores ($216 \cdot 15 = 3240$), and the rook boundary supplies the missing 12 six-set fibres ($12 \cdot 15 = 180$).

(ii) The* 660 *carrier. *At an apex cell (r, c) of the same 6×6 grid, remove the incident row and column. The apex plus the five nonincident rows and five nonincident columns gives*

$$11 = 1 + 5 + 5.$$

Crossing these eleven labels with the verified 60-element Clifford A_5 selector gives

$$11 \cdot A_5 = 11 \cdot 60 = 660 = kN_{\mathrm{eff}}.$$

This is a carrier theorem only: it does not assert an internal $\mathrm{PSL}(2, 11)$ action on $W(3, 3)$.

(iii) The* 96 *tomotope half-flags. *In the D_4-root model of the 24-cell, every edge lies in a unique central hexagon plane. The two endpoint axes determine the third, missing axis in that plane, so each edge maps to a Reye incidence (missing axis, hexagon plane). Each of the 48 Reye incidences has exactly two edge lifts:*

$$96 = 2 \cdot 48.$$

Thus the tomotope omnitruncated half-flag count is literally the 24-cell edge lift of the common tomotope/Reye spine.

The hexagon clue supplies the conceptual bridge to the 11-cell. Each central 24-cell hexagon is a six-root cycle C_6. Completing that cycle to a full K_6 gives 15 duads, exactly the hemi-icosahedral skeleton count of the 11-cell cell. Across the 16 central hexagons,

$$16 \binom{6}{2} = 16 \cdot 15 = 240,$$

the W33 edge count and the E_8 root count. The tested dot profile of these 240 duad slots is

$$96_{\langle x,y \rangle = 1} \ + \ 96_{\langle x,y \rangle = -1} \ + \ 48_{\langle x,y \rangle = -2},$$

namely live 24-cell edges, mirror pairs, and repeated opposite-axis Reye incidences. Witnesses: BT840, BT841, and BT842; executable scripts are in `analysis/`.

13 Build sheet and falsifiable witnesses

Components (all standard quantum optics): one heralded single-photon source; one polarizing beam splitter; one symmetric 3-port coupler (tritter); a $0/\tau/2\tau$ delay ladder; one bin-synchronous electro-optic modulator; one polarization rotator set to $\arccos(-2/3)$ in the recirculation loop; single-photon detectors.

Witnesses (kill criteria, all parameter-free):

witness	predicted value	substrate form
trace–Choi visibility of F_3	$1/3$	$1/q$
trace–Choi visibility of X, Z	0	—
Kochen–Specker classical budget	$36/40$	$(q!)^2/v$
beacon-mesh pair visibility	$1/3$ (all 21 pairs)	$1/q$
Werner separability threshold	$3/4$	q/μ
BC-drive gap census at n=30	2 gap lengths	$h(E_8)$ ring
Hashimoto transport phases (one tick)	$\arctan\sqrt{10},\ \pi - \arctan(\sqrt{7}/2)$	Φ_4, Φ_6; $\lvert u\rvert^2 = 11$
key-agreement rate	$13/40$	Φ_3/v
D_{12} mirror-slot stabilizer	order 12	projective bus
runtime factorization	$24 \cdot 45 \cdot 48$	$\lvert\mathrm{Sp}(4, \mathbb{F}_3)\rvert$
one recursive route digit	≤ 8 reversible hops	$3 + 5$
level-n leaves	40^n	v^n
level-n $W(3, 3)$ instances	$(40^n - 1)/39$	geometric tower
durable commit clock	$4(7^g - 1)$ ticks	tomotope/Csaszar
compiled stress route	$47 < 48$ moves	BT828 level six
cover-lifted packet capacity	$48k^3$ slots	BT832 storage gauge
Bell–compass prefix router	$648 + 324 + 162 + 162$	code $0, 10, 110, 111$
sentinel-aware stress energy	$4 \to 2$	BT833 reroute
full route sync criterion	$2n \mid 7^n - 1$	BT834 guard band
tomotope Wythoff runtime ladder	$4, 12, 24, 48, 96$	BT838 packet growth
GC operation Euler charges	$-11, -57, -4$	BT839 table audit
sentinel projector trace	15	g sector
first full-route desync	$n = 5$ remainder 24	f guard band
tomotope cover ABI	48 blocks / 192 flags	cover-invariant

A measured deviation in any row falsifies the corresponding layer.

14 Implications, corrections, and the ethos

This section states what the architecture means — for physics, for computing, for networking, and beyond the laboratory. Every claim below is either (i) a theorem with an executable witness in this paper's ledger, or (ii) an implication of such a theorem, flagged as such. Nothing here floats free of the verification chain.

14.1 Physics

The vacuum is a gauge choice among five. The decomposition $40 = 1 + 12 + 27$ (self, gauge, matter) that organizes the entire Standard-Model derivation is *one of five* maximal decompositions, indexed by the Schläfli inventory of the cubic surface: 27 registers, 36 schedules, $40 + 40$ building parabolics, 45 polar pairs. The five vacua are connected by an explicit 5×5 double-coset transition matrix. Physically: what looks like "the" vacuum is a parabolic gauge choice, and the theory knows its own alternative phases and the exact combinatorial cost of moving between them. No continuum field theory has this property; here it is finite, exact, and enumerated.

Matter is magic, with three octane grades. The machine's non-classical resource (magic, in the resource-theoretic sense) coincides *exactly* with its matter sector: the 36 entangled Witting rays are the matter shell, and the 27 matter contexts stratify $[1, 8, 12, 6]$ — the cube's cell census. The three magic grades — deep $(2 + \sqrt{3})/6$ on $8 = 2^q$ rays, mid $(5 + 2\sqrt{3})/12$ on $24 = f$, shallow $3/4 = q/\mu$ (the Werner threshold) on $4 = \mu$ — mean that "what matter is" and "what fuels quantum advantage" are one census. A physicist reads this as a structural identification of the fermionic sector with the non-stabilizer sector; an engineer reads the same integers as a fuel gauge. Both readings are forced by the same double cosets.

The Standard-Model spine: one transvection. The preceding paragraphs assemble into a single statement, verified end-to-end in one pass (`bt886_standard_model_spine.py`): *from the pair* $(W(3,3),\ R)$ *— the symplectic quadrangle and one long-root transvection — the discrete Standard Model follows with zero free parameters.* The transvection R fixes the 13-point gauge perp-plane and acts freely (nine orbits) on the 27 matter shell; its centralizer $C(R)$ (order 648) is the gauge group, with module $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8} = U(1) \times SU(2) \times SU(3)$; its centre $Z(C(R)) = \langle R \rangle = \mathbb{Z}_3$ is the three generations; its normalizer gives the flavor group S_3 and charge-conjugation; the W/Q duality gives parity; the matter shell grades 9+9+9 with the $\mathbb{Z}_3$ Yukawa rule; and the connection is flat $(\mathbb{Z}_3^2)$ along collinear edges, $2T$-curved on the matter graph Q with a quaternionic field strength. The gauge group, the generations, the flavor symmetry, the Yukawa texture, chirality, parity, and the gauge dynamics are not separate inputs — they are the centralizer, centre, normalizer, duality, grading, and commutator structure of one element of $\mathrm{Sp}(4, \mathbb{F}_3)$, replicated over the 40 points. The gauge group even carries its color/electroweak split: $C(R) = 3^{1+2} : SL(2,3) = 27{:}24$, with the normal Heisenberg radical (order $q^q = 27$, the color part) fixing the electroweak $\mathbf{1} \oplus \mathbf{3}$ pointwise (the $W/Z/\gamma$ are color-blind, constant on the four lines through p_0) and acting on the gluon octet $\mathbf{8}$, and the Levi $SL(2,3)$ (order $f = 24$, the electroweak part) carrying the $\mathbf{1} \oplus \mathbf{3}$ — color $\rtimes$ electroweak as a parabolic, with substrate orders q^q and f. And the color radical is *the same* Heisenberg group $3^{1+2} = O_3$ that acts simply transitively on the 27 matter shell (the matter torsor): **color is the matter-shell Heisenberg group**, so matter carries color precisely because the 27-shell is a torsor under the color group — color rotations and matter-shell motions are one action of $q^q = 27$. Under the full gauge group the matter shell is the homogeneous space $27 = C(R)/SL(2,3)$ (gauge group over the electroweak Levi): $C(R)$ is transitive with point-stabiliser the electroweak $SL(2,3)$, rank 6 (suborbits $[1, 1, 1, 8, 8, 8]$), module $\mathbb{C}[27] = \mathbf{1} \oplus \mathbf{6} \oplus \mathbf{8} \oplus \mathbf{12}$, and the three electroweak-fixed points form a color-singlet (lepton-like) 3-set while the three color-8 suborbits carry the colored (quark-like) content $(27 = 3 + 24)$. One generation's fermion multiplets are the $C(R)$-module structure of the 27.

Three generations, matter-blind and gauge-visible. Because the matter register is the Steinberg module (Corollary, §10) and the Steinberg character vanishes on every 3-singular element, *any* order-3 symmetry of the substrate splits matter into exactly $27 + 27 + 27$: the three generations carry identical matter/Steinberg structure — they are indistinguishable to the protected register, which is why generations share gauge charges. Yet the four order-3 conjugacy classes *are* distinguished in the gauge (40-point) sector: the transvections grade it $22 + 9 + 9$ and fix a whole $\mathrm{PG}(2,3)$ plane of $13 = \Phi_3$ points, while the regular classes grade it $16 + 12 + 12$ and fix only $\mu = 4$ points. The *physical* texture triality (Pillar 68's Yukawa map R) is identified exactly: it is the long-root transvection (a 40-class element = the centre of the point-parabolic's Heisenberg radical), fixing the 13-point gauge perp-plane $p_0^\perp$ pointwise and acting on the 27-point matter shell with 9 free orbits of 3 — the canonical "fix the gauge, stir the matter" generator. Its eigengrading of $\mathbb{C}[27]$ is $9 \oplus 9 \oplus 9$, and $\mathbb{Z}_3$ Clebsch–Gordan then *forces* the Pillar-68 Yukawa selection rule $T[a, b, v] = 0$ unless the three grades sum to 0 mod 3 (exactly 9 of 27 grade-triples survive); the matter coupling lives on the 36 within-shell tritangent triples (Pillar 69's Heisenberg-twisted triads, $45 = 9 + 36$). And the gauge side closes it: the centralizer $C(R) = \mathrm{Stab}(p_0)$ of order 648 (the gauge group is what commutes with generation-cycling) acts on the $12 = k$ gauge neighbours with permutation rank 3, decomposing the gauge module as

$$\mathbb{C}[12] = \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8} = U(1) \oplus SU(2) \oplus SU(3),$$

the Standard-Model gauge group adjoint (the $\mathbf{1}$+$\mathbf{3}$ from the A_4 action on the four lines through p_0, the $\mathbf{8}$ the within-line gluon octet). So R fixes the gauge sector (12 = 8+3+1, generation-blind) and grades the matter sector (27 = 9+9+9, three generations): the gauge group and the generations are the two eigen-data of one long-root transvection. The gauge sector even carries a parity statement: $C(R)$ acts on the four lines through p_0 as A_4 inside PSp, but as the full S_4 in $\mathrm{PGSp}(4, \mathbb{F}_3) = W(E_6)$ — the odd (parity-violating) 4-line permutations are *exactly* the anti-symplectic W/Q-duality coset, so gauge parity is the duality/chirality $\mathbb{Z}_2$ and lives only in the full $W(E_6)$. The flavor sector even has its charge conjugation: $N_{W(E_6)}(\langle R\rangle)/C(R) = \mathbb{Z}_2$, so there is an element C with $CRC^{-1} = R^{-1}$ that fixes generation grade-0 and swaps grade-1 $\leftrightarrow$ grade-2 — the discrete shadow of CP, conjugating the generation $\mathbb{Z}_3$ exactly as the temporal Bell qutrit's CPT conjugates $\omega \mapsto \bar{\omega}$. The full Standard-Model discrete flavor/parity data — gauge group $1 \oplus 3 \oplus 8$, three generations $\mathbb{Z}_3$, generation C, matter chirality, gauge parity — is the normalizer geometry of one long-root transvection. The grading R and its conjugation C together generate the **flavor group** S_3 ($\langle R, C\rangle$, order 6, non-abelian — the minimal non-abelian flavor symmetry of BSM model-building), under which the matter shell decomposes as $\mathbb{C}[27] = 6{\cdot}\mathbf{1} \oplus 3{\cdot}\mathbf{1}' \oplus 9{\cdot}\mathbf{2}$, the standard doublet appearing $q^2 = 9$ times. And the flavor–gauge relationship is exact: R is central in the gauge group $C(R)$, with $Z(C(R)) = \langle R\rangle \cong \mathbb{Z}_3$ — **the three generations are the centre of the gauge group**, acting trivially on the adjoint exactly as the $\mathbb{Z}_3$ centre of colour $SU(3)$ acts on the gluon octet (so the $SU(3)$ centre *is* the generation $\mathbb{Z}_3$). Globally there are 40 such local gauge groups — one per point, a single conjugacy class (homogeneous spacetime) — and the 40 generation-centred long-root $\mathbb{Z}_3$'s generate all of $\mathrm{PSp}(4, \mathbb{F}_3)$: **the 40 points of $W(3, 3)$ are the space of local gauge groups**, each $SU(3){\times}SU(2){\times}U(1)$ with its generation centre, and the global symmetry is the closure of the local generation symmetries. The connection between adjacent local gauge groups is exact: for collinear points the generation centres commute ($\langle R_p, R_{p'}\rangle = \mathbb{Z}_3 \times \mathbb{Z}_3$), so the gauge connection is **flat along the 240 edges**; for non-collinear points $\langle R_p, R_{p'}\rangle = SL(2, 3) = 2T$ (the binary tetrahedral / 24-cell group), so it is **curved across the 27-per-point matter shell**. Gauge curvature lives exactly on the matter graph Q — flat = collinearity, curved = non-collinearity = matter, with holonomy the 24-cell group. The curvature 2-form $F(p, q) = [R_p, R_q]$ makes this quantitative: it vanishes on collinear (causal) directions and is an *order-4 quaternion unit* of $2T$ (one of $\pm i, \pm j, \pm k$, with $F^2 = -I$ the centre) on every matter pair — an **$\mathbf{su}(2)$-valued field strength supported exactly on the matter graph Q**. Flat causality, quaternionic curvature on matter. The integrated flux (Wilson loop $R_a R_b R_c$) confirms it: all 160 collinear triangles carry order-3 abelian flux (the flat $\mathbb{Z}_3$ line grading), while the 3240 matter triangles of Q carry non-abelian flux of orders $\{2, 4, 6, 12\}$ (counts 180, 180, 1440, 1440) up to $12 = k$ — gauge energy lives on the matter graph. The matter register (Steinberg) couples to this flux only through its non-3-singular part: by Steinberg vanishing, $\chi_{\mathrm{St}}(W) = 0$ on every order-3, 6, 12 loop (3040 triangles, matter-invisible) and is nonzero only on the order-2 loops ($\chi_{\mathrm{St}} = +q^2 = 9$, the polar-pair chirality involution of BT869) and order-4 loops ($\chi_{\mathrm{St}} = -q = -3$); the total coupling $\sum \chi_{\mathrm{St}}(W) = 180 \cdot 9 - 180 \cdot 3 = 1080 = 540 \cdot 2$ is exactly the chart double-cover count — gauge dynamics and the routing fabric meet at 1080. This is the Standard-Model pattern as a representation- theoretic theorem: generations are identical in gauge charge (matter-blind) and differ only through gauge-sector–visible couplings (the Yukawa/mass structure), here the eigenvalue degeneracy of an order-3 symmetry rather than a modeling input. *Witnesses: bt863_generations_from_steinberg_vanishing.py, bt864_triality_census.py.* Order-6 symmetries see *both* internal gradings at once: an order-6 element factors as g^2 (generation, order 3) times g^3 (chirality, order 2 — the sign-twisted carrier of H_2), and since χ_{St} vanishes on every 3-singular power, its $\mathbb{Z}_6 = \mathbb{Z}_3 \times \mathbb{Z}_2$ eigengrading factors cleanly: each chirality half $(81 \pm \chi(g^3))/2$ carries 3 equal generations. For the dominant involution type ($\chi(g^3) = q^2$) the chirality split is $\mathbf{45} + \mathbf{36}$ — the substrate's own tritangent and double-six counts — so the left/right matter split is tied to Schläfli geometry; the other type gives $39 + 42$. Generation $\times$ chirality $= 3 \times 2 = 6 = q!$. The dominant chirality involution is identified exactly: $\mathrm{PSp}(4, \mathbb{F}_3)$ has just two involution classes, and the 45-class (eight fixed points carrying the cube Q_3, centralizer $2T \times 2T$ of order 576) is the *central involution of a hyperbolic polar pair* $L \leftrightarrow L^{\perp}$ — so the left/right split of matter, $45 + 36$, is sized by the substrate's own tritangent-plane and double-six counts, and the matter chirality $\mathbb{Z}_2$ *is* the 24-cell polar-pair swap. *Witnesses: bt868_joint_generation_chirality_grading.py, bt869_involution_chirality_classes.py.*

Time has three hands and an irrational escapement. The internal $\mathbb{Z}_{12}$ rectangle clock, the external $\mathbb{Z}_7$ (Császár) and $\mathbb{Z}_{13}$ (Singer) references, and the Boerdijk–Coxeter drive $\arccos(-2/3)$ together realize a discrete time quasicrystal: a drive that never repeats yet has exactly bounded gap structure (two gap lengths at $n = 30 = h(E_8)$). The arrow of time appears as the unique self-loop fixed point I/q with its two-trit forgetting cost: irreversibility is priced, not postulated. Implication: any laboratory system realizing this architecture is also a clock whose long-term stability is protected by a number-theoretic irrationality rather than by feedback.

The exceptional polytopes are the theory's siblings, not its parts. The Grünbaum–Coxeter arc (Theorems 12.2–12.4) shows the substrate carries the complete local geometry of the 11-cell and 57-cell — hemi-icosahedra and hemi-dodecahedra in every schedule, with exact incidence — while universality plus Lagrange forbid their global crystallization inside $\mathrm{Sp}(4, \mathbb{F}_3)$. The same local amalgam $A_5 *_{D_{10}} A_5$ closes as $L_2(9)$ (the substrate's schedule core), $L_2(11)$ (the 11-cell), $L_2(19)$ (the 57-cell), with $J_1 \times L_2(19)$ — a sporadic group — atop the universal ladder. The physical moral is delocalization: the substrate *cannot* host a "GC condensate," so the icosahedral order is necessarily spread over the 36 schedules as compass structure. This is the same mathematical mechanism by which frustrated phases in condensed matter refuse local order parameters — realized here exactly, with the obstruction an integer $(11, 19 \nmid 25920)$.

Spectral ghosts. The primes that refuse to divide the group still govern its analysis: $11 = k - 1$ is the Ihara prime of the zeta critical circle $|u| = 1/\sqrt{11}$, 19 enters through the Heawood genus $H(19) = 20 =$ the BC ring count, and the chart-web eigenvalue fields are $\mathbb{Q}(\sqrt{\Theta})$ and $\mathbb{Q}(\sqrt{\Phi_{12}})$ ($\Theta = 10$, $\Phi_{12} = 73$, both parameter-closure constants). A spectroscopist's summary: the machine's transport spectrum speaks the tier-ladder dialect of the physics paper, and the forbidden primes appear as poles of the zeta function rather than as symmetries.

Gravity counts the matter register. The discrete gravitational bridge is the Matrix-Tree action $S = \frac{M_P^2}{2} \ln \tau(G)$, τ the spanning-tree count, and for the substrate it is exact: the Laplacian spectrum $\{0, 10^{24}, 16^{15}\}$ gives

$$\tau(W(3,3)) = \frac{10^{24} \cdot 16^{15}}{40} = 2^{81} \cdot 5^{23},$$

where the exponent of 2 is $81 = q^4$ = the matter-sector (Steinberg) dimension, the base $5 = F_5$ is the tier-ladder prime ($r = 27/80$), and the exponent $23 = f - 1$. The complement matter graph $Q = \mathrm{SRG}(40, 27, 18, 18)$ gives $\tau(Q) = 2^{66} \cdot 3^{39} \cdot 5^{23}$ with 3-exponent $39 = q\Phi_3$ = the gauge-sector dimension. So the discrete-gravity partition function carries the Standard-Model sector sizes as its prime-exponent charges — matter (q^4) in the power of 2, gauge ($q\Phi_3$) in the power of 3, cosmological (F_5) shared — and the entropy of the spanning-tree ensemble *counts the registers.* This is not a separate fact from the transport spectrum: the Ihara zeta $\zeta^{-1}(u) = (1-u^2)^{200}(1-u)(1-11u)(1-2u+11u^2)^{24}(1+4u+11u^2)^{15}$ (verified directly by the 480×480 Hashimoto operator) has its non-trivial zeros on the graph-RH circle $|u| = 1/\sqrt{11}$ (the transport sector, $p_{\mathrm{Ih}} = k - 1 = 11$) and its Perron zero at $u = 1$, where the Bass matrix $I - A + 11I = 12I - A$ *is* the Laplacian, so the Matrix-Tree leading coefficient $10^{24}16^{15} = 2^{84}5^{24} = v \cdot \tau$ recovers exactly the gravity count. Transport and gravity are the off-critical and critical behavior of one zeta: the prime 11 is the transport critical norm, the matter dimension $q^4 = 81$ is the gravity entropy. The substrate is in fact a *pair* of zetas: the complement matter graph $Q = \mathrm{SRG}(40, 27, 18, 18)$ has its own Ihara prime $26 = 2\Phi_3$ (Hashimoto Perron, all complex eigenvalues on $|u| = 1/\sqrt{26}$) and its Perron-point gravity $\tau(Q) = 2^{66}3^{39}5^{23}$ puts the *gauge* dimension $39 = q\Phi_3$ where $W(3,3)$ puts the matter dimension — complementation (collinearity $\leftrightarrow$ non-collinearity) swaps the matter and gauge entropies, with the cosmological charge 5^{23} shared. The non-backtracking walk census confirms the dynamics: $N_3 = \mu|E| = 960$, $N_5 = |E|\, q^q\, n_{\mathrm{even}} = 181440$, $N_n \sim 11^n$. *Witnesses:* *`bt870_spanning_tree_gravity.py`*, *`bt872_ihara_zeta_spanning_bridge.py`*, *`bt873_dual_zeta_walk_census.py`*.

Falsifiability is retail, not wholesale. Each physics reading above is pinned to a tabletop witness with an exact rational value: trace–Choi visibility $1/3 = 1/q$ exactly (not approximately); Kochen–Specker budget $36/40 = (q!)^2/v$ exactly, with deficit μ; contextual fraction $1/10 = 1/\Phi_4$; Werner threshold $3/4 = q/\mu$; two gap lengths — not three — at drive step 30. A single wrong integer kills the corresponding layer. This is a stronger falsifiability posture than parameter-fitting physics can offer, because there are no parameters to refit.

14.2 Computing

What the machine is. One photon carries a 4-dimensional path–polarization register (states) and a 9-dimensional past–future time-bin register (operators), realizing the same $W(3,3)$ geometry twice; the operator–operand duality makes program and data literally the same incidence object. Universality is a two-ingredient theorem: three passive optical elements (tritter, phase plate, EOM) generate the full 51840-element Clifford symplectic closure — *exactly*, not approximately — and the matter shell supplies the magic states that promote Clifford to universal. There is no cryostat, no vacuum chamber, no trapped ion: the exotic resource is geometry.

What it costs. The build sheet is a heralded single-photon source, one polarizing beam splitter, one symmetric tritter, a three-rail delay ladder, one electro-optic modulator, one recirculation loop with a fixed irrational rotation, and single-photon detectors. Every component is catalog optics. The machine's benchmarks are integer-valued: a laboratory can certify the Clifford layer by counting to 51840 and certify the fuel by hitting 36/40. Self-testing is native because the specification is arithmetic.

The software stack exists and is finite. Programs are braid words (with $\sigma^5 = Z$ exact); the instruction set is the BT828 compiler ABI; the operating system is the timetable library (36 schedules; overlap law 1-or-4 with switching cost 6); synchronization is the beacon-heptad mesh (2880 frames, themselves forming an exchange network); memory protection is the Steinberg sector ($81 = q^4$, a single irreducible — protected by Schur's lemma, the strongest no-leakage statement representation theory can make). The state/program compiler makes that register directly addressable: $\mathbb{F}_3^3$ supplies three commuting copies of the 27-slot program shell, H_{27} supplies three noncommuting copies of the state shell, and its center supplies the native generation flag $27 \subset 54 \subset 81$. The oriented timetable header above that memory splits as $5_\omega + 5_{\omega^2} + 30$: the full Weyl runtime fuses the handed pair into a 10-channel and leaves one 30-channel with an outer parity choice. Cache lookup is a distinct split layer: two copies of H/C_4 share an 81-route H/D_8 base, and their 324 slots form 81 four-state D_8 address fibers. The non-split ternary transport operator then updates the selected state; address choice and temporal memory cannot alias because their minimal polynomials differ. The watchdog is the BC loop. Each layer is a theorem; the stack compiles because double cosets compose.

Redundancy is built in, not bolted on. The newest results sharpen the fault story: each pentad core stores its timetable *three independent ways* (lit-chart transversals, the K_5 edge labeling, dark-sector shadows), every skew pair has six Petersen homes and two matching covers, and the dark sector carries its own chart matching and two chiral tetrad partitions (Theorem 12.4 block). Erasure of any single description leaves the schedule reconstructible — redundancy appears as a counting identity, not as an engineering allowance. The middleware's two packet phases (tetrahedral/hemioctahedral, the tomotope's $2_{\{0,1,2\}}$ class) confine phase changes to cell hops, which is where the commit boundaries already sit, and the runtime's vertex layer is a free $\mathbb{F}_2^2$-torsor (the Cayley structure), so address arithmetic is XOR end to end.

What it is not. No claim is made here of asymptotic speedups over error-corrected gate-model machines, of fault-tolerance thresholds, or of scalability beyond the fractal substitution rule (40^n leaves, diameter $8n$, commit clock $T(n) = 4(7^n - 1)$) — those are stated as exact combinatorial scaling laws, and their physical realization at depth $n > 2$ is an open engineering problem with its first hard boundary already computed (the $n = 5$ desynchronization with remainder $24 = f$). The honest claim is narrower and stranger: a complete, finite, self-verifying computational architecture exists inside one photon's worth of quantum optics, with zero adjustable parameters.

14.3 Networking

The network is the computer, twice over. The thesis is not rhetorical: the atlas's transport groupoid and the register's gate groupoid are the same groupoid, so moving a packet and applying a gate are one operation. The new library-geometry theorem (Theorem 12.2) doubles the statement at the control plane: the 36 timetables form $\mathrm{SRG}(36, 15, 6, 6)$, the *other* rank-3 geometry of the same group, so the control plane and the data plane ($\mathrm{SRG}(40, 12, 2, 4)$) are the two classical geometries of one symmetry. Configuration management inherits strongly-regular guarantees: any two timetables, near or far, have exactly six common near-neighbors — worst-case reconfiguration paths are bounded by the geometry itself.

Routing, addressing, switching. Local routing is e-cube XOR on Q_3 charts (540 of them; the atlas is glued along the 1620 apartments of the Tits building, diameter 5). Addresses are Gray-coded $\mathbb{F}_2^3$ words; the antipode map is bit complement. Timetable switching obeys the overlap law:

cheap hops preserve $4 = \mu$ of 10 calibrated contexts, the switch graph is the $U_4(2)$ rank-3 geometry, and every skew pair lies in exactly 3 schedules, 6 Petersen homes, and 2 pentad-matching covers — the router's spare-path inventory is an exact census, not a provisioning estimate.

Synchronization without a master. The beacon heptads (2880 complete K_7 interference meshes, all pairwise overlaps 1/3) are reference frames that form their own exchange network — the network's clock distribution is itself a network, with no distinguished master node. External discipline comes from the $\mathbb{Z}_7/\mathbb{Z}_{13}$ clocks (both decimal period 6) and the BC quasicrystal drive; internal phase order from C_{12}. A distributed-systems reader will recognize the structure: consensus by geometry, where the "leader election" problem dissolves because the frame family is a single transitive orbit.

Security posture (implication, flagged). Three structural facts have security readings, stated here as directions rather than theorems. (i) Contextuality is tamper-evidence: the KS budget 36/40 is exact, so any channel intervention that classicalizes measurement statistics moves an integer and is detectable in principle. (ii) Gate teleportation stores a unitary in the photon's self-entanglement and applies it by Bell projection against its own past — a store-and-forward primitive for quantum repeaters that never exposes the gate classically. (iii) The timetable redundancy (three independent reconstructions) is an erasure-code posture against denial: losing a description class does not lose the schedule. Turning these into security proofs requires adversary models this paper does not supply.

Scaling. The fractal holonet substitutes a full 40-node substrate for each leaf: 40^n leaves, $(40^n - 1)/39$ substrate instances, reversible diameter $\leq 8n = 8\log_{40} N$, durable commit clock $T(n) = 4(7^n - 1)$, with the first guard band an arithmetic necessity at $n = 5$. Because every child inherits the same finite protocol, scaling adds no new control logic — the protocol stack is depth-invariant. This is the architectural property that datacenter fabrics approximate statistically and this geometry has exactly.

14.4 The physics-to-architecture dictionary

The dictionary is the deepest implication: one set of integers reads simultaneously as physics and as engineering, because both readings are projections of the same incidence theorems.

object	physics reading	architecture reading
$1 + 12 + 27$ split	Bell line + meeting + disjoint	pole / I-O ports / fuel+store
240 edges	interaction qutrits	bandwidth = memory
$81 = q^4$ Steinberg	protected sector (Schur)	logical register
dual O_3 restrictions	Heisenberg state phases	state/program compiler
$H_2 = 5 + \bar{5} + 30$	oriented context chirality	timetable header $10 + 30^{\pm}$
$324 = 81 \times 4$, commutant D_8	split cache multiplicity	four-state address fiber
$0 \to 81 \to 162 \to 81 \to 0$	ternary temporal extension	non-split state update
36 spreads	measurement bases	timetable OS
$\mathrm{SRG}(36, 15, 6, 6)$	basis-overlap law	control-plane fabric
five vacua	phases of the theory	boot modes
magic strata 8+24+4	fermion shell grading	fuel octanes
540 skew pairs	two-fold null systems	hypercube charts
1620 apartments	Weyl frames	inter-chart trunks
2880 heptads	equiangular frames	sync beacons
BC drive $\arccos(-2/3)$	quasicrystal time	watchdog clock
compasses (216+216)	icosahedral order, delocalized	needle registers
pentads (chiral 216+216)	maximal partial spreads	F_5 chart caches
dark sector zoo	hidden-half structure	erasure store
tomotope ($2_{\{0,1,2\}}$, Cayley/V_4)	sphere/projective alternation	packet middleware
11-cell, 57-cell, J_1	universal completions (blocked)	the covers the hardware refuses

The first row carries the deepest identification, drawn from the companion paper *Universal Quantum Computation Through a Single Photon*: the split $1+12+27$ is not merely the local SRG anatomy of a point — it is the **Bell-line shell** of the photon's own self-entanglement seed. The temporal Bell qutrit $|\Omega\rangle$ stabilizes one line of $W(3,3)$; the 40 lines then decompose as 1 (the Bell line itself) $+12$ (lines meeting it — the gauge sector) $+27$ (lines disjoint from it — the matter sector), with substrate reading $1 + k + q^q$. The vacuum decomposition that organizes the Standard-Model derivation is literally the photon's view of its own now-context: *self = the seed of self-entanglement*. The Bell-line stabilizer has order $51840/40 = 1296 = \mu^2 q^{q+1}$ — the same $1296 = 6^4$ that counts the compass D_{10}-incidence pairs of Theorem 12.4 (36 schedules $\times$ 36 core pairs), but Theorem 2.3 proves that this is not a regular-action identification. Instead, the projective Bell stabilizer resolves the incidence carrier as 648+324+162+162, the exact mirror/schedule/left-cache/right-cache prefix decoder, while the point parabolic supplies two persistent 648 address sheets. The two cache words 110 and 111 are now resolved more finely by Theorem 10.6: they name two copies of the same H/C_4 cache, not two inequivalent internal chiral modules. Both double-cover one canonical H/D_8 base of 81 routes, and the complete cache quarter is an exact D_8-over-81 square bundle. This reversible address plane is provably not the non-split $81 \to 162 \to 81$ transport plane, despite their common 162 count. The companion also supplies the master equation at Hilbert-space level ($q^2 = q + q!$: the 9-dimensional past$\otimes$future space splits into 3 diagonal "now" histories and $6 = q!$ context-changing ones), and a ledger identity: temporal gate teleportation consumes exactly 2 classical trits of memory — the same 2-trit cost that prices the arrow of time in the self-loop fixed point of §11. Storing a state for one's future self and forgetting one's past cost the same two trits. The 27 itself now has a dual decoding: the 27 non-collinear *points* form a torsor under the genuine Heisenberg group (the normal 3^{1+2}_{+} of the point parabolic, acting simply transitively), whose invariant relations reconstruct the whole classical 27-zoo — the 9-triangle fibration, $\mathrm{GQ}(2,4) = \mathrm{SRG}(27,10,1,5)$, and the Schläfli graph $\mathrm{SRG}(27,16,10,8)$ of the E_6 27-lines world — while the 27 disjoint *lines* of the Bell shell form a torsor under elementary $\mathbb{F}_3^3$ (the line parabolic's O_3), a flat 3-trit register cube carrying *no* invariant strongly regular graph. Curvature for states, flatness for programs: the matter sector is a Heisenberg manifold on the state side and a ternary address space on the operational side. The register reading is exact: assigning each Bell-shell context its 3-trit address, the four pair relations $[4, 4, 6, 12]$ are the difference-vector classes, and the geometry is a function of register arithmetic alone — meeting contexts (±-paired 4-classes) share exactly 1 transversal onto the Bell line, near-skew (12) exactly 2, far-skew (6) exactly 0: base-3 register incidence for matter contexts, twin to the charts' base-2 Gray arithmetic. *Witnesses:* `bt858_heisenberg_shell_torsors.py`, `bt860_bell_shell_register_arithmetic.py`.

Three rows deserve emphasis. The *Steinberg row* is the strongest protection statement available: the register is one irreducible representation, so *any* equivariant leakage operator is zero by Schur's lemma — protection by symmetry, not by redundancy. The *compass rows* say that what physics sees as delocalized icosahedral order, engineering sees as a navigation instrument: 12 needles per schedule, each carrying Petersen addressing, five-chart caches, and a dark-sector mirror. The *universal-polytope row* states the program's sharpest structural lesson: the hardware keeps the local physics of the exceptional objects and refuses their global rigidity; the refusal (an integer non-divisibility) is precisely why the architecture is distributed.

14.5 Beyond the laboratory

A different contract between theory and experiment. Every layer of this machine is specified by integers and verified by scripts that re-derive those integers from first principles in seconds. The cultural implication is a publishing model in which papers *run*: a claim is not a sentence but a reproducible computation, and a correction is a commit. This program has already operated under that contract (see the ethos below) — twice it corrected its own published claims at the source, and the corrected values were more natural than the originals. If the broader physics literature adopted executable claims at even a small scale, entire classes of irreproducibility would become type errors.

Quantum hardware without a building. Architectures that need millikelvin cryostats centralize by economics: only institutions can own them. A universal architecture whose parts list is catalog optics — source, splitters, delay fiber, modulator, detectors — decentralizes by the same economics. The realistic near-term role is not supremacy-class computation but *certified* small-scale quantum information: teaching laboratories that can verify 51840 and 36/40 on a bench; metrology nodes whose clock is protected by an irrational rotation; network testbeds where gates and routes are the same hardware. A machine whose benchmarks are exact integers is also the natural pedagogical machine: a student can check the whole specification.

Communication and computation stop being separate utilities. In this architecture there is no boundary at which computation hands off to networking — the gate groupoid and the transport groupoid coincide, and the fractal substitution makes the computer/network identity scale-invariant. The long-run implication, if any physical system realizes this pattern at depth, is infrastructural: bandwidth and compute become one provisioned quantity, scheduling becomes group theory, and the spectrum-allocation problem for the network's measurement contexts is solved by a finite library (36 timetables with a known switching metric) rather than by auction or contention.

Society-scale stakes of a zero-parameter physics. Separately from the machine, the substrate program's physics claims — 54+ observables, 14 falsifiable predictions, no adjustable parameters — carry a specific societal meaning: they are maximally exposed. A theory with no dials cannot retreat; the 2027 neutrino-hierarchy determination, the proton-decay window, the CMB measurements will each either hit the predicted integers or end the corresponding pillar. Public trust in fundamental science is built by exactly this posture — predictions that can die — and the machine half of the program extends the posture to hardware: the device specification is its own audit.

What could go wrong, said plainly. The architecture is exact; its physical realization is not yet demonstrated beyond the component level. The fractal scaling laws are combinatorial theorems, not engineering schedules. The security readings are directions, not proofs. And the identification of the substrate with fundamental physics stands or falls with the prediction table, not with the elegance of the mathematics. This paragraph exists because the program's ethos (below) requires the failure modes to be listed next to the claims.

14.6 The ethos

Several pre-existing corpus claims failed under exact computation and were corrected at their sources during this program: the independence number of $W(3,3)$ is 7, not 10 (no ovoid exists; the "perfect graph" block is withdrawn); the Kochen–Specker optimum is 36/40, not 34/40; the pentad-sharing count of BT844 was corrected by the chiral-orbit census of BT845 within one working session; and the tomotope's framing was corrected from "non-polytopal maniplex" to "uniform polytope with non-polytopal minimal covers" when the literature was read precisely; and the "$\mathbb{Z}_3$ Berry phase as a discrete second Chern class" was corrected (BT871) — the uniform Berry 2-cochain is a coboundary over $\mathbb{F}_3$, so the physical $\mathbb{Z}_3$ is the order-3 action on the Steinberg register, not a curvature class. In every case the corrected value is *more* substrate-natural than the claim it replaced ($\alpha = \Phi_6$; budget $= (q!)^2/v$; $432 = 216 + 216$ chiral; infinitely many covers as the actual pathology; the $\mathbb{Z}_3$ as a group action rather than a cocycle) — the pattern a real structure shows when probed honestly. The program treats corrections as first-class results: they are committed, logged, and cited like theorems, because a corpus that cannot heal is a corpus that cannot be trusted.

A Verification ledger

Every theorem above is backed by an executable script in `analysis/` with results in `data/`; the chain BT739–BT839 was developed and pushed incrementally with exact arithmetic (GF(p) elimination, cyclotomic integers, complete branch-and-bound, GAP witnesses for group-theoretic facts).

layer	primary witnesses
duality / carriers	bt817, bt820, bt821
atlas / building / web	bt773, bt777, bt744, bt779
mirror middleware / tomotope runtime	bt814, bt815, bt826
braids / registers	bt740, bt741
vacua / geography / transitions	bt808–bt813, bt816
fuel / contextuality	bt818, bt822, bt823
memory / Steinberg	bt742, bt744
clocks / quasicrystal / arrow	bt774, bt819, bt820, bt823
sync / schedules / overlap law	bt817, bt819, bt835
Grünbaum–Coxeter hemicells / library geometry	bt836, bt837, bt839
two compasses / $4 \cdot K_{10} = 12$ Petersens / chart double cover	bt843–bt845
pentad anatomy / schedule reconstruction / dark dodecahedra	bt846, bt847
amalgamation ladder / universality / J_1	bt848
maniplex barren rung / class $2_{\{0,1,2\}}$ / Cayley over V_4	bt849–bt851
seventeen universals / Aut(tomotope) universal	bt852
dark orbit zoo / shadow bijection / K_5-complete dark sector	bt853–bt855
dark charts = mirror bus (G-set iso)	bt856
Heisenberg/$\mathbb{F}_3^3$ shell torsors, Schläfli decoding	bt858
Bell-shell register arithmetic ($[4, 4, 6, 12]$ named)	bt860
code register = Steinberg ($[[240, 81, 4, 3]]_3$)	bt861
H_2 = sign-twisted lines (homological dictionary)	bt862
three generations = Steinberg vanishing	bt863
triality census (matter-blind, gauge-visible)	bt864
joint generation $\times$ chirality grading	bt868
chirality = polar-pair involution $(45 + 36)$	bt869
spanning-tree gravity $\tau = 2^{81}5^{23}$	bt870
Berry-phase correction ($\mathbb{Z}_3$ is the Steinberg action)	bt871
Ihara zeta = spanning-tree gravity bridge	bt872
two-zeta duality / walk census	bt873
texture triality = long-root transvection	bt874
Yukawa rule = $\mathbb{Z}_3$ grade conservation	bt875
gauge group $1{\oplus}3{\oplus}8$ from $C(R)$	bt876
gauge parity = duality ($A_4{\to}S_4$)	bt877
generation charge-conjugation ($N/C = \mathbb{Z}_2$)	bt878
flavor group S_3 ($\mathbb{C}[27] = 6{\cdot}1{+}3{\cdot}1'{+}9{\cdot}2$)	bt879
generations = centre of the gauge group	bt880
spacetime = space of 40 local gauge groups	bt881
gauge connection flat on edges, $2T$-curved on Q	bt882
curvature 2-form (quaternionic, on Q)	bt883
gauge flux (Wilson loops, $\mathbb{Z}_3$/lines, $\leq 12/Q$)	bt884
YM–matter coupling $\sum \chi_{\mathrm{St}} = 1080$	bt885
SM spine (one transvection, master theorem)	bt886
gauge group = color : electroweak (27:24)	bt887
color = the matter-shell Heisenberg group	bt888
fermion content $27 = C(R)/SL(2, 3)$ (lepton/quark)	bt889
dual-torsor Steinberg compiler / native generation flag	bt865
oriented H_2 irreducible spectrum / outer lift	bt866
cache D_8-over-81 fibration / split–non-split boundary	bt867
chirality bookkeeping / Bell–compass parabolic router	bt857, bt859
universality	bt825
fractal computer/network	bt827
runtime compiler / sentinel / commit / cover boundary	bt828–bt834, bt838
corpus corrections	bt818, bt823, bt824

B Architecture Completeness and the Three Physical Residuals

The holonet machine is now specified end to end: the 40-point substrate register and its $W(3,3)$ commuting-projector Hamiltonian; the $[[240, 81, 4]]_3$ Steinberg code as the native error-correcting register; the local gate set and transversal/holonomy operations; the gauge connection (flat along the 240 edges, curved on the matter graph with binary-tetrahedral $2T$ holonomy) as the physical wiring of multi-qubit interaction; and the discrete dynamics, in which the physical on-shell states are exactly the harmonic forms $\ker D$ (vacuum $\oplus$ Steinberg matter $\oplus$ line module $= 1 \oplus 81 \oplus 40$) — the least-action ground space of the finite spectral action $\mathrm{Tr}\, f(D^2)$, whose value reproduces the spanning-tree gravity invariant. *As a universal computer the architecture is complete*: universality, the code, the gate set, and the runtime compiler are established (cf. the substrate ledger above).

What remains is not computational but *physical*, and the three residuals of the master manuscript's completeness boundary (`w33_paper.tex`) are now each resolved or reduced:

- **(R1) the gauge-lattice embedding — resolved.** The machine's symmetry register carries the canonical symplectic $E_8/2E_8$ datum mod 2, and that datum admits a *unique* $\mathrm{PSp}(4,3)$-invariant quadratic refinement, of plus type; hence $\mathrm{PSp}(4,3) \subset O_8^+(2) = \mathrm{Aut}(E_8/2E_8)$, and the canonical positive-definite lift is E_8 (up to a central $\{\pm 1\}$). The gauge lattice is fixed.
- **(R2) the symmetry / matter register — datum supplied, bridge built.** The quintic moonshine traces $\mathrm{Tr}(2A|V_5) = 74428120$, $\mathrm{Tr}(2B|V_5) = 184024$ are computed; and the matter register is realized as a finite spectral triple (a Hilbert–Schmidt bimodule of dimension 81, first-order condition verified, gauge content $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8}$). The residual is now the explicit physical fermion-multiplet assignment, a representation- selection task.
- **(R3) the emergent-spacetime limit — theorem-governed.** On a shape-regular (edge-wise) refinement tower the full gravity-plus-matter spectral action converges term by term to the continuum Einstein–Hilbert + Standard-Model action — by Cheeger–Müller–Schrader (curvature), higher-order Regge (higher-derivative), classical lattice gauge, and the exact finite moments $\{440, 1920, 16320\}$; the spectral action is moreover continuous for Latrémolière's spectral propinquity. The emergent-spacetime limit is thus the *application* of established convergence theorems, not an open analytic obstruction.

None of the three is a gap in the machine's *operation* — the holonet runs, corrects, and computes universally as specified — and none is now an open problem in principle: a fixed gauge lattice (E_8), a matter register awaiting only its physical labelling, and an emergent spacetime with a theorem-governed continuum limit. This is the precise sense in which the architecture is complete: a finite, exact, universal machine whose physics closes onto the Standard Model and general relativity through results that are now either proved or reduced to the application of known theorems.

Reading guide for what follows. The remaining subsections develop the architecture in four movements, each a chain of witnessed theorems. *(A) The logical layer is substrate-forced*: the machine is the substrate's symplectic continuum (the oscillator representation), its fault-tolerant code is the lattice tower $A_2 \subset D_4 \subset E_8$, its universal gate set is the degree-2 symplectic plus degree-3 cubic invariants, the gate group equals the code's logical Clifford group $\mathrm{Sp}(4,3)$, and the code lattices sit on the $E_8 \to$ Leech $\to$ Monster moonshine ladder. *(B) The physical carrier is one massless self-protected photon*: the demonstrator versus the fault-tolerant machine, gates as geometric ($2T$) holonomies, self-error-correction from the substrate-fixed quasicrystal drive (topological, $|C| = q - 1$), why the primitive must be a single massless photon (zero proper time, Wigner helicity λ), and the carrier as its own geon inside a fractal of nested shells. *(C) The physics and the computer are one object*: the quantum power is the gauge curvature on the matter graph, that discrete curvature is a finite-group lattice gauge theory whose continuum limit is the spectral action's gauge-plus-gravity field equations (with an exact mass gap f), the whole network is a holographic code — black-hole entropy is the code distance, and the substrate carries a discrete AdS/CFT — and the runtime closes it: the clock is a harmonic oscillator whose frequency $\sqrt{\lambda}$ is the top mass, and the supercycle is the gauge group $\mathrm{Sp}(4,3)$, so running the machine is the physics. *(D) It is testable now*: a parameter-free near-term falsification protocol on the single-photon demonstrator. Throughout, one substrate number recurs — $q-1=\lambda=2$ sets the drive angle, the topological protection, and the photon's transverse helicity count — and no constant is fitted.

B.1 The architecture is the substrate's symplectic continuum

There is a structural reason the machine and the spacetime are different objects built from the same $W(3,3)$: the substrate has *two* continuum limits. The arithmetic tower $W(3,q)$ (the symplectic quadrangle over $\mathbb{F}_q$, $q=3^n$) is spectrally rigid — exactly three eigenvalues at every q, no Weyl law — so it never becomes a smooth spacetime; the metric continuum needs the external $K3$ seed (the physics side, `w33_paper.tex`, Prop. on the two continua). The *intrinsic* continuum of $W(3,3)$ is instead *symplectic*: its Heisenberg matter shell 3^{1+2} carrying the $\mathrm{Sp}(4,3)$ action is a Weil-representation datum, whose $q\to\infty$ limit is the metaplectic (oscillator) representation of $\mathrm{Sp}(4,\mathbb{R})$ on $L^2(\mathbb{R}^2)$ — precisely the continuous-variable, Fock/oscillator computation this paper realizes. So the holonet *is* the symplectic continuum of the substrate, exactly as the spacetime is its metric continuum: the universal computer and the universe are the two faces of one finite geometry's continuum limit. (Witness: `analysis/w33_two_continua_symplectic_metric.py`.)

B.2 The fault-tolerant layer is the substrate's lattice tower

Because the holonet's logical layer is a continuous-variable (CV) oscillator computation, its canonical fault-tolerant encoding is the Gottesman–Kitaev–Preskill (GKP) code, which embeds a qudit in an oscillator. A *multimode* GKP code *is* a symplectic lattice in phase space: stabilisers are displacements by lattice vectors, and error correction is closest-point decoding in the symplectic dual lattice. The code quality is set entirely by the lattice, and the established optima are exceptional lattices: the *hexagonal* lattice A_2 for one mode, the densest 4-dimensional lattice D_4 for two modes (strictly better than any product of one-mode codes), and E_8 for four modes [23, 24].

This is not a free design choice for the holonet: the substrate *fixes* the lattice at every scale. The machine is two qutrits, hence two oscillator modes ($L^2(\mathbb{R}^2)$, phase space $\mathbb{R}^4$), so its optimal GKP code lattice is D_4 — and D_4 is exactly the substrate's matter shell, with $|W(D_4)|=192$ the tomotope/flag symmetry and D_4 triality the three-generation structure. The single-mode optimum A_2 is the $q=3$ (hexagonal, $SU(3)$) lattice of a single qutrit, and two modes stacked, $D_4\oplus D_4$, sit primitively inside E_8 — the substrate's mod-2 homology lift (R1), which is also the $K3$ gauge lattice. Thus the tower

$$A_2 \subset D_4 \subset E_8$$

is *simultaneously* the optimal 1-, 2-, 4-mode GKP code lattices (the architecture's quantum error-correction layer) and $W(3,3)$'s physical lattice tower (single-qutrit $SU(3)$, matter shell, gauge E_8). The error correction of the computer and the gauge structure of the universe are one and the same lattice tower; the QEC code is not chosen, it is the substrate. (Witness: `analysis/w33_gkp_lattice_architecture.py`, verifying the kissing numbers $6, 24, 240$ and the embedding $D_4\oplus D_4\subset E_8$.)

This also fixes the lattice part of the fault-tolerance threshold. The substrate lattices are *isodual* (symplectically self-dual — the balanced-GKP condition, $\hat{q}$ and $\hat{p}$ protected equally) and densest in their dimensions, so they maximise the nominal coding gain $\gamma = d_{\min}^2/\det^{1/n}$ among symplectic lattices of that rank:

$$\gamma_{\mathrm{dB}}(A_2) = 0.6, \qquad \gamma_{\mathrm{dB}}(D_4) = 1.5, \qquad \gamma_{\mathrm{dB}}(E_8) = 3.0$$

(and 6.0 for the Leech lattice). So the holonet's two-mode code D_4 already buys ~ 1.5 dB, and the four-mode code $E_8 \sim 3$ dB, of squeezing-margin over the trivial square code, for free. The absolute threshold still needs the noise model, the finite-squeezing GKP-state quality and the syndrome-extraction protocol — which we do not fix here — but the lattice contribution, the part that depends only on the code, is the substrate's and is optimal. (Witness: `analysis/w33_gkp_coding_gain.py`.)

B.3 The universal gate set is degree two plus degree three

The code fixes the fault-tolerant layer; the gate set completes the architecture, and it too is substrate-forced. The continuous-variable universality theorem [25] states that the Gaussian unitaries — those generated by Hamiltonians at most *quadratic* (degree 2) in the mode operators: beamsplitters, phase shifters, squeezers, i.e. the CV Clifford group $\mathrm{Sp}(2n, \mathbb{R})$ — are *not* universal, but adjoining any single generator of degree ≥ 3 (canonically the cubic phase gate $e^{i\gamma\hat{q}^3}$) generates the full unitary group on the oscillators. A universal CV machine thus needs exactly

$$\underbrace{\text{degree 2}}_{\text{Gaussian / Clifford}} \quad + \quad \underbrace{\text{degree 3}}_{\text{cubic, non-Gaussian}} \quad .$$

The substrate supplies precisely these, as its two lowest matter invariants and no others below degree three: the *degree*-2 symplectic form on $\mathbb{F}_3^4$ that $\mathrm{Sp}(4,3)$ preserves — whose oscillator-rep limit is the Gaussian group $\mathrm{Sp}(4,\mathbb{R})$ that the photon's tritter, phase plate and modulator already realise (§6.5) — and the *degree*-3 E_6 Cartan cubic on the matter $27 = 3{\otimes}3{\otimes}3$, the ε-cubic preserved by $\mathfrak{sl}(3,\mathbb{F}_3)^3$ (`w33_paper.tex`, the 27-cubic and its 45 triads). So the paper's earlier statement that "the matter shell supplies the magic" is, in the CV picture, exact: the magic is the degree-3 E_6 cubic, the minimal non-Gaussian element, matching the Lloyd–Braunstein criterion on the nose. The substrate's invariant degrees $\{2,3\}$ *are* the universal CV generating set $\{\text{Gaussian}, \text{cubic}\}$. Together with the lattice tower $A_2 \subset D_4 \subset E_8$ this closes the architecture: $W(3,3)$ fixes both the fault-tolerant code and the universal gate set, leaving no free choice in the logical layer. (Witness: `analysis/w33_cv_universality_cubic.py`.)

Finally, the code and the gate set are not two independent facts: they are welded by a single group. The logical Clifford group of a GKP code on n qudits of dimension d is the symplectic group $\mathrm{Sp}(2n, \mathbb{Z}/d)$ over the logical phase space. For the holonet ($n = 2$, $d = 3$) this is

$$\mathrm{Sp}(4, \mathbb{Z}/3) = \mathrm{Sp}(4,3), \qquad |\mathrm{Sp}(4,3)| = 51840,$$

which is *exactly* $\mathrm{Aut}(W(3,3))$, the 2-qutrit Clifford group mod Pauli, and the Clifford group the photon's own tritter, phase plate and modulator already generate (§6.5, "symplectic closure = 51840"). So the substrate's symmetry group, the machine's realised gate group, and the GKP code's logical gate group are *one group*: the gates act on the code by construction. The code-preserving (transversal) Gaussian gates are the lattice automorphisms $\mathrm{Aut}(D_4) = W(F_4)$ of order $192 \cdot 6 = 1152$, whose S_3 factor is D_4 triality — the three-generation structure — and the remaining $[\mathrm{Sp}(4,3) : \mathrm{Aut}(D_4)] = 45$ logical Cliffords are teleported using the cubic resource. The logical layer of the holonet is thus a single coherent object, every part of it fixed by $W(3,3)$. (Witness: `analysis/w33_gkp_clifford_coherence.py`.)

B.4 The code lattices are the moonshine ladder

The same lattices carry one last structure that ties the machine to the deepest arithmetic of the substrate. The holonet's modes are free bosons, and a GKP code is a free-boson theory compactified on its stabiliser lattice; an even lattice Λ of rank r is in turn the lattice vertex operator algebra (VOA) V_Λ of central charge $c = r$ (Frenkel–Kac–Segal), while the stabiliser code itself fixes the corresponding (Narain) code CFT [26, 27]. The holonet's GKP code lattices are the substrate tower A_2, D_4, E_8 of ranks $2, 4, 8$, hence the lattice VOAs of central charge

$$c(A_2) = 2, \qquad c(D_4) = 4, \qquad c(E_8) = 8.$$

The top rung, the E_8 VOA at $c = 8$, is exactly the base of the moonshine ladder that $W(3,3)$ already realises on the physics side,

$$V_{E_8}\ (c=8) \ \longrightarrow\ V_{\Lambda_{24}}\ (c=24) \ \longrightarrow\ V^\natural\ (c=24,\ \mathrm{Aut}=\mathbb{M}),$$

with $V^\natural$ the Frenkel–Lepowsky–Meurman Monster module (24 free bosons on the Leech torus, $\mathbb{Z}/2$-orbifolded), and $\Lambda_{24} \supset E_8^3$ the gauge lattice thrice. So the *same* E_8 is the holonet's optimal 4-mode GKP code, the substrate's homology/gauge lattice, and the $c = 8$ chiral VOA that begins moonshine; and the top central charge $c = 24 = f = \chi(K3)$ is the $W(3,3)$ matter count. The computer's error-correcting code and the Monster module are one vertex-algebra tower — the architecture and the moonshine residual (R2) are the bottom and top of a single ladder of substrate lattices. (Witness: `analysis/w33_gkp_voa_bridge.py`.)

B.5 What is physically needed: demonstrator versus fault-tolerant machine

It is worth being exact about what we are building, because the design is now complete but the *physical* realisation has two genuinely different layers. The single self-entangled photon of the build sheet — polarisation + time-bin qutrits, a tritter (F_3), a delay ladder and an EOM, read out by single-photon detectors — realises the *ideal, un-encoded* logical layer: the $\mathrm{Sp}(4,3)$ gate set, the routing network, and every parameter-free witness. It is buildable with today's quantum optics and it demonstrates the *logic*. But a single photon has fixed photon number; it cannot carry a GKP grid state (a many-photon quadrature lattice). So the photonic demonstrator is not, and cannot be, fault-tolerant.

Fault tolerance is the standard two-layer continuous-variable stack — and here both layers are substrate-fixed, with nothing left to choose:

$$\underbrace{\text{240 squeezed modes}}_{\text{physical}} \to \underbrace{\text{120 } D_4 \text{ GKP pairs}}_{\substack{\text{inner: analog}\to\text{digital} \\ \text{gain 1.5 dB}}} \to \text{240 GKP qutrits} \to \underbrace{[[240, 81, 4]]_3}_{\text{outer: Steinberg}} \to \text{81 logical qutrits.}$$

The inner code is the GKP code on the matter-shell lattice D_4 (converting continuous displacement noise into discrete qutrit errors, with the 1.5 dB coding gain that eases the squeezing budget); the outer code is the $[[240, 81, 4]]_3$ Steinberg code on the substrate's 240 edges. What remains is purely physical, and it is the *universal* CV fault-tolerance bottleneck, not a $W(3,3)$-specific one:

1. squeezed light at threshold ($\sim$ 10 dB; protocol-dependent, 7–20 dB in the literature), lowered by the D_4/E_8 coding gain (1.5/3.0 dB);

2. GKP qutrit *state generation* — measurement-based breeding from squeezed/cat states with photon-number-resolving detection, or matter-assisted preparation: the hard step;

3. the cubic non-Gaussian resource — the degree-3 E_6 "matter-shell magic" — via a $\chi^{(3)}$ medium or magic-state injection;
4. a programmable beamsplitter/phase/squeeze network implementing the $\mathrm{Sp}(4,3)$ Clifford (mature integrated photonics);
5. homodyne detection for GKP and Steinberg syndrome readout.

This is the precise, honest statement of what the holonet is and what it still needs: a concatenated $\mathrm{GKP}(D_4) \circ \mathrm{Steinberg}\, [[240, 81, 4]]_3$ continuous-variable qutrit computer in which $W(3,3)$ fixes *both* code layers, the gate set ($\mathrm{Sp}(4,3)$ Gaussian + E_6 cubic) and the network — leaving *zero design freedom above the hardware.* The only open problem is the same one the entire CV fault-tolerance field faces, GKP-state generation at threshold squeezing, and the substrate's optimal lattices ease rather than remove it. (Witness: `analysis/w33_holonet_physical_stack.py`.)

B.6 The gates are geometric: holonomic Clifford from the gauge connection

There is a third, gate-level layer of robustness that the substrate supplies for free, and it explains the name. The substrate gauge connection is flat along the 240 edges and curved on the matter graph, with binary-tetrahedral $2T$ holonomy. That holonomy group is not arbitrary:

$$2T \;\cong\; \mathrm{SL}(2,3) \;=\; \mathrm{Sp}(2,\mathbb{Z}/3), \qquad |2T| = 24,$$

the centre being $\{\pm I\}$ and the quotient $2T/\{\pm I\} = A_4$; and $\mathrm{Sp}(2,\mathbb{Z}/3)$ is precisely the single-qutrit Clifford group modulo Pauli. So the gauge connection's holonomy *is* the qutrit Clifford group, and the gates may be realised as *non-abelian holonomies* (Wilczek–Zee geometric phases) traced in the connection [4, 5]. Holonomic gates depend only on the geometric path, not on pulse timing or shape, so they are intrinsically robust to control error: this is fault tolerance at the *gate* level, complementary to the GKP code (analog level) and the Steinberg/color outer code (logical level). The full two-qutrit Clifford $\mathrm{Sp}(4,3)$ (order 51840) is generated by the matter-graph holonomy on each qutrit together with the entangling network.

The deeper point is the identity of the connection. The *same* connection whose *curvature* furnishes the substrate's gauge fields furnishes, through its *holonomy*, the computer's gates: the gauge interactions of the physics and the logic gates of the machine are one and the same non-abelian holonomy. This is the literal content of "holo-net". (Witness: `analysis/w33_holonomic_gates.py`, constructing $\mathrm{SL}(2,3)$ and verifying order 24, symplecticity, centre $\{\pm I\}$, quotient A_4, and the order spectrum $\{1, 1, 8, 6, 8\}$ of $2T$.)

B.7 The quantum power is the gauge curvature

Read together with the physics tier (§14: the gauge connection is flat on the 240 collinear edges and $2T$-curved on the matter graph Q, with curvature $F(p,q) = [R_p, R_q]$ a quaternion unit), the holonomic identification sharpens into a single statement: *the gates are the gauge generators, and the entangling power is the curvature.* The 40 long-root transvections R_p — one per point, each the generation-$\mathbb{Z}_3$ source of the Standard-Model spine — are exactly the elementary gates (Wilson loops of the substrate connection), and they generate the whole gate group:

$$\langle R_p : p \in W(3,3)\rangle = \mathrm{Sp}(4,3), \qquad |\cdot| = 51840.$$

Each point has $12 = k$ *flat* (collinear, causal) neighbours, where $[R_p, R_q] = I$ and $\langle R_p, R_q\rangle = \mathbb{Z}_3 \times \mathbb{Z}_3$ is abelian and classically simulable, and $27 = q^q$ *curved* (matter-graph Q, non-collinear) neighbours, where the curvature $F = [R_p, R_q]$ is an order-4 quaternion unit and $\langle R_p, R_q\rangle = 2T = \mathrm{SL}(2,3)$ is non-abelian. So the non-Clifford/entangling resource sits *exactly* on the curved directions: a flat pair gives only the abelian $\mathbb{Z}_3 \times \mathbb{Z}_3$ (classically simulable), and the per-gate non-abelian content requires curvature. Quantitatively, a triangle $p \to q \to r \to p$ carries a Wilson loop $W = R_p R_q R_r$ whose order is the discrete curvature flux: the 160 collinear triangles all carry the uniform abelian flux of order 3, while the 3240 matter triangles of Q carry non-abelian flux of orders $\{2, 4, 6, 12\}$ (counts $180, 180, 1440, 1440$, up to $12 = k$), and the discrete Yang–Mills action $S = \sum_\triangle \bigl(1 - \frac{1}{\operatorname{ord} W}\bigr)$ concentrates $\approx 96\%$ on the matter graph. In the physics reading the same curvature on Q is the $su(2)$ field strength and, through its $\mathbb{Z}_3$ Yukawa grading, the fermion-mass texture — so on the matter graph,

$$\text{curvature} = \text{entanglement} = \text{magic} = \text{mass},$$

all one object, while flatness is causality. *(Honest scope: this is a field-strength statement. Both the flat and the curved Wilson-loop* sets *generate all of* $\mathrm{Sp}(4,3)$ *— the order-*3 *flat loops are genuine gates — so the curved/flat split is the per-plaquette curvature and the location of the Yang–Mills action, not a global classical-vs-quantum partition of the group.)* The machine's quantum advantage is not added to the substrate; it lives in the curvature of the Standard-Model gauge field. (Witnesses: `analysis/w33_gauge_curvature_is_computation.py` (40 order-3 transvections, 12 flat / 27 curved per point, curvature order 4 on Q, $\langle R_p\rangle = \mathrm{Sp}(4,3)$); `analysis/w33_yang_mills_computation.py` (Wilson-loop census, the 96% matter-graph action).)

B.8 The fuel is contextuality: the register is a Wigner phase space and magic is its negativity

There is a second, resource-theoretic answer to "where does the quantum power come from," and it is exact on the substrate. Howard, Wallman, Veitch and Emerson proved that for odd-prime-dimensional qudits *contextuality* — equivalently, negativity of the Gross discrete Wigner function — is the necessary resource that lifts Clifford computation to universal quantum computation by magic-state injection [10]. The qutrit $d = q = 3$ is the smallest, cleanest case, and the substrate realizes the whole picture geometrically. The discrete phase space of two qutrits is $\mathbb{Z}_3^4$ with $3^4 = 81 = q^4$ points — *exactly* the logical register $H_1 = 81$ of the $[[240, 81, 4]]_3$ code (irreducible under $\mathrm{PSp}(4,3)$); the Wigner function lives on the same 81 points the machine computes on. The 40 vertices of $W(3,3)$ are the $(81-1)/2 = 40$ nonzero rays of that phase space — the two-qutrit Pauli operators up to phase — so the substrate *is* the two-qutrit Pauli geometry. The Clifford group acts as $\mathrm{Sp}(4,3) = \mathrm{Aut}(W(3,3))$, permuting phase points and preserving Wigner-positivity (the classically simulable sector, Gottesman–Knill). And the non-Clifford states the holonet injects — the Hesse/cubic/T-type magic of the universal gate set — are exactly the Wigner-*negative* states: a stabilizer state such as $|0\rangle$ or $|+\rangle$ has $W \geq 0$, while the qutrit "strange" state $(|1\rangle - |2\rangle)/\sqrt{2}$ has $\min W = -\frac{1}{3}$ and mana $\ln \frac{5}{3} > 0$. So the machine is quantum precisely where the substrate's Wigner function goes negative; the magic it spends is the contextuality of the geometry, and the corpus already measures the geometry's contextual fraction as $4/40 = 1/\Phi_4$. Curvature (the gauge picture) and Wigner-negativity (the resource picture) are two readings of the same fuel. (Witness: `analysis/w33_contextuality_is_the_fuel.py`.)

B.9 From the matter-graph lattice to continuum Yang–Mills: the loop closes

The discrete curvature of the previous subsection is not a metaphor for a gauge theory — it *is* one, and its continuum limit is the gauge sector of the physics. Each matter edge carries the curvature $F = [R_p, R_q]$, an order-4 element of $2T = \mathrm{SL}(2,3) \subset \mathrm{SU}(2)$; realised as the binary tetrahedral group of 24 unit quaternions, these are exactly the quaternion units $\pm i, \pm j, \pm k$ (the six order-4 elements, SU(2)-trace 0). So the matter graph Q carries a genuine *finite-group* ($2T$) lattice gauge theory with an $su(2)$-valued field strength — the computer's gate fabric, read as a Wilson lattice. The standard Wilson plaquette action $\frac{\beta}{2}\mathrm{Tr}\{1-P\}$ assigns each plaquette the curvature energy $1-\frac{1}{2}\mathrm{Re}\,\mathrm{Tr}\,P$, maximal ($=1$) on a curved ($\mathrm{Tr}=0$) plaquette and zero on a flat one, and Wilson's theorem gives its continuum limit $\frac{\beta}{2}\mathrm{Tr}\{1-P\} \to \frac{a^4}{2g^2}F^b_{\mu\nu}F^b_{\mu\nu}$. On the shape-regular edgewise tower (the same refinement that carries the finite spectral action to the continuum) the matter-graph action converges to the continuum $su(2)$ Yang–Mills integral $\frac{1}{4g^2}\int \mathrm{Tr}\,F^2$, which is precisely the gauge-kinetic a_4 term of the $W(3,3)$ spectral action. Together with the a_2 Einstein–Hilbert term — whose metric variation gave the Einstein field equations (`w33_paper.tex`) — and the a_0 cosmological term, varying $a_0 + a_2 + a_4$ yields the coupled Einstein + $su(2)$ Yang–Mills + Higgs field equations. The loop therefore closes: *the computer's gate curvature (a $2T$ lattice gauge theory on Q) and the physics gauge-plus-gravity field equations (the continuum spectral action) are the lattice and continuum descriptions of one $W(3,3)$ connection* — discrete gates below, smooth fields above, the edgewise tower between. *(Honest scope: the discrete $2T$ structure and Wilson density are computed; the continuum limit is the established Wilson and edgewise spectral-action results, not re-derived numerically; the local field strength is $su(2)$ per edge while the global holonomy group is* Sp(4, 3).*)* (Witness: `analysis/w33_lattice_to_continuum_ym.py`.)

And the discrete theory is already *gapped*, exactly: a gauge theory on a finite graph has the graph Laplacian as its kinetic operator, so its mass gap is the algebraic connectivity of the matter graph. Since $Q = \mathrm{SRG}(40, 27, 18, 18)$ is an expander, that gap is large and substrate-fixed — the adjacency spectrum is $\{27^1, 3^{15}, (-3)^{24}\}$, so the Laplacian $L_Q = 27I - A_Q$ has spectrum $\{0^1, 24^{15}, 30^{24}\}$ and

$$\text{mass gap} = 24 = f,$$

the smallest nonzero eigenvalue, at multiplicity $g = 15$ (the gauge-sector dimension), with the top eigenvalue 30 at multiplicity $f = 24$ (the matter dimension). The Laplacian multiplicities are the Standard-Model sector sizes and the gap is the matter count; confinement is structural, forced by the expander geometry of Q, with no continuum limit needed to certify that the gap is nonzero. *(This is the gap of the exact finite theory — the substrate's shadow of the Yang–Mills mass gap, not a continuum proof.)* (Witness: `analysis/w33_mass_gap_from_matter_graph.py`.)

B.10 Why "holo-net": the network is a holographic code

The name is earned, not decorative: the architecture is a holographic quantum error-correcting code in three exact senses, each read straight off the substrate spectrum, and each tying a physics fact to a machine component. *(i) Black-hole entropy is the code distance.* A horizon is a stabilizer edge-cut, and the Bekenstein–Hawking law $S_{\mathrm{BH}} = A/4$ has its universal $\frac{1}{4}$ *equal to* the code's distance bound $1/d_Z$: the $W(3,3)$ CSS code has $d_Z = 4 = \mu$, so $S_{\mathrm{BH}} = A/d_Z = A/\mu$. The black-hole quarter is the QEC distance. *(ii) Discrete AdS/CFT.* A holographic duality maps a negatively curved bulk to a conformal boundary. The collinearity graph has adjacency spectrum $\{12, 2^{24}, (-4)^{15}\}$; its single *negative* (hyperbolic) eigenvalue -4 has multiplicity $g = 15 = \dim \mathrm{SO}(4,2)$, the 4-dimensional conformal group. The 15 negative-curvature bulk modes match the 15 conformal-boundary generators exactly — discrete AdS/CFT with no continuum strings. (Dually, the matter graph's Laplacian $\{0, 24^{15}, 30^{24}\}$ puts the same g/f multiplicities on the gauge mass-gap spectrum.) *(iii) The fractal is the holographic principle.* The nested-shell fractal of §7 is boundary-encodes-bulk made literal: each outer shell's $[[240, 81, 4]]_3$ code holographically contains the logical information of the inner layers, exactly the concatenated holographic code structure (the Pastawski–Yoshida–Harlow–Preskill picture), with the substrate fixing the dictionary. So the holonet is a holographic code whose entropy law, conformal boundary, and bulk–boundary map are all substrate arithmetic — which is the literal content of "holo-net". (Witness: `analysis/w33_holographic_architecture.py`.)

B.11 The boundary central charge is $f = 24$: the Monster CFT is the holonet's holographic dual

The holographic boundary even has a definite central charge, and it is the substrate's master invariant. The code lattices climb the moonshine ladder established above — A_2 at $c = 2$, D_4 at $c = 4$, E_8 at $c = 8$, then Leech and the Monster module $V^\natural$ at $c = 24$ — and the top rung fixes the boundary:

$$c \;=\; 24 \;=\; f \;=\; \chi(K3) \;=\; \mathrm{rank}(\Lambda_{24}) \;=\; 2k \;=\; q^q - q,$$

the matter count, the $K3$ Euler number, and the Leech rank, all one number. At $c = 24$ the boundary is the *extremal* holomorphic CFT, whose partition function is $Z = j(\tau) - 744 = q^{-1} + 196884\,q + \cdots$ with graded dimensions $V_0 = 1,\ V_1 = 0,\ V_2 = 196884 = 1 + 196883$ (the McKay vacuum-plus-smallest-irrep identity), and the Monster is its unique symmetry. By Witten's analysis of pure three-dimensional gravity this extremal $c = 24$ CFT *is* the dual of pure AdS_3 gravity [8], so its graded dimensions count the AdS_3 black-hole microstates — and those dimensions are exactly the quintic moonshine data of the substrate (e.g. $\dim V_5 = 333\,202\,640\,600$ from the master manuscript). Finally, Almheiri–Dong–Harlow showed that AdS/CFT *is* a quantum error-correcting code [9]; the holonet is that code, with bulk $W(3,3)$ and boundary the Monster CFT. So one substrate number, $c = f = 24$, ties the holographic boundary, the moonshine module, the matter sector, and pure 3D gravity into a single object: the holonet is the quantum error-correcting code that AdS/CFT always was, and its boundary is monstrous moonshine. (Witness: `analysis/w33_holographic_central_charge.py`.)

Two consequences sharpen this. First, the entropy is a microstate count: the graded dimensions $\dim V_n$ are the boundary state numbers, so the black-hole entropy is $S(n) = \ln \dim V_n$, and it follows the Cardy law $S \sim 2\pi\sqrt{cn/6} = 4\pi\sqrt{n}$ at $c = 24$ — the Bekenstein–Hawking area law of the BTZ black hole. The leading area term captures the entropy to $\sim 5\%$ (the slow Cardy approach) and the Petersson–Rademacher-corrected form to ~ 2–3% by $n \sim 5$, so the moonshine dimensions are a Strominger–Vafa-style microstate count of the substrate's own horizon, with $S = A/\mu$. Second — and this is why the holographic story is not a separate layer — the holographic *redundancy is the error correction.* A holographic code over-encodes each bulk qudit on many boundary regions, and that over-encoding *is* the code distance; on the holonet the identical redundancy appears three ways at once: the fractal nested shells (each outer $[[240, 81, 4]]_3$ shell re-encoding the inner layers), the timetable triple-storage (§12), and the holographic boundary. So the network's holographic/fractal redundancy and the computer's fault tolerance are one mechanism — the precise sense in which the computer is the network. (Witness: `analysis/w33_microstate_count.py`.)

The central charge even explains the generation number. The matter sector's affine symmetry is the WZW model $\mathfrak{sp}(4)$ at level $k = 12$ — the level is the graph degree — with $c = 10 \cdot 12/(12+3) = 8 = \mathrm{rank}(E_8)$, the same c as $(E_8)_1$. So the matter rung carries one E_8 of central charge, and the boundary's $c = 24 = f$ is exactly

$$24 = q \cdot 8 = q \cdot \mathrm{rank}(E_8),$$

three generations of that matter E_8 rung; the jump from the matter rung to the full boundary *is* the generation number $q = 3$. The lattice picture makes it concrete: among the $24 = f$ Niemeier (even unimodular rank-24) lattices — the boundary's possible realizations, their count equal to the central charge itself — the Leech realization is the extremal Monster CFT, while E_8^3 is literally three copies of the matter E_8, the three-generation realization. "Why three generations" and "why $c = 24$" are one fact: the boundary is q copies of the matter E_8. (Witness: `analysis/w33_wzw_generation_ladder.py`.)

And that single matter E_8 rung already contains the grand unification and the cosmological constant. The genuine conformal embeddings of $c = 8$ algebras into $(E_8)_1$ are exactly the substrate's physics: $(E_6)_1 \times (A_2)_1$ is the grand-unified $E_8 \to E_6 \times \mathrm{SU}(3)_{\mathrm{color}}$ ($248 = 78 + 162 + 8$, the 27 matter and the 8 gluons), while $(D_8)_1 = \mathrm{SO}(16)_1$ is the *cosmological-constant* sector — $\dim \mathrm{SO}(16) = 120 = vq$, with $E_8 = \mathrm{SO}(16) \oplus \mathbf{128}$. (The affine $\mathfrak{sp}(4)_{12}$ shares $c = 8$ but is a distinct CFT, not a conformal subalgebra of E_8 — the central charge matches, the embedding does not.) That SO(16) sector is where the holography meets thermodynamics: the substrate is discrete de Sitter ($\kappa = 2/k = 1/6 > 0$, with $\sum_{\mathrm{edges}} \kappa = v$ by Gauss–Bonnet), its horizon radiates at the Gibbons–Hawking temperature $T_H = k/2\pi$ — which is exactly the temperature of the modular-flow clock — and its entropy is A/μ, so *time, temperature and the cosmological constant are one thermal fact*, the arrow of time being the de Sitter expansion. On the matter/topological side the same unification recurs: the double $D(2T)$ has 6 abelian and $36 = 4 \cdot 9$ non-abelian anyons (total quantum dimension $f = 24$), but $2T$ is solvable, so braiding alone is non-universal — the magic that lifts it to universality is precisely the Wigner negativity of §B.8. Gauge curvature, contextuality, and topological magic are three names for one resource. (Witnesses: `analysis/w33_e8_conformal_embeddings.py`, `analysis/w33_thermal_cosmology.py`, `analysis/w33_topological_magic.py`.)

Three further consequences close the holographic picture, and they share a single constant $\mu = 4$. *(a) The dark sector is the* $\mathbf{128}$. The visible half of E_8 is SO(16) ($\dim = 120 = vq$, the gauge/cosmological-constant sector); the remainder is the spinor $\mathbf{128} = 2^{\Phi_6}$, and under $\mathrm{SO}(10) \times \mathrm{SU}(4)$ it is $(\mathbf{16}, \mathbf{4}) \oplus (\overline{\mathbf{16}}, \overline{\mathbf{4}})$ — Standard-Model families (the $\mathbf{16}$) replicated under a *hidden* $\mathrm{SU}(4) = \mathrm{SO}(6)$ with $\mu = 4$ dark colors, so $E_8 = \mathrm{SO}(16) \oplus \mathbf{128}$ is a visible/dark split and the holographic boundary predicts dark matter. *(b) Ryu–Takayanagi is exact and discrete*: a boundary region A has entanglement entropy $S(A) = \delta(A)/\mu$, its minimal edge-cut over the Bekenstein factor (a point gives q, a GQ line gives q^2), with a symmetric Page curve; by the JLMS equality of bulk and boundary modular Hamiltonians the machine reconstructs its own bulk, the entanglement-wedge threshold being the code distance $d_Z = \mu$. *(c)* $q = 3$ *is selected thermodynamically*: the expanding de Sitter horizon (temperature $T_H = k/2\pi$, curvature $\kappa = 2/k$, entropy A/μ) closes its first-law/Gauss–Bonnet condition $2(q-1)(q^2+1) = (1+q)(1+q^2)$ *only* at $q = 3$ — the thermal clock is self-consistent only there. The unifying thread: the same $\mu = 4$ is the Bekenstein factor, the QEC distance, the RT $1/\mu$ and its reconstruction threshold, the de Sitter entropy factor, *and* the number of dark colors — one holographic quantum. (Witnesses: `analysis/w33_dark_sector_128.py`, `analysis/w33_rt_bulk_reconstruction.py`, `analysis/w33_desitter_q3_selection.py`.)

Pushing the dark sector further collapses it into the bulk itself. *(a)* The hidden SU(4) that carries the $\mathbf{128}$ is a confining gauge theory: with $N_c = \mu = 4$ dark colors and the $\mathbf{16}$-spinor's $N_f = 16$ flavors its one-loop coefficient is $\beta_0 = (44 - 32)/3 = 4 = \mu > 0$ (versus the visible $\beta_0 = \Phi_6 = 7$), so it is asymptotically free and *confines* — dark matter is a colorless *dark hadron* at the dark confinement scale, a falsifiable alternative to a fundamental WIMP (the relic mass needs UV matching and is not derived). *(b)* That same $\mathrm{SU}(4) = \mathrm{SO}(6)$ has dimension $4^2 - 1 = 15 = g$, and its non-compact real form is $\mathrm{SO}(4, 2)$, the 4D conformal group — the discrete-AdS/CFT bulk isometry (the 15 negative-curvature modes). So *one* dimension-$15 = g$ symmetry is at once the dark color, the Pati–Salam unifier (lepton number as the fourth color), and the holographic bulk, with its fundamental $\mathbf{4} = \mu$ being both the dark-color count and the spacetime dimension: the dark colors are the spacetime directions. *(c)* And the information story closes inside that bulk: the discrete RT entropy $S(A) = \delta(A)/\mu$ of a growing radiation region rises, peaks at the balanced cut, and falls, the *island* (the complementary matter region) taking over at the Page point $|A| = v/2 = 20$ — the $[[40, 20, 11]]$ code — so evaporation is manifestly unitary and information is recovered. Dark matter, the holographic bulk, grand-unified color, the dimension of spacetime, and the Page curve are facets of the one $SU(4) = SO(6) = 15 = g$. (Witnesses: `analysis/w33_dark_matter_mass.py`, `analysis/w33_su4_is_spacetime.py`, `analysis/w33_page_curve_island.py`.)

Three last threads make the dark/bulk SU(4) quantitative, dynamical, and temporal. *(a) A dark-matter scale from* E_8. Matching the dark SU(4) and visible SU(3) couplings at the unification scale gives $\Lambda_{\text{dark}} = M_{\text{GUT}}(\Lambda_{\text{QCD}}/M_{\text{GUT}})^{\beta_0^{\text{vis}}/\beta_0^{\text{dark}}}$; exponentially sensitive to β_0^{dark}, it lands the dark hadron at *tens of GeV* when $\beta_0^{\text{dark}} = 8 = k - \mu$ (the gluon count, $N_f = \Phi_4 = 10$), matching the corpus's ~ 22.8 GeV dark-matter value — a *confining* alternative to the 308 GeV WIMP ($g\, v_{EW}/k$) the corpus also records. *(b) The expanding holography is dS/CFT.* Because the substrate is de Sitter, the bulk Laplacian modes $\{0^1, 10^{24}, 16^{15}\}$ read as vacuum + matter ($f = 24$, the boundary central charge) + the $15 = g$ de Sitter isometries (the adjoint of $SO(4, 2) = SU(4)$, the same dark/bulk group), with the Monster $c = 24$ CFT at future infinity — the expanding counterpart of the corpus's discrete AdS/CFT. *(c) Time is dark braiding.* The clock $2T$ sits at the bottom of the chain $2T < SU(2) < SU(4)$, so one tick is a topological twist of the $D(2T)$ dark anyons — the modular T-matrix, of order $\text{lcm}(4, 6) = 12 = k$, the clock period. So dark matter, the holographic bulk, the spacetime, and the arrow of time are nested in one $SU(4) = SO(6) = 15 = g$: the dark colors are the spacetime directions, and time is the dark anyons braiding. (Witnesses: `analysis/w33_dark_lambda_gut.py`, `analysis/w33_dscft_modes.py`, `analysis/w33_clock_is_dark_braiding.py`.)

The same expanding boundary is also the origin of the cosmic microwave background. A de Sitter substrate *is* an inflating universe, and dS/CFT organizes its perturbations: of the bulk modes, exactly one is light — the complementary-series vacuum mode ($m^2 = 0 < \frac{9}{4}$), the inflaton — while the matter ($m^2 = 10$) and isometry ($m^2 = 16$) modes are heavy principal-series modes that decay. So there is a unique scalar to drive inflation, and its spectrum is near-scale-invariant precisely because of the de Sitter conformal symmetry of the boundary. The Starobinsky e-fold count is $N = 2(v - \Phi_4) = 2(40 - 10) = 60$, giving

$$n_s = 1 - \tfrac{2}{N} = \tfrac{29}{30} = 0.9667, \qquad r = \tfrac{k}{N^2} = \tfrac{1}{300} = 0.00333,$$

matching Planck ($n_s = 0.9649 \pm 0.0042$) with the tensor amplitude carrying the degree $k = 12$. And the primordial two-point function is a boundary CFT correlator on the *same* Monster $c = 24$ future-infinity boundary that carries the dark/bulk SU(4) — so the CMB fluctuations are correlations in the moonshine module. Inflation, dark matter, and the arrow of time share one expanding dS/CFT boundary. (Witness: `analysis/w33_inflation_dscft.py`.)

These pieces assemble into one early-universe history, with the **128**'s right-handed neutrino as the linchpin. *Inflation* ends and reheats, producing the heavy N_R — the SO(10)-singlet inside each family's **16** in the dark spinor $\mathbf{128} = (\mathbf{16}, \mathbf{4}) \oplus (\overline{\mathbf{16}}, \overline{\mathbf{4}})$. *Seesaw:* N_R has a Majorana mass at the dark/GUT scale $M_R \sim 10^{15}$ GeV $\sim M_{\text{GUT}}$, so the type-I seesaw $m_\nu = m_D^2/M_R \sim 0.05$ eV gives the light neutrino masses (corpus $\sum m_\nu = 58$ meV, cyclotomic PMNS). *Cogenesis:* the *same* N_R decays out of equilibrium with CP violation, seeding both the baryon asymmetry by leptogenesis ($\log_{10} \eta_B = -|E|/(v - k - \lambda) = -240/26 = -9.23$, observed -9.21) and an equal dark hidden-SU(4) asymmetry — so dark matter is asymmetric and $\Omega_{\text{DM}} \sim \Omega_b$ is automatic, with $\Omega_{\text{DM}}/\Omega_b = 82/15$. *CMB:* the single light inflaton mode makes inflation single-field, so the non-Gaussianity is the consistency value $f_{\text{NL}}^{\text{local}} = \frac{5}{12}(1 - n_s) = \frac{5}{6N} = \frac{1}{72}$, with the bispectrum *shape* the Monster $c = 24$ boundary three-point function. So one substrate runs the whole sequence — inflation, the moonshine CMB, the neutrino masses, the matter–antimatter asymmetry, and the dark matter — off the dS/CFT boundary and the **128** spinor, joined at the right-handed neutrino. (Witnesses: `analysis/w33_cmb_nongaussianity.py`, `analysis/w33_cogenesis.py`, `analysis/w33_neutrino_seesaw_128.py`.)

This whole history hangs on a substrate *scale ladder* and ends in falsifiable numbers. The energy scales descend as closed forms — $M_{\rm Pl} > M_{\rm GUT} = v_{EW}10^{2\Phi_6} > M_R \sim 10^{15} > T_{\rm RH} > v_{EW} = |E|+2q > \Lambda_{\rm dark} > \Lambda_{\rm QCD}$ — and the thermal history walks down them (inflation, reheating, the N_R lepto/cogenesis, electroweak, dark and QCD confinement). The ladder corrects a loose expectation: $M_R \gg T_{\rm RH}$, so *leptogenesis is non-thermal* (N_R from inflaton decay), and $T_{\rm RH}$ is a different rung from $\Lambda_{\rm dark}$. At the top, the primordial suite is one moonshine sheet from $N = 2(v - \Phi_4) = 60$: $n_s = \frac{29}{30}$, $r = \frac{1}{300}$, $f_{\rm NL} = \frac{1}{72}$, and the new running $\mathrm{d}n_s/\mathrm{d}\ln k = -2/N^2 = -\frac{1}{1800}$. At the seesaw rung, the cyclotomic PMNS angles ($\sin^2\theta_{12} = \frac{4}{13}$, $\sin^2\theta_{23} = \frac{7}{13}$, $\sin^2\theta_{13} = \frac{2}{91}$) with the $\mathbb{Z}_3$ Majorana phases $(120°, 240°)$ give the neutrinoless-double-beta mass $m_{\beta\beta} \approx 2.3$ meV and the Dirac phase $\delta_{CP} = 14\pi/13 \approx 194°$ — the same phase that drives the cogenesis. So the substrate's cosmology is a falsifiable program: tiny $f_{\rm NL}$ and running for CMB-S4/LiteBIRD, and $m_{\beta\beta} \approx 2.3$ meV for nEXO/LEGEND. (Witnesses: `analysis/w33_cmb_moonshine_suite.py`, `analysis/w33_scale_ladder.py`, `analysis/w33_neutrinoless_betabeta.py`.)

B.12 The falsifiability ledger, and the demonstrator as an experiment on the theory

A theory of everything is worth only the experiments that can refute it, so the program collects into one ledger: $r = \frac{1}{300}$ and $f_{\rm NL} = \frac{1}{72}$ and the running $-\frac{1}{1800}$ (CMB-S4/LiteBIRD), $m_{\beta\beta} \approx 2.3$ meV and $\sum m_\nu = 58$ meV and $\delta_{CP} \approx 194°$ (nEXO/LEGEND, DESI, DUNE), a confining tens-of-GeV dark hadron (LZ/XENONnT) with a dark-confinement gravitational-wave background peaking in the LISA band ($f \sim 10^{-4}$ Hz, a fourth GW source beyond the primordial/electroweak/cosmic-string ones), and the collider/proton-decay handles — each killable by a named experiment in 2028–2035.

The sharpest point is that the loop closes: the tabletop single-photon demonstrator *is* one of these experiments, and it can run *now*. It reads the substrate primitives directly as quantized or protected optics. The flagship is the topological frequency-conversion pump: the photon's three modes form a spin-1, and a two-tone drive pumps energy quanta between the tones at the rate $(C/2\pi)\hbar\omega_1\omega_2$ with C the Floquet-band Chern number; for spin-1 the bands carry $\{+2, 0, -2\}$, so the pump quantum is the *topologically protected* integer $|C| = 2S = \lambda = q-1 = 2$. A Kochen–Specker test on the 40 Pauli rays gives contextual fraction $4/40 = 1/\Phi_4 = \frac{1}{10}$; the BC clock beats at $\arccos(-\frac{2}{3}) = 131.8°$; the oscillator gate spectrum is $q \pm \sqrt{\lambda} = 3 \pm \sqrt{2}$. So the demonstrator measures λ (twice, once protected), q, and Φ_4 on a bench: if the pump quantum is not 2, or the contextual fraction is not 1/10, the $W(3,3)$ substrate is falsified at the tabletop. The machine we are building is an experiment on the theory of everything. (Witnesses: `analysis/w33_falsifiability_ledger.py`, `analysis/w33_dark_confinement_gw.py`, `analysis/w33_demonstrator_measures_substrate.py`.)

Three results sharpen this to a theorem, a protocol, and a single picture. *The pump is protected.* A $(2S{+}1)$-level qudit driven by two incommensurate tones has extreme Floquet-band Chern number $2S$ (verified for $S = \frac{1}{2}, 1, \frac{3}{2}$ with $|C|_{\max} = 1, 2, 3$), so the qutrit's pump quantum is $2S = q - 1 = \lambda = 2$ for *any* drive realizing the winding, protected by a finite Floquet gap $\Delta_{\min} \sim \mathcal{O}(1)$ that fixes the spec (drive rate and coherence/temperature below the gap). *The contextuality is a protocol.* The 40 Pauli rays carry the 40 self-dual $\mathrm{GQ}(3,3)$ lines as measurement contexts (four commuting Paulis each, four contexts per ray); a state-independent Kochen–Specker test over them has contextual fraction $4/40 = 1/\Phi_4 = \frac{1}{10}$, a second tabletop falsifier. *And the whole theory is one graph.* Every claim in this paper — the Standard Model, the exceptional/moonshine tower with Monster $c = 24$, the holographic boundary and code, the dark $\mathrm{SU}(4)/\mathbf{128}/N_R$ sector, the cosmology ledger, the runtime, and the bench falsifiers — forms one connected acyclic dependency graph whose unique root is the selection $q! = 2q$: every node is reachable from $q = 3$. The theory of everything is the unfolding of a single integer, and it ends on numbers a bench and a telescope can check. (Witnesses: `analysis/w33_pump_protection_theorem.py`, `analysis/w33_kochen_specker_protocol.py`, `analysis/w33_grand_dependency_map.py`; graph: `docs/w33_grand_dependency_map.dot`.)

B.13 The geometric origin of the clock: the triangulation ladder $3 \to 4 \to 7$

Where does the clock come from, geometrically? From the same place as the substrate integers: closing simplices. The master quadratic factors as $(x - \lambda)(x - \mu) = (x - 2)(x - 4)$ from the seed $q! = 2q$, and the two simplices that build everything are the *triangle* and the *tetrahedron*. Three points close a loop into a 2-simplex; closing that third edge fixes one orientation — it "saves" a single $q = 3$ trit (the Klee–Irwin cycle-clock / trit-saving picture), so the minimal closed cell carries exactly one trit. Four points ($\mu = q + 1$) close the 3-simplex, the tetrahedron — self-dual (K_4 vertices *and* K_4 faces, both maximally adjacent), genus 0, and the building block of the Boerdijk–Coxeter helix whose stack twist $\arccos(-\frac{q-1}{q}) = \arccos(-\frac{2}{3})$ is the time-quasicrystal clock.

The torus is born when these two combine. The Ringel–Youngs genus of the complete polyhedral graph is $g(K_n) = \lceil (n-3)(n-4)/12 \rceil$, with denominator $12 = k$. At $n = \Phi_6 = 7$ the numerator is

$$(n-3)(n-4) = \mu \cdot q = 4 \cdot 3 = 12 = k, \qquad \text{so} \qquad g = \frac{\mu\, q}{k} = \frac{k}{k} = 1,$$

the torus — and the "3" and "4" in that numerator are precisely the triangle ($n - 4 = q$) and the tetrahedron ($n - 3 = \mu$). The genus is 1 exactly because triangle $\times$ tetrahedron equals the degree. (K_4, the tetrahedron itself, gives $(1)(0)/12 = 0$, the sphere.) The two genus-1 realizations are the only "complete" toroidal polyhedra: the Szilassi polyhedron, with 7 mutually edge-adjacent faces (the tetrahedron and Szilassi are the *only* polyhedra in which every pair of faces touches), and its dual the Császár polyhedron, with 7 mutually adjacent vertices (K_7). They share $21 = \binom{7}{2}$ edges — the Fano flags — and their incidence is the Heawood graph ($14 = 7+7$ vertices, 21 edges), which is exactly the Fano clock whose Laplacian middle shell gives the oscillator frequency $\omega = \sqrt{\lambda} = \sqrt{2}$ of §B.14. So the two toroidal polyhedra *are* the clock oscillator, and the tetrahedral BC helix is its genus-0 companion: one triangulation ladder $3 \to 4 \to 7$, the triangle and the tetrahedron, builds the clock at both genera. (Witness: `analysis/w33_genus_ladder_clock.py`.)

And the ladder takes us all the way home: the same simplices generate the surfaces, the code, and the clock at once. *Surfaces (climbing to genus 3).* Above the sphere (tetrahedron, $g = 0$) and the torus (Fano/Heawood, $g = 1$) the next maximally-symmetric rung is the Klein quartic, the lowest-genus Hurwitz surface ($g = 3$): a regular map of 24 heptagons / 56 triangles / 84 edges with automorphism group $\mathrm{PSL}(2,7)$, $|\mathrm{PSL}(2,7)| = 84(g-1) = \lambda k \Phi_6 = \Phi_6 f = 168$. Its numbers are substrate integers ($24 = f$, $56 = v+k+\mu = \dim E_6^{\mathrm{fund}}$ analogue, $84 = k\Phi_6$), and that $\mathrm{PSL}(2,7)$ is exactly the clock/readout layer of §B.14 — so the heptagon $\Phi_6 = 7$ threads the whole ladder, the Fano heptad becoming the genus-3 heptagonal surface. *Code (trit-saving is encoding).* Closing a triangle saves one $q = 3$ trit: on the $W(3,3)$ chain complex $(40, 240, 160, \partial^2 = 0)$ the 160 triangles are the weight-3 CSS Z-checks (rank 120) and the vertices the X-checks (rank 39), so the trits no triangle can save are $H_1 = (240-39)-120 = 81 = q^4$, the logical register of the $[[240, 81, 4]]_3$ code — closing a loop and encoding a qutrit are one act. *Clock (the helix is a quasicrystal).* The tetrahelix twist $\arccos(-\frac{2}{3})$ is Niven-irrational, so the Boerdijk–Coxeter helix never closes in flat 3D — the aperiodic, two-incommensurate-frequency time-crystal clock — yet closes at exactly 30 tetrahedra in S^3 (the 600-cell $= 20 \times 30$), and $30 = h(E_8)$ is the middle factor of the supercycle $51840 = 24 \cdot 30 \cdot 72 = |\mathrm{Sp}(4,3)|$. So from the single integer $q = 3$ — three points closing a triangle, saving a trit — unfold the surfaces (genus $0, 1, 3$), the error-correcting code ($H_1 = 81$), and the quasicrystal clock (supercycle $|\mathrm{Sp}(4,3)|$): geometry, code, and time are one triangulation. (Witnesses: `analysis/w33_klein_quartic_genus3.py`, `analysis/w33_trit_saving_code.py`, `analysis/w33_bc_helix_quasicrystal.py`.)

B.14 The clock is a mass and the supercycle is the gauge group: running the machine is the physics

The movement closes on the runtime itself. The holonet's clock is not an external metronome — it is a *topological harmonic oscillator* living on the Heawood graph (the incidence graph of the Fano plane PG(2, 2), the same 7 points and 7 lines that the time-bin guard shell uses). Its Laplacian spectrum is $\{0, (q-\sqrt{\lambda})^6, (q+\sqrt{\lambda})^6, 2q\} = \{0, (3-\sqrt{2})^6, (3+\sqrt{2})^6, 6\}$, and on the 12-dimensional middle shell

$$(L_H - qI)^2 = \lambda I, \qquad \omega = \sqrt{\lambda} = \sqrt{2}, \qquad E_\pm = q \pm \sqrt{\lambda} = 3 \pm \sqrt{2},$$

two six-mode Cl(1, 1) branches — an exact discrete oscillator whose frequency is $\sqrt{\lambda}$. That frequency is not abstract: it is a mass. The heaviest fermion sits at

$$m_{\mathrm{top}} = \frac{v_{EW}}{\sqrt{\lambda}} = \frac{246}{\sqrt{2}} = 173.95 \text{ GeV},$$

so the machine's clock *rate* is the top-quark mass scale. And the runtime stack $8 \to 48 \to 24 \to 72 \to 2160 \to 51840$ tops out at the full Clifford supercycle

$$51840 = 720 \cdot 72 = 24 \cdot \underbrace{30}_{h(E_8)} \cdot \underbrace{72}_{q^2 \cdot 8} = |\mathrm{Sp}(4,3)|,$$

with the 72-tick frame $= q^2\times$ the eight-tick word and the 2160 mirror bus $= h(E_8)\times$ frame. The supercycle is the order of the gauge group. So executing the machine for one supercycle is one complete traversal of Sp(4, 3), ticked by the oscillator whose frequency is the top mass: the holonet does not *simulate* the physics on a clock, its clock is the mass-setting oscillator and one run of its instruction stream is the gauge dynamics. Running the machine and enacting the physics are the same act. (Witness: `analysis/w33_machine_clock_is_mass.py`.)

Yet the clock is a *separate* module, and its arithmetic says why. The clock/readout layer is the Fano plane PG(2, 2) with symmetry $\mathrm{PSL}(2,7) = \mathrm{PSL}(3,2)$ of order $168 = 2^3 \cdot 3 \cdot 7$ (the same layer that carries the $168 = 7 \cdot f$ detector-bin weld), while the gauge/state layer is Sp(4, 3) of order $51840 = 2^7 \cdot 3^4 \cdot 5$. The clock prime $7 = \Phi_6(3)$ — the external cyclotomic prime that also fixes β_0^{QCD} and the atmospheric angle — *does not divide* the gauge order, so PSL(2, 7) is no subgroup of Sp(4, 3): the clock's Fano symmetry can never be a gauge operation, which is the group-order reason the Heawood clock is a separate homology module rather than a $W(3,3)$ subgraph (the girth obstruction). The two layers are coprime except for

$$\gcd(51840, 168) = 24 = f,$$

and both realize that shared scale as an order-f subgroup — the single-qutrit Clifford $2T = \mathrm{SL}(2,3)$ inside Sp(4, 3), and the Fano point-stabilizer S_4 ($= 168/7$) inside PSL(2, 7). So the clock layer $= 7 \times f$ and the gauge layer $= (h(E_8) \cdot 72) \times f$ are welded only at the matter count $f = 24$, which is exactly where the clock–matter spectral resonance lands. The clock and the readout are one Fano layer, coprime to the computation and bonded to it at the 24-cell core. (Witness: `analysis/w33_clock_gauge_two_layer.py`.)

B.15 Four further faces of the machine: time, topological order, scrambling, and periodic orbits

Four short, exact results round out the physical reading of the machine. *(i) Time is the modular flow, and only the clock can source it.* By the Connes–Rovelli thermal-time hypothesis, physical time is the modular (Tomita–Takesaki) automorphism flow of the state, generated by $K = -\ln \rho$. On the maximally mixed gauge/matter sector $\rho = I/81$ one has $K \propto I$, so the modular flow is trivial: the H_1 register is *timeless*. On the Fano/Heawood clock the Gibbs state gives $K = \beta L_H + (\ln Z)I$, so the modular flow is generated by the clock Laplacian and ticks at frequency $\sqrt{\lambda} = \sqrt{2}$ — time emerges from the state, and only the clock layer sources it (KMS verified). *(ii) The matter sector is a non-abelian topological order.* The $2T$ gauge theory on Q is the quantum double $D(2T)$: 7 flux sectors (the conjugacy classes of $2T$), 42 anyons $=$ Catalan $C_5 = 2q\,\Phi_6$, and total quantum dimension $D = |2T| = 24 = f$. Topological protection is then concrete — logical information is anyon fusion/braiding, with total quantum dimension the matter count. *(iii) The machine is a near-optimal scrambler/decoder.* Because $W(3,3)$ is Ramanujan, its random walk mixes in the minimum time (total-variation mixing time $= 2 =$ diameter) and operator growth runs at the maximal non-backtracking rate $\sqrt{k-1} = \sqrt{11}$; as a holographic code it is therefore a Hayden–Preskill decoder, with a bulk perturbation recoverable from $O(\log)$ boundary modes essentially at once. *(iv) The spectrum is a periodic-orbit trace formula.* The closed non-backtracking geodesics (particle worldlines) are counted by $N_m = \mathrm{Tr}(B^m)$ on the Hashimoto operator ($N_3 = 960 = 6 \cdot 160$); the Ihara zeta factorizes with all non-trivial zeros on the critical circle $|u| = 1/\sqrt{11}$ — an exact graph Riemann Hypothesis, equivalent to the Ramanujan property, with topological entropy $\ln(k-1) = \ln 11$. (Witnesses: `analysis/w33_thermal_time_clock.py`, `analysis/w33_anyons_from_2T.py`, `analysis/w33_scrambling_decoder.py`, `analysis/w33_periodic_orbit_spectrum.py`.)

These four are not separate curiosities; cross-read against the corpus they are one thermal object, unified by the Hawking temperature $T_H = k/2\pi$. The modular flow of (i) is a KMS state at exactly that T_H, so time, temperature and the clock coincide; the scrambling time of (iii) is $\lambda = 2$ with Page entropy $S_{\max} = E = 240$, the same fast-scrambler the corpus records. The topological order of (ii) is the *non-abelian* Drinfeld double $D(2T)$ ($42 = C_5 = v + \lambda$ anyons, total quantum dimension f), complementary to the *abelian* double $D((\mathbb{Z}/3)^2)$ of dimension $q^\mu = 81$ that is the logical register itself — one double per sector. And the zeta of (iv), counting non-backtracking geodesics $\mathrm{Tr}(B^m)$, certifies the same Ramanujan / graph Riemann Hypothesis as the corpus's point-counting $\mathrm{Tr}(A^n)$, from the complementary walk operator. So the new thermal-time/modular layer is what binds the pre-existing thermodynamic, topological, and arithmetic pillars into a single structure. (Witness: `analysis/w33_thermal_synthesis.py`.)

B.16 A near-term falsification test: the demonstrator as an experiment on the substrate

The single-photon demonstrator of the build sheet is buildable now and needs no GKP states or squeezing, yet the holonomic-gate and quasicrystal-clock structure make it a *discriminating, parameter-free* test of the substrate hypothesis itself. Three signatures, all measurable on that machine:

1. **Geometric-gate timing-independence (the discriminator).** If the gates are Wilczek–Zee holonomies (the substrate claim), the gate unitary depends only on the *path* traced in control space, not on the schedule. Reparametrising the loop — any speed, any pulse shape with the same path — leaves the gate, read out through the trace–Choi visibility

$V(U) = |\mathrm{Tr}\, U|/3$, *invariant*; a generic dynamical gate would change. A measured timing-dependence falsifies the holonomic claim.

2. **Holonomy-group fingerprint.** Composing the geometric gates must close into $2T = \mathrm{SL}(2,3)$: order 24, element-order spectrum $\{1,1,8,6,8\}$, and a single-qutrit Clifford visibility spectrum $V \in \{0, \frac{1}{3}, \frac{1}{\sqrt{3}}, 1\}$ (with $V(F_3) = \frac{1}{3}$ and $V(X) = V(Z) = 0$). Gate-set tomography reads this out; any other group falsifies the gauge-connection-holonomy identification.

3. **Provably aperiodic clock.** The Boerdijk–Coxeter drive rotates by $\theta = \arccos(-\frac{2}{3}) \approx 131.81^\circ$ per pass. By Niven's theorem [18] the only rational $\cos(r\pi)$ with r rational are $\{0, \pm\frac{1}{2}, \pm 1\}$, and $-\frac{2}{3}$ is none of them, so θ/π is irrational and the drive *never recurs* — a provably aperiodic time-quasicrystal clock. A measured recurrence at any finite pass falsifies it.

Each signature is fixed by the geometry with no free parameter, and each is within reach of the demonstrator's existing optics. Together they turn the buildable machine into an experiment *on the substrate*: not a test of whether the device computes, but of whether the gauge connection, the holonomy group, and the quasicrystal drive are the ones $W(3,3)$ predicts. (Witness: `analysis/w33_near_term_falsification_test.py`.)

B.17 BT1402 runtime-frontier handoff: readout, port, and certificate

The post-BT1377 runtime frontier makes the physical story sharper. The deterministic packet machine is a protected Clifford machine first: the 8-tick optical word, Q6/tomotope packet edge, S_3 phase synchronization, central C_3 Steinberg scheduler, and 51840-window $\mathrm{Sp}(4,3)$ supercycle are all finite Clifford data. Universal computation enters at the non-Clifford boundary, not by pretending that this Clifford kernel is already universal by itself.

The near-term single-photon demonstrator now has a concrete readout contract. Its Bell-qutrit signatures are

$$V(I) = 1, \qquad V(F_3) = \frac{1}{3}, \qquad V(X) = V(Z) = 0,$$

and route-controlled Clifford transport deliberately hides route coherence if the Bell legs are discarded: the reduced route register has uniform probabilities, purity 1/3, and zero ℓ_1 coherence. Coherence is restored only by the quantum eraser on the Bell branch,

$$\frac{|\Omega\rangle + |Z\Omega\rangle + |X\Omega\rangle}{\sqrt{3}},$$

which succeeds with probability 1/3, restores route ℓ_1 coherence 2, and routes the interferometer to a single output port. The current parametric noise envelope keeps this falsifiable: the baseline readout has $\gamma = 0.92887$ and port-0 probability 0.95258, while the conservative case keeps $\gamma = 0.79198$ and port-0 probability 0.86132.

The non-Clifford resource has likewise become an ABI rather than a slogan. The Hesse-SIC/T token has nine outcomes, is consumed at a Clifford packet boundary, and feeds a Clifford correction plus one T-frame parity bit into the next 8-tick word. Over one 51840-tick Clifford window there are 720 microframes. In the current queue envelope the baseline heavy case still has positive slack (about 55.75 successful tokens per window), and the conservative medium case still has positive slack (about 89.51 tokens per window). This is an engineering envelope, not a claim that the SIC optics or magic-state yield are solved.

Finally, the correction frontier is now stated in certificate language. The best current S_3 gauge witness has score 210 and is locally strict through the checked radius, but the MaxSAT frontier remains witness-only: the example optimality certificate exercises the verifier path and is explicitly synthetic. A project-level global optimum requires an imported solver upper-bound proof matching the witness score. This is the honest architecture boundary after BT1401: the single-photon experiment tests the Bell/route physics, the Clifford machine runs the protected finite kernel, the Hesse-SIC/T port supplies the non-Clifford ABI, and the remaining global optimality claim is a certificate import problem.

Witnesses: `tools/bt1394_reduced_qutrit_demonstrator.py`; `tools/bt1396_qutrit_quantum_erasure_readout.py`; `tools/bt1385_hesse_sic_t_port_abi.py`; `tools/bt1391_hesse_sic_t_queue_model.py`; `tools/bt1395_s3_maxsat_bound_pathway.py`; `tools/bt1400_qutrit_erasure_noise_sensitivity.py`.

B.18 BT1403 eraser-lift port: the Hesse boundary is the readout boundary

The Hesse port is not a second computer attached to the holonet after the fact. It is the quantum-eraser measurement boundary lifted by one qutrit phase label. BT1396 gives the first half of the boundary: the route register is incoherent if the Bell legs are traced out, and coherence returns only when the Bell branch is erased by the projector

$$\frac{|\Omega\rangle + |Z\Omega\rangle + |X\Omega\rangle}{\sqrt{3}}.$$

This is a three-branch measurement, with branch labels

$$\Omega, \quad Z\Omega, \quad X\Omega.$$

BT1385 gives the non-Clifford half: the Hesse-SIC/T token has nine outcomes. The ABI statement is therefore the exact lift

$$9 = 3 \times 3, \qquad h = 3r + p,$$

where $r \in \{0, 1, 2\}$ selects the route/Bell branch and $p \in \{0, 1, 2\}$ is the added phase label. Equivalently, the nine Hesse outcomes are exactly 3 route branches times 3 phase labels. The feed-forward record maps the outcome to a two-trit Clifford correction $X^r Z^p$ plus the one T-frame parity bit needed by the non-Clifford injection.

This is the physical meaning of the non-Clifford port in the single-photon architecture. The same interferometric readout that proves the route qutrit is coherent also supplies the boundary at which a non-stabilizer qutrit resource can be consumed. The Clifford runtime remains an 8-tick packet machine; the Hesse measurement writes a side record, restores the packet ABI, and hands the next word back to the protected scheduler. Thus the architecture is not "Clifford machine plus mysterious magic box" but

$$\text{Bell-branch eraser} \quad \longrightarrow \quad \text{Hesse } 3 \times 3 \text{ lift} \quad \longrightarrow \quad \text{next Clifford packet word.}$$

The boundary remains honest: this identifies the ABI and alphabet of the port, not the optical SIC implementation or magic-state factory yield. (Witness: `tools/bt1403_hesse_port_eraser_lift.py`.)

B.19 BT1404 holonet scope: one Hesse grid is one microframe

BT1404 makes the eraser-lift port visible as a packet oscilloscope. The timing identity is

$$9 \text{ Hesse outcomes} \times 8 \text{ packet ticks} = 72 \text{ ticks},$$

exactly one BT1385/BT1391 microframe. In plain terms: 9 Hesse outcomes times 8 packet ticks fill the whole scope. Thus the entire Hesse outcome alphabet fits in one microframe, with each cell $h = 3r + p$ occupying one 8-tick return word.

The scope word is:

$$\begin{array}{r|l} 0 & \text{erase Bell branch} \\ 1 & \text{record route trit } r \\ 2 & \text{record phase trit } p \\ 3 & \text{apply } X^r \text{ frame correction} \\ 4 & \text{apply } Z^p \text{ frame correction} \\ 5 & \text{store } T\text{-frame bit } (r + p \bmod 2) \\ 6 & \text{restore Clifford ABI} \\ 7 & \text{hand off next packet word.} \end{array}$$

This is the first machine-surface form of the non-Clifford boundary: not just a nine-outcome label, but a one-microframe live trace compatible with the 720 microframes in a 51840-tick Clifford window. The generated static scope is `docs/bt1404_holonet_scope.html`. It remains an ABI display, not a claim of physical SIC-optics hardware.

B.20 BT1405 continuous Q6 path router

BT1374 turned packet digits into executable Q6 edge addresses, but deliberately left one boundary open: an address list is not yet a continuous walk. BT1405 closes that boundary at the tomotope ABI level. For every compiled BT828 packet program, orient the packet edges to minimize the total Hamming connector length, insert shortest Q6 connector walks between consecutive packet edges, and map every traversed edge back through the BT1371 table.

The six-digit stress route keeps the six BT1374 packet edges

$$175, \ 133, \ 56, \ 37, \ 142, \ 77$$

but now they are waypoints on a single hypercube walk:

$$6 \text{ packet edges plus } 10 \text{ connector edges} = 16 \text{ Q6 steps}.$$

In words: 6 packet edges plus 10 connector edges give the continuous carrier trace. Since the tomotope body budget is 48 ticks, the continuous trace leaves

$$48 - 16 = 32$$

slack ticks inside the same body. Thus the route is no longer merely a set of six distinct Q6 addresses; it is a contiguous Q6 path from `110111` to `010011`, with every traversed edge carrying a tomotope flag and block. This is still an ABI route certificate, not a waveguide layout or detector-loss model. (Witness: `tools/bt1405_continuous_q6_path_router.py`.)

B.21 BT1406 tomotope-body edge pulse scheduler

BT1406 uses the slack exposed by BT1405 as timing structure. Each Q6 edge in the continuous route is not a bare edge label; it expands into the ternary microcycle

$$\begin{array}{r|l} 0 & \text{LOAD\_FLAG: load tomotope flag, block, and transversal} \\ 1 & \text{FLIP\_Q6\_AXIS: perform the one-bit Q6 traversal} \\ 2 & \text{LATCH\_VERTEX: latch the target vertex and ABI record.} \end{array}$$

Therefore the stress route obeys

$$16 \text{ Q6 edge traversals} \times 3 \text{ pulse phases} = 48 \text{ body ticks.}$$

In words: 16 Q6 edge traversals times 3 pulse phases equals 48 body ticks. The old 32-tick slack becomes the timing shell: the stress route has no idle body tick after expansion. Its packet-edge load ticks are

$$0,\ 6,\ 12,\ 21,\ 27,\ 45,$$

so the final packet edge occupies ticks $45, 46, 47$: load tomotope flag 180 in block 45, flip 010111 $\to$ 010011, and latch the target vertex. This is a timing ABI, not a calibrated optical pulse-width, detector-jitter, or loss model. (Witness: `tools/bt1406_tomotope_body_edge_pulse_scheduler.py`.)

B.22 BT1407 microframe transaction composer

BT1407 closes the whole oscillator frame. BT1406 fills the first 48 tomotope-body ticks. The remaining BT1300 local-lift epilogue is

$$24 = 3 \times 8,$$

exactly three Hesse return words. In words: 48 Q6 body pulse ticks plus 3 Hesse return words equals 72 ticks.

For the six-digit stress route, the final target digit is 4. Hence $4 \bmod 3 = 1$, so the selected Hesse route branch is $r = 1$. The epilogue is the phase fanout along that branch:

frame ticks	h	(r, p)	correction
48–55	3	$(1, 0)$	$X^1 Z^0$
56–63	4	$(1, 1)$	$X^1 Z^1$
64–71	5	$(1, 2)$	$X^1 Z^2$.

Thus ticks 0–47 execute the Q6/tomotope carrier body, while ticks 48–71 execute the Hesse return words that erase the branch, record route and phase, apply the Pauli-frame correction, store the T-frame bit, restore the Clifford ABI, and hand off the next packet word. This is a full-frame ABI transaction, not a SIC-optics implementation. The witness is the BT1407 generator and JSON certificate.

B.23 BT1408 Witting contextual communication bridge

BT1408 imports the Witting-polytope communication scheme into the same runtime ABI. The external protocol uses 40 Witting rays and 40 orthogonal tetrads. Around any selected ray the other side of the delayed-query round splits as

$$1 \text{ same ray} + 12 \text{ compatible orthogonal rays} + 27 \text{ incompatible rays} = 40.$$

In words: 1 same ray plus 12 compatible orthogonal rays plus 27 incompatible rays make the whole Witting communication desk. Hence the accepted key-agreement rate is

$$\frac{1+12}{40} = \frac{13}{40},$$

and the rejected sector is 27/40, the same q^q matter shell that the holonet already uses as contextual fuel.

The four slots of an accepted Witting tetrad are not a new bus. They are the four local residues in the packet ABI:

$$\text{tomotope\_flag} = 4\,\text{tomotope\_block} + (\text{mirror\_slot} \bmod 4).$$

Thus an accepted communication round has the handoff

$$\text{Witting query pair} \to \text{common tetrad} \to \text{mirror slot} \to \text{Q6 body} \to \text{Hesse epilogue}.$$

The contextuality audit remains the corrected BT823 ceiling 36/40, with deficit 4 and contextual fraction 1/10; the older 34/40 value from the communication paper is a useful historical warning, not the local theorem. BT1408 is an ABI bridge, not a security proof or a ququart hardware certificate.

B.24 BT1409 Witting duplex admission scheduler

BT1409 separates the two acceptance clocks inside the Witting communication bridge. The state-query clock is the BT1408 shell:

$$\frac{13}{40} = \frac{1 \text{ same ray} + 12 \text{ compatible rays}}{40}.$$

The basis-query clock is smaller. A selected Witting ray lives in exactly four tetrads, hence

$$\frac{4}{40} = \frac{1}{10}$$

is the witness aperture. In words: 13/40 is communication throughput, while 1/10 is contextual witness aperture. The complementary basis shadow is

$$\frac{36}{40},$$

which matches the corrected BT823 noncontextual ceiling. Thus the old "36/40" number is not a communication success rate; it is the basis retry shadow and the classical boundary.

For one selected ray, serving every compatible state as a BT1407 transaction uses

$$13 \times 72 = 936$$

frame ticks. Auditing only the four witness apertures uses

$$4 \times 72 = 288$$

frame ticks. Rejected choices need not consume a full frame; they are classified as retry shadow or matter-shell context unless an implementation chooses to log them. This is a scheduler certificate, not a cryptographic security proof.

B.25 BT1410 Witting delayed-query frame compiler

BT1410 makes the delayed-query step operational. The protocol in [28] first lets the two sides query Witting rays, then delays the final basis measurement until the queried states have been compared. The holonet reading is a compiler:

$$(\text{Alice ray}, \text{Bob ray}) \longmapsto \text{common Witting tetrad} \longmapsto \text{BT1407 frame}.$$

If the two rays share no tetrad, the pair is a retry-shadow event and no full frame is opened. If they are distinct compatible rays, they share exactly one tetrad, so the selected basis is forced. If they are the same ray, they share four tetrads; a two-bit aperture selector chooses which witness basis to audit.

There are two tables, and confusing them loses the physics. The logical ordered pair table has

$$40 \cdot 40 = 1600$$

rows, split as

$$40 \text{ same-ray pairs} + 480 \text{ compatible distinct pairs} + 1080 \text{ retry-shadow pairs}.$$

Thus the accepted logical table has 520 rows and rate $520/1600 = 13/40$. The physical frame table is basis-local:

$$40 \text{ tetrads} \times 4 \text{ Alice query slots} \times 4 \text{ Bob query slots} = 640.$$

Equivalently, every Witting tetrad is a 4×4 frame tile with four diagonal witness apertures and twelve off-diagonal data handshakes. Across all forty tetrads this is

$$160 \text{ diagonal witness records} + 480 \text{ off-diagonal data records}.$$

The extra $640 - 520 = 120$ basis-local records are not extra throughput. They are the same-ray contextual aperture: each of the 40 rays has four basis contexts, so the physical compiler carries three extra context choices per ray above the single logical same-ray identity.

Once the common basis is selected, the later four-outcome measurement slot is just the packet slot:

$$\texttt{tomotope\_flag} = 4\,\texttt{tomotope\_block} + (\texttt{mirror\_slot} \bmod 4).$$

Therefore the Witting desk is now a packet admission ROM. Off-diagonal entries open communication frames; diagonal entries open contextual audit frames; both land in the same BT1407 72-tick body-plus-Hesse-epilogue transaction. This matches the feed-forward/programmable-loop physical idiom of time-bin photonic qudit processors [29], but BT1410 itself remains a finite compiler certificate rather than a loss model, detector calibration, or cryptographic security proof.

B.26 BT1411 Witting basis analyzer unitaries

BT1411 supplies the physical analyzer that BT1410 deliberately left abstract. For every Witting tetrad

$$B = (r_0, r_1, r_2, r_3)$$

define U_B by taking row j to be the conjugate ray $r_j^\dagger$. Then

$$U_B\, r_j = e_j,$$

so detector slot j is exactly the packet residue `mirror_slot` mod 4. This is the standard linear-optics measurement move: an analyzer unitary rotates the chosen orthonormal basis to the detector basis, after which fixed detectors read the slots. General finite-dimensional unitaries are optically synthesizeable by beam-splitter/phase-shifter meshes [30, 31]; BT1411 shows the Witting case is much more structured than a generic 4×4 mesh.

The forty analyzers have the exact hardware split

$$1 + 12 + 27.$$

There is one computational direct-rail analyzer, twelve one-direct-rail plus complement-tritter analyzers, and twenty-seven four-three-rail contextual analyzers. Equivalently, the nonzero-entry histogram is

$$4^1, \quad 10^{12}, \quad 12^{27},$$

where the exponent records how many analyzers have that many nonzero matrix entries. No Witting analyzer uses all 16 entries of a generic dense four-mode unitary. The entry alphabet is only

$$0, \quad 1, \quad \frac{\pm 1}{\sqrt{3}}, \quad \frac{\pm \omega}{\sqrt{3}}, \quad \frac{\pm \omega^2}{\sqrt{3}}, \qquad \omega^3 = 1.$$

Thus the physical reading is sharper than "program a universal interferometer." The Witting communication desk is a sparse analyzer ROM:

$$\begin{aligned} \text{delayed-query tetrad} &\longmapsto \text{four-mode Witting analyzer} \\ &\longmapsto \text{detector slot} \longmapsto \text{BT1374 packet residue.} \end{aligned}$$

The twelve middle analyzers literally have one bypass rail and a three-rail tritter-like complement; the twenty-seven contextual analyzers are balanced three-rail rows, each dark on one rail. This is a physics certificate for the single-photon ququart analyzer bank, not a chip calibration, loss budget, or beam-splitter-angle synthesis.

B.27 BT1412 toroidal Q4 oscillator boundary

BT1412 returns to the 4×4 toroidal square because that square is not a decorative drawing of the local packet router. With ordinary boundary conditions the knight graph has only 24 edges and mixed degree; with toroidal wraparound it is exactly Q_4. The existing closed toroidal knight tour is therefore a 16-tick Gray Hamilton clock on the four-cube: each packet tick flips one Q_4 bit, and the board parity $(r + c)$ mod 2 is the same as the Q_4 word parity. Hence parity alternates on every clock edge.

The new splice is that the two-dimensional cells of this same local router have the right oscillator boundary:

$$\#F_2(Q_4) = \binom{4}{2} 2^2 = 24 \text{ Q4 square faces} = q(q-1)(q+1).$$

BT802 already proved that the tetrahedral oscillator is allowed exactly at the mod-12 marks

$$\{0, 3, 4, 7\}$$

and that the neighborly genus level $h = q = 3$ is forbidden by the discriminant 145. Lifting those four marks onto the 24-plaquette Q_4 shell gives the exact aperture

$$\frac{4}{24} = \frac{1}{6},$$

which is the local "one-sixth" selector window in the clock face, not a fitted probability. Removing the forbidden genus level then lands on the toroidal dual edge invariant:

$$24 - 3 = 21.$$

That 21 is not numerology. It is the edge channel preserved by the dual toroidal polyhedra:

$$\text{Császár} : (V, E, F) = (7, 21, 14), \qquad \text{Szilassi} : (V, E, F) = (14, 21, 7).$$

Császár is the maximal vertex-adjacency boundary, Szilassi is the maximal face-adjacency boundary, and geometric duality swaps V and F while keeping $E = 21$ intact. In this exact local reading the tetrahedral oscillator sits between the two toroidal boundaries: the Q_4 plaquette shell supplies the 24-slot face clock, the forbidden $q = 3$ genus gap removes the non-realizable handle, and the surviving 21-channel edge bus is precisely the dual boundary shared by the two toroidal polyhedra.

The error-code boundary is also now sharper. A Hamilton cycle in a labeled hypercube is a cyclic Gray code, but a snake-in-the-box code is an induced path or cycle constraint [14, 15]. The full BT1412 Q_4 Gray clock is not claimed to be such a snake: it uses 16 cycle edges while Q_4 has 32 edges, so there are 16 non-cycle chords. The protected projection is instead the every-other-tick even layer

$$0,\ 6,\ 3,\ 5,\ 9,\ 15,\ 10,\ 12,$$

an eight-word parity-even code whose cyclic steps have Hamming distance 2 and whose pairwise minimum distance is 2. Thus the physical story is: toroidal Q_4 Gray clock for packet motion, every-other parity projection for single-bit-error detection, and the 1/6-aperture oscillator boundary tying the 24-plaquette shell to the 21-edge Császár/Szilassi channel. BT1412 is an ABI certificate, not a waveguide layout, a calibrated oscillator, or a new snake-code optimum. (Witness: `tools/bt1412_toroidal_q4_oscillator_boundary.py`.)

B.28 BT1413 Q4 plaquette-tomotope compiler

BT1413 makes the 24-plaquette boundary executable. BT1363 had already shown that Q4 face-edge incidence modulo the antipodal translation is the tomotope/Reye medial layer, with 48 middle blocks. BT1371 had already shown that the 192 tomotope flags are a bijective Q6 edge address table. The BT1413 compiler is just the composed map:

$$24 \text{ Q4 plaquettes} \cdot 4 \text{ edge incidences}/2 = 48 \text{ middle blocks},$$

$$48 \text{ middle blocks} \cdot 4 \text{ flag residues} = 192 \text{ tomotope/Q6 flags}.$$

The generated certificate checks that each middle block has exactly two Q4 lifts, that the three ternary sheets carry 64 flags each, that each sheet hits all 16 tomotope face labels exactly once at block level, and that every flag is a one-bit Q6 edge address. The result is not an optical layout. It is the local address compiler that lets the toroidal Q4 shell talk directly to the tomotope/Q6 packet ABI.

B.29 BT1414 Csaszar-Szilassi dual port

BT1414 uses the active/ground split that BT1317 already found in the tomotope packet:

$$192 = 168 + 24.$$

The active part is now a concrete dual analyzer port:

$$21 \text{ shared toroidal edge channels} \cdot 2 \text{ orientations} \cdot 4 \text{ residues} = 168.$$

In Császár mode the seven analyzers are vertex-neighborhood checks: each vertex of K_7 sees six incident edge channels, and each channel is seen by two vertex analyzers. In Szilassi mode the same 21 channels are read as the dual complete face-adjacency checks: seven face analyzers, again six channels each, again two analyzers per channel. The current BT1318 metric-axis records fix Császár vertex 6 and Szilassi face 4, so the crossed calibration channel is the shared edge $\{4, 6\}$.

The remaining 24 flags are not discarded. They are the BT1412/BT1413 Q4 plaquette guard band. Thus the local physical port has a sharp shape:

$$21 \cdot 2 \cdot 4 \quad \text{active dual-boundary slots} \quad + \quad 24 \quad \text{Q4 plaquette guards.}$$

B.30 BT1415 even-projection Steinberg/CSS syndrome ledger

BT1415 finishes the front end by making the every-other Q4 projection into a check ledger. The allowed clock states are the eight even words

$$0, \ 6, \ 3, \ 5, \ 9, \ 15, \ 10, \ 12.$$

They form a binary distance-2 parity layer: every allowed word has parity zero, and any one-bit Q4 clock fault flips the syndrome bit. This check is binary front-end logic; the memory register is still the existing $\mathbb{F}_3$ Steinberg/CSS object.

BT1375 supplies the memory clock. The central C_3 action on the Steinberg register has 27 three-cycles and nilpotent rank profile

$$54, \ 27, \ 0.$$

Therefore the exact syndrome ledger is

$$27 \text{ Steinberg central cycles} \cdot 8 \text{ even Q4 states} = 216,$$

and the BT1414 guard band completes the existing W33 CSS edge ledger:

$$216 + 24 = 240.$$

This is the first complete local front end in the paper:

$$\begin{aligned} \text{Q4 plaquettes} &\to \text{tomotope/Q6 flags} \\ &\to \text{Császár/Szilassi port} \\ &\to \text{even-parity syndrome} \\ &\to \text{Steinberg/CSS edge ledger.} \end{aligned}$$

I also checked the external Golden Quartic/Moebius-ball electron line of work. Otto's 2022 paper connects a golden quartic, icosahedral coefficients, chirality, and a proposed Moebius-ball electron geometry [36]; his 2025 preprint asks whether AI can compare such extended electron models [37]. A 2025 review frames this family as Zitterbewegung/field-dynamics modelling rather than established particle physics [38]. The holonet use is deliberately conservative: closed-loop topology and chirality are good design warnings, but the repo's exact quartic frontier remains the two independent D_4 quartic atoms in the local minimal-magic audit. No electron mass, spin, or waveguide claim depends on the Moebius-ball model. (Witnesses: `tools/bt1413_q4_plaquette_tomotope_face_compiler.py`, `tools/bt1414_csaszar_szilassi_dual_physical_port.py`, and `tools/bt1415_even_projection_steinberg_syndrome_layer.py`.)

B.31 BT1416 sparse CSS intertwiner

BT1416 turns the BT1415 ledger from a count into matrices. It rebuilds the canonical W33 edge-chain CSS carrier:

$$H_X = d_1 : C_1 \to C_0, \qquad H_Z = d_2^T : C_1 \to C_2,$$

over $\mathbb{F}_3$. The generated sparse certificate has shapes

$$H_X \in \mathbb{F}_3^{40\times 240}, \qquad H_Z \in \mathbb{F}_3^{160\times 240},$$

and verifies

$$\operatorname{rank}(H_X) = 39, \qquad \operatorname{rank}(H_Z) = 120, \qquad H_X H_Z^T = 0.$$

Thus the carrier is still the exact edge code

$$[[240,\ 240 - 39 - 120,\ 3]]_3 = [[240, 81, 3]]_3.$$

The new map is the typed 240×240 ledger interwiner:

$$\Pi(e_i) = e_i, \qquad 0 \le i < 240.$$

Rows $0, \ldots, 215$ are the $\mathbb{F}_2$ Q4 parity front end, with check vector $(1, 1, 1, 1)$, rank 1, and the eight even Q4 words as a distance-2 kernel. Rows $216, \ldots, 239$ are the guard tail addressing the same $\mathbb{F}_3$ edge carrier. This is deliberately not a field coercion: the binary clock syndrome and ternary CSS stabilizer live in different fields, and the interwiner is the finite scheduling boundary between them. (Witness: `tools/bt1416_css_sparse_intertwiner_matrices.py`.)

B.32 BT1417 linear-optical dual-port primitive synthesis

BT1417 converts the BT1414 port into an objectwise single-photon optical primitive list. The active dual port is not a vague optical metaphor; it has the exact inventory

21 edge-channel couplers, 42 oriented phase latches,
168 active detector bins, 24 guard apertures.

The 21 couplers are the shared K_7 edge channels. The 42 latches choose the two directions on each channel. The four residue bins behind each oriented latch are the tomotope flag residue. The 24 Q4 guard apertures are separate from the active mesh.

The analyzer matrices are 7×21 incidence matrices. In Császár mode a row is a vertex star; in Szilassi mode a row is the dual face star. In both modes the Gram law is

$$AA^T = 5I_7 + J_7,$$

so every analyzer sees six channels and any two analyzers overlap in exactly one channel. This is the chip-level object language the previous ABI was missing: coupler, direction latch, residue demultiplexer, detector bin, and guard aperture. It is still not a calibrated waveguide mask, loss budget, or detector-efficiency model. (Witness: `tools/bt1417_linear_optical_dual_port_primitives.py`.)

B.33 BT1418 finite D4-quartic magic injection

BT1418 closes the non-Clifford aperture without importing the continuum Moebius-ball electron hypothesis. The repo-local minimal-magic audit already found the exact signed frontier: two independent irreducible D_4 quartic atoms, with no shared quadratic subfield and splitting field

$$D_4 \times D_4.$$

The BT1415/BT1416 guard band fits those atoms exactly:

$$2 \text{ quartic atoms} \cdot 4 \text{ algebraic branches} \cdot 3 \text{ qutrit phases} = 24 \text{ guard apertures}.$$

Then the internal D_4 orientation torsor expands each aperture to the tomotope flag bus:

$$24 \text{ guard apertures} \cdot 8\ D_4 \text{ orientations} = 192 \text{ oriented tomotope tokens}.$$

This is the exact finite counterpart of the golden-quartic topology that is safe to use in the paper. For one quartic atom, the Steinberg lift gives

$$4 \text{ branches} \cdot 27 \text{ central cycles} \cdot 8\ D_4 \text{ orientations} = 864,$$

which is exactly the repo's Golden D_4/Weyl shell. With both independent atoms the shell is 1728, but the two atoms stay independent; BT1418 does not collapse them into one continuum object. The physical reading is that the 24 guard apertures are the non-Clifford injection rail, while the 216 parity rows remain the cheap binary front-end syndrome and the $\mathbb{F}_3$ CSS carrier keeps its commutation proof. (Witness: `tools/bt1418_d4_quartic_magic_injection_frontier.py`.)

B.34 BT1697 typed packet ABI

Reading the paper back as an architecture exposes the missing promoted object. The header introduced in §4

$$(\text{payload}, \text{Pauli frame}, \text{chart word}, \text{apartment hop}, \text{mirror slot}, \text{clock phase})$$

is not only a description of fields. It is a fixed-width transaction ABI whose verified projections are all the packet layers above. The master frame identity is

$$72 = 48 + 24 = 16 \cdot 3 + 3 \cdot 8.$$

The first 48 ticks are the continuous Q6/tomotope body:

$$16 \text{ Q6 edge traversals} \times 3 \text{ pulse phases} = 48,$$

and the last 24 ticks are the selected Hesse branch:

$$3 \text{ Hesse return words} \times 8 \text{ ticks} = 24.$$

The same transaction header receives the Witting communication desk as an admission ROM,

$$1600 = 40^2 = 520 + 1080, \qquad 520/1600 = 13/40,$$

while the basis-local physical table is

$$640 = 40 \cdot 4 \cdot 4 = 480 + 160.$$

The extra $160 - 40 = 120$ physical rows are the same-ray contextual aperture, not extra throughput. At the physical front end the dual toroidal port is

$$192 = 168 + 24 = 21 \cdot 2 \cdot 4 + 24,$$

and at the memory front end the CSS edge ledger is

$$240 = 216 + 24 = 27 \cdot 8 + 24.$$

Finally the non-Clifford guard rail is typed by the finite quartic atoms,

$$24 = 2\ D_4 \text{ atoms} \cdot 4 \text{ branches} \cdot 3 \text{ qutrit phases}, \qquad 192 = 24 \cdot 8.$$

Thus 24 is not one untyped object renamed four times. It is the common guard boundary at which Hesse epilogue, Q4 guard flags, CSS guard-tail rows, and D_4-quartic magic apertures meet while retaining distinct source certificates and field semantics. Globalizing the local 48-block packet then recovers the mirror runtime:

$$2160 = 45 \cdot 48, \qquad 51840 = 24 \cdot 2160 = 24 \cdot 45 \cdot 48.$$

So the packet header is the finite operating-system object of the holonet: compute, route, admit, check, inject, and commit are typed projections of one transaction. BT1697 is an ABI theorem, not a calibrated optical layout. (Witness: `analysis/bt1697_holonet_typed_packet_abi.py`.)

B.35 BT1698–BT1700 execution stack

The ABI can now be executed, lowered, and recursively scaled. BT1698 reads the existing microframe rows as a deterministic state machine. The body is not a bag of 48 ticks; it is exactly 16 chained Q6/tomotope edges, each with the three-phase instruction word

`LOAD_FLAG` → `FLIP_Q6_AXIS` → `LATCH_VERTEX`.

Every committed target vertex is the next edge source. The epilogue is exactly three Hesse return words, each with the eight-tick Clifford word

`ERASE`, `ROUTE`, `PHASE`, `X-CORR`, `Z-CORR`, `T-BIT`, `RESTORE`, `NEXT`.

Thus the packet is a finite transition system:

$$72 = (16 \cdot 3) + (3 \cdot 8).$$

BT1699 then lowers the same ticks onto the physical single-photon frame already used by the Witting/Fano automaton:

ticks	hardware role
$0 \ldots 7$	source switch
$8 \ldots 31$	program/delay
$32 \ldots 47$	analyzer or OAM body
$48 \ldots 63$	detector or Hesse handoff
$64 \ldots 71$	dark-reference closeout.

Each tick has a symbolic loss hook, and the last eight ticks carry the dark reference. The guard boundary is welded objectwise:

$$\text{port guard flag }(168+i) = \text{ CSS edge row }(216+i) = D_4\text{ magic aperture }i, \qquad 0 \le i < 24.$$

This proves that the repeated 24 is a typed interface between physical port, memory syndrome, and non-Clifford resource rail.

BT1700 is the fractal compiler law. Substituting a W(3,3) packet fabric at each site gives, at depth n,

$$N(n) = 40^n, \qquad T(n) = 72 \cdot 40^n,$$
$$B(n) = 48 \cdot 40^n, \qquad G(n) = 24 \cdot 40^n, \qquad B(n) = 2G(n),$$

and the route/scheduler clocks

$$R(n) = 8n, \qquad S(n) = 2160 \cdot 40^n, \qquad C(n) = 51840 \cdot 40^n.$$

The local operating system never changes: every scale keeps the same packet ABI, the same 48/24 body/guard split, and the same typed guard handoff. This is the computer/network architecture in one sentence: a single-photon packet operating system recursively tiled by W(3,3). (Witnesses: `analysis/bt1698_holonet_packet_state_machine.py`, `analysis/bt1699_holonet_abi_to_hardware_lowering.py`, and `analysis/bt1700_recursive_holonet_packet_compiler.py`.)

B.36 BT1701–BT1703 runtime safety layer

The execution stack has three immediate runtime consequences. First, BT1701 renders the packet as an actual trace artifact at `docs/bt1701_holonet_packet_trace_visualizer.html`. The page is generated from the verifier data, not drawn by hand: every tick in the 72-tick packet is colored by the physical stage

$$8 + 24 + 16 + 16 + 8,$$

and the same page records the 48/24 split, the 24-row guard weld, and the recursive compiler law.

Second, BT1702 proves the scheduler boundary. The collision-free recursive key is

$$\text{extended slot} = \text{packet address} \cdot 2160 + \text{polar sheet} \cdot 48 + \text{body slot}, \qquad \text{phase} = \text{body slot mod } 3.$$

With this key, recursive packets do not collide. If one intentionally reuses a single finite 2160-slot physical bus, the packet address must become a time slice:

$$(\text{packet time slice}, \text{local mirror slot}, \text{phase}).$$

This is the correct engineering boundary. The recursive compiler gives a collision-free addressing fabric; it does not turn one finite bus into simultaneous infinite parallel hardware.

Third, BT1703 classifies symbolic faults at the ABI boundary:

fault	finite exit
loss	retry frame or local reprogram retry
dark click	local dark-reference termination
parity fault	CSS syndrome handoff.

The verifier injects 72 loss hooks, 8 dark-reference clicks, and 24 guard-weld parity faults. Every symbolic fault has one finite exit. This is still not a stochastic noise model; it is the routing table that a calibrated noise model must plug into. (Witnesses: `analysis/bt1701_holonet_packet_trace_visualizer.py`, `analysis/bt1702_holonet_scheduler_collision_audit.py`, and `analysis/bt1703_holonet_fault_propagation_simulator.py`.)

B.37 BT1704–BT1706 replay and retry economics

The next layer turns runtime safety into an operational model. BT1704 replays the 72-tick event log twice from the same initial state. The two executions produce identical tick logs and identical final registers. The final cursor is

`HESSE_WORD:5:DONE`,

with final corrections X^1, Z^2, and time-frame bit 1. This is the determinism test the packet ABI needed: the trace is not merely ordered; it is replayable.

BT1705 simulates one shared 2160-slot mirror bus under FIFO time division. One packet consumes one 2160-slot service slice. Three profiles are tested:

profile	packets	max queue
depth-1 burst	40	40
depth-2 wavefront	1600	1561
depth-2 sequential	1600	1.

All profiles are collision-free, every packet is served once, and Jain fairness is exactly 1. The result is deliberately engineering-facing: the finite bus is fair and schedulable, but bursty recursive traffic pays finite queueing latency.

BT1706 attaches symbolic rates to the BT1703 fault exits. For the nominal profile

$$p_{\text{loss}} = 10^{-5}, \qquad p_{\text{dark}} = 10^{-6}, \qquad p_{\text{parity}} = 5 \cdot 10^{-6},$$

the expected retry/reprogram load per packet is

$$64\, p_{\text{loss}} = \frac{2}{3125},$$

and the expected CSS handoff load is

$$24\, p_{\text{parity}} = \frac{3}{25000}.$$

The verifier also includes an engineering 10%-guard profile and the exact guard-budget limit profile. These are not experimental thresholds; they are the exact ABI economics once a bench supplies rates. (Witnesses: `analysis/bt1704_holonet_packet_replay_runner.py`, `analysis/bt1705_holonet_shared_bus_time_division_simulator.py`, and `analysis/bt1706_holonet_retry_economics.py`.)

C The exceptional skeleton and its physics

The architecture above shows that a single photon on $W(3,3)$ is a universal computer, network, and clock. This section turns from the *machine* to the *world*: it shows that the same $q = 3$ substrate carries an exact tower of exceptional structures — and a layer of falsifiable physics.

Roadmap. The argument is one ascending ladder, each rung an executable witness.

1. *Why* $q = 3$ (§C.14): three independent selections — geometric ($q! = 2q$), quantum-resource (minimal magic dimension), and holographic ($c = 24$) — all force $q = 3$.

2. *The qubit/qutrit crossover and the genus tower*: the multi-qubit contextuality ladder meets the $\{3,7\}$ Hurwitz tower at the genus-7 Macbeath surface; the $\{3,8\}$ octagon tower is its qubit twin; and the substrate selects the vertex figure $n-6 \mid k=12$.
3. *The Eisenstein body*: the Witting polytope (240 E8 roots, 2160 bus edges, 40 $W(3,3)$ rays) is the substrate over $\mathbb{Z}[\omega]$, welded to E8 two ways (Eisenstein $\omega = C^{10}$ and the 120 icosians).
4. *The exceptional rungs* $\mathrm{E}_6 \to \mathrm{E}_7 \to \mathrm{E}_8$: the Hessian polytope (27 lines), the Klein quartic (28 bitangents), and the icosian 600-cell; their Weyl groups telescope $51840 \to \times 56 \to \times 240$.
5. *Above E8*: the complex Leech (Eisenstein, $12 = k$), the Suzuki chain from the three-qubit $\mathrm{G}_2(2)$, and the Monster at $c = 24 = f$.
6. *The physics* (§C.11): the Standard Model from trinification, gauge unification, the neutrino sector, and the falsifiable scorecard — with one honest tension and its resolution.

The whole climb is drawn in `docs/w33_exceptional_tower.svg`, and its honest epistemic status (framework / derived / measurable) is kept by `analysis/w33_exceptional_tower_ledger.py`: most rungs are exact theorems wearing substrate labels, a smaller set is computed from $q=3$, and only the measurable layer below falsifies.

Results at a glance. The substrate's falsifiable predictions, all fixed by $q=3$ arithmetic with no free parameters ($N = 2(v-\Phi_4) = 60$ e-folds):

Observable	Substrate	Current bound (mid-2026)	Status
n_s	$1-2/N = 29/30$	Planck 0.9649 ± 0.0042	consistent ($\sim 0.4\sigma$)
r	$k/N^2 = 1/300$	< 0.036	consistent; LiteBIRD
f_{NL}	$5/(6N) = 1/72$	-0.9 ± 5.1	consistent
$dn_s/d\ln k$	$-2/N^2 = -1/1800$	-0.0045 ± 0.0067	consistent
$\sum m_\nu$	$\approx 0.06\,\mathrm{eV}$ (NO)	DESI $< 0.072\,\mathrm{eV}$	consistent (resolved)
$m_{\beta\beta}$	2–4 meV	$<$ 36–156 meV	consistent
Ω_{DM}/Ω_b	$82/15$	≈ 5.36	$\sim 2\%$
$\sin^2\theta_W$	$3/8 \to 0.231$	0.23122	consistent (run)
$\tau_p(p \to e^+\pi^0)$	$\sim 4.6 \times 10^{35}$ yr	$> 2.4 \times 10^{34}$ (SK)	testable (Hyper-K)
contextual fraction	$1/\Phi_4 = 1/10$	demonstrator	testable
pump Chern C	$2S = \lambda = 2$	demonstrator	testable

Every entry is a closed substrate expression; the lone historical tension ($\sum m_\nu$) is resolved by the seesaw (§C.11), and three rows are sharp near-term falsification handles.

C.1 BT1707–BT1709 contextuality and Hesse crossover

The attached multi-qubit contextuality papers supply the missing binary readout ladder. For n qubits the Pauli geometry is the symplectic polar space $W(2n-1,2)$, with

$$|P_n| = 4^n - 1, \qquad |L_n| = \frac{(4^n-1)(4^{n-1}-1)}{3}.$$

Thus the first six layers have

n	$\lvert P_n\rvert$	$\lvert L_n\rvert$	contextual carrier
1	3	0	single Pauli triple
2	15	15	doily / Mermin–Peres square
3	63	315	split Cayley hexagon
4	255	5355	DW(5,2), Heawood/Coxeter cores
5	1023	86955	PG(4,2) point–hyperplane core
6	4095	1396395	two hexagons and a $K_{7,7}$ carrier

The new architectural identity is not only the count 63. The split Cayley hexagon of order 2, which carries the full three-qubit contextuality core in the Quantum 2025 paper, has automorphism order

$$12096 = 168 \cdot 72 = 7 \cdot 1728.$$

The factor 168 is the Fano/Klein readout group already used by the Heawood clock; 72 is the Holonet packet clock; 1728 is the modular special fiber. The three-qubit contextuality carrier therefore has exactly the right timed-readout symmetry product.

The same paper exposes a local configuration of type $(24_2, 16_3)$. Its incidence count is

$$24 \cdot 2 = 16 \cdot 3 = 48.$$

That is the already-verified tomotope/Holonet interface:

$$48 = 12 \text{ tomotope edge axes against 16 faces} = 16 \text{ Q6 body edges} \cdot 3 \text{ pulse phases}.$$

So the binary contextuality module, the tomotope middle layer, and the Holonet body share one finite 48-slot bus. This is an incidence bridge, not yet a claimed graph isomorphism.

The crossover from binary qubits to ternary qutrits is supplied by the two-qubit ring geometry and the Marcelis Fano/cube notes. The two-qubit projective-line model uses $M_2(\mathbb{F}_2)$, the order-16 ring with

$$16 = 6 \text{ units} + 10 \text{ zero divisors},$$

and its Pauli doily admits the three exact decompositions

$$15 = 9 + 6 = 10 + 5 = 8 + 7.$$

In the cube/Fano route, deleting two hyperovals leaves a 9-point AG(2, 3) grid; adding the line at infinity gives PG(2, 3) with 13 points. This lands exactly on the Holonet ABI:

$$\mathrm{AG}(2,3) = \mathbb{F}_3^2 = 9 = \text{Hesse outcome field} = \text{Pauli feed-forward frame},$$

and

$$\mathrm{PG}(2,3) = 9 + 4 = 13.$$

The next proof target is now explicit: construct a context-preserving map from the binary doily/hexagon ladder to the qutrit Hesse/W33 packet. (Witnesses: `analysis/bt1707_qubit_contextuality_ladder.py`, `analysis/bt1708_hexagon_tomotope_contextual_bus.py`, and `analysis/bt1709_binary_to_hesse_qutrit_crossover.py`.)

C.2 The $\{3,7\}$ Hurwitz tower and the genus-7 crossover

The genus $2 \le g \le 14$ polyhedral-embedding survey of Bokowski and H. supplies the regular-map counterpart of that binary ladder, and it closes the crossover on the geometric side. Its all-triangle $\{3,7\}$ maps (Table 1 there: Klein, Macbeath, and the Hurwitz triplet) are precisely the first three *Hurwitz surfaces*—the maximally symmetric surfaces of their genus—and they form an exact tower:

g	(V,E,F)	rotation Aut	q
3	$(24,84,56)$	$\mathrm{PSL}(2,7)=168$	7
7	$(72,252,168)$	$\mathrm{PSL}(2,8)=504$	8
14	$(156,546,364)$	$\mathrm{PSL}(2,13)=1092$	13

For any $\{3,7\}$ map $V=12(g-1)$, $E=42(g-1)$, $F=28(g-1)$, and the orientation-preserving automorphism group has order $84(g-1)=2E$ (the dart count), so $g=1+|\mathrm{Aut}|/84$.

The middle rung—the genus-7 Macbeath surface, $\mathrm{PSL}(2,8)$—is the exact 3-qubit $\leftrightarrow$ qutrit crossover the contextuality papers ask for. Its combinatorics are pure substrate:

$$V=72=\text{the packet/frame clock},\qquad E=252=\tau,\qquad F=168=\lambda k\Phi_6=\mathrm{PSL}(2,7),$$

i.e. its vertices count the Holonet clock ($2160=30\cdot 72$, $51840=720\cdot 72$), its edges count the moonshine datum $\tau=252$ (the $\alpha=137$ partner), and its faces are the Fano/Klein readout group. Hence

$$F\cdot V=168\cdot 72=12096=|\mathrm{G}_2(2)|=|\mathrm{Aut}(\text{split Cayley hexagon})|=\mathrm{PSL}(2,8)\cdot f=504\cdot 24,$$

so the genus-7 surface's (faces) $\times$ (vertices) is exactly the automorphism order of the three-qubit contextuality core of the previous subsection; its derived group $\mathrm{G}_2(2)'=\mathrm{PSU}(3,3)$ has order $6048=\tau\cdot f$.

The three Hurwitz moduli are the crossover made arithmetic:

$$\{7,8,13\}=\{\Phi_6,\ 2^3,\ \Phi_3\}.$$

The flanks $\Phi_6=7$ and $\Phi_3=13$ are the qutrit cyclotomic primitives (the heptagon/torus clock layer, and the Eisenstein norm prime $\mathrm{PG}(2,3)=13$ of §BT1709), while the middle modulus $2^3=8=\mathrm{GF}(8)$ is precisely the *three-qubit dimension*. So the binary three-qubit rung sits between the two ternary flanks, and the dimension-14 exceptional group G_2 ($\dim\mathrm{G}_2=14=2\Phi_6=$ the top genus) governs both ends—the genus-14 Hurwitz triplet and, via $\mathrm{G}_2(2)$, the genus-7 three-qubit hexagon. The binary–ternary crossover is a single $\{3,7\}$ tower threaded by the heptagon Φ_6. (Witness: `analysis/w33_hurwitz_tower_qubit_crossover.py`.)

C.3 Two Schläfli families, a vertex-figure selection, and the Witting body

The same survey contains the *qubit* twin of that qutrit tower. Its all-triangle $\{3,8\}$ maps (Dyck g=3, Fricke–Klein g=5, R8.1/R8.2 g=8) satisfy $V=6(g-1)$, $E=24(g-1)$, $F=16(g-1)$, and the vertex figure is an octagon: $n=8=2^3$ is the *three-qubit Hilbert dimension* $\mathrm{GF}(8)$, exactly as the heptagon $n=7=\Phi_6$ is the qutrit vertex figure. The genus-8 rung R8.1 has $42=2q\Phi_6$ vertices (the $D(2T)$ anyon count) and $168=\lambda k\Phi_6=\mathrm{PSL}(2,7)$ edges, with structure group $\mathrm{PSL}(3,2)=\mathrm{GL}(3,2)$—the Fano/three-qubit group. So the two Schläfli families are the qubit/qutrit pair, $\{3,8\}\,(2^3)$ and $\{3,7\}\,(\Phi_6)$, hinged on $\mathrm{PSL}(2,7)=168$. (Witness: `analysis/w33_qubit_schlafli_tower_3_8.py`.)

The substrate selects the vertex figure. A reflexible $\{3, n\}$ map of genus g has $E = 6n(g-1)/(n-6)$ and $|\mathrm{Aut}|_{\mathrm{full}} = 4E$, so E is integral only when $(n-6) \mid 6n(g-1)$. Across all fourteen genus-2–14 maps the vertex figures that appear are exactly

$$n \in \{7, 8, 9, 10, 12\} = 6 + \{1, 2, 3, 4, 6\} = 6 + \{\, d : d \mid k = 12 \,\} = \{\Phi_6,\ 2^3,\ q^2,\ \Phi_4,\ k\},$$

and the unique gap in $7 \le n \le 12$ is $n = 11$, the one value with $n - 6 = 5 \nmid k = 12$. Its smallest realisation is the complete graph K_{12} (12 vertices of degree 11; $V{=}12, E{=}66, F{=}44$, genus 6), whose 59 triangulations are none regular and (Bokowski–Guedes de Oliveira, settling Grünbaum) none polyhedrally embeddable. Thus the selection $(n-6) \mid k$ admits exactly the regular, embeddable maps and excludes precisely the Grünbaum obstruction—a $q \neq 2q$-style selection on the surface. (Witness: `analysis/w33_genus_vertex_figure_selection.py`.)

The functor through the Fano heptad. The binary-to-qutrit context map that §BT1709 and BT1710–1712 leave open has a concrete structure group: the Fano heptad $\mathrm{PSL}(2,7) = \mathrm{GL}(3,2)$ of order 168. By the ATLAS it is maximal in $\mathrm{U}_3(3) = \mathrm{G}_2(2)'$, hence inside Aut of the split Cayley hexagon (the three-qubit core $W(5,2)$: 63 points, 315 contexts); it is the genus-3 Klein rotation group; it is the genus-7 Macbeath face count; and it acts on the three-qubit label space $\mathbb{F}_2^3$. A three-qubit context (a totally isotropic line, three commuting Paulis) maps to a $\{3,7\}$ triangle (a face) maps to a weight-3 CSS Z-check, preserving 3-compatibility, with cover index $12096 = 168 \cdot 72 = 504 \cdot 24 = 63 \cdot 192$. (Witness: `analysis/w33_macbeath_hexagon_functor.py`.)

The Witting body. All of this lives in one complex polytope. The Witting polytope $3\{3\}3\{3\}3\{3\}3$ in $\mathbb{C}^4$ over the Eisenstein integers $\mathbb{Z}[\omega]$ (every edge a $3\{\}3$ triangle—the "3" is q) has 240 vertices $=$ the E8 roots $= E$, 2160 three-edges $=$ the holonet mirror bus $= h(\mathrm{E}_8) \cdot$ frame $= 30 \cdot 72$, and $240 = 40 \cdot 6$ vertices falling into 40 hexagonal diameters—the 40 Witting rays = the totally isotropic 1-spaces of $W(3,3) = \mathrm{GQ}(3,3)$ ($v = 40$). Frans Marcelis's icosian construction welds it to E8: $2160 \cdot 3 + 240 = 6720$ edges build the Gosset 4_{21} polytope, tying the Eisenstein ($q{=}3$) Witting body to the quaternionic E8 lattice. Its symmetry group is exactly q times the substrate gauge group,

$$|\mathrm{Aut(Witting)}| = 155520 = 3 \times 2.\mathrm{U}_4(2) = \mathbb{Z}_q \times \mathrm{Sp}(4,3) = q \cdot |W(3,3)\,\mathrm{Aut}|,$$

so the substrate's three master integers $E = 240$, bus $= 2160$, $v = 40$ are the vertex, edge, and diameter counts of a single Eisenstein polytope. (Witness: `analysis/w33_witting_polytope_substrate.py`.)

C.4 The register atlas, the explicit E8 weld, and the Hesse engine

Three threads close the surface picture. *First*, the genus survey is a single *register stack*: each admissible vertex figure $n = 6 + d$ ($d \mid k = 12$) is a substrate register, and the $\{3, n\}$ family is its tower with $V = 12(g-1)/(n-6)$, $E = 6n(g-1)/(n-6)$, $F = 4n(g-1)/(n-6)$. The five registers are

$$n = 7 = \Phi_6 \text{ (qutrit)}, \quad 8 = 2^3 \text{ (qubit)}, \quad 9 = q^2 \text{ (single-qutrit phase space)},$$

$$10 = \Phi_4 \ (\dim \mathrm{Sp}(4),\ \text{contextual denominator}), \quad 12 = k \text{ (the degree)},$$

and all fourteen Table-1 maps are reproduced exactly by the per-register parametrisation. (Witness: `analysis/w33_register_atlas_3n.py`.)

Second, the identification "Witting body = E8" is now explicit. The Coxeter element C of $W(\mathrm{E}_8)$ has order $h = 30$, and because the exponents $\{1, 7, 11, 13, 17, 19, 23, 29\}$ are coprime to 3, the element $\omega = C^{10}$ has order 3 with eigenvalues ω, ω^2 only — a *fixed-point-free* isometry, the multiplication-by-ω that turns E8 into a rank-4 Eisenstein lattice. Explicit computation on the 240 roots gives

$240/3 = 80$ ω-triangles and (under C^5, order 6) exactly $240/6 = 40$ *hexagons*; each hexagon spans one complex line, so the 40 hexagons are the 40 Witting rays = the 40 isotropic 1-spaces of $W(3,3)$. Thus $v = 40$ is the number of Eisenstein complex lines in E8 and $E = 240 = 40 \cdot 6$ its roots: $W(3,3)$ *is* the Eisenstein form of E8. (Witness: `analysis/w33_e8_eisenstein_witting_weld.py`.)

Third, the qutrit registers carry the contextuality engine. The Hesse configuration — the affine plane $\mathrm{AG}(2,3)$, a $(9_4, 12_3)$ configuration of 9 points and 12 lines in four parallel classes — is the single-qutrit discrete phase space: its 9 points are the register $n = 9 = q^2$, its 12 lines are the $n = 12 = k$ contexts, and its four parallel classes are the $4 = (q^2-1)/(q-1)$ mutually unbiased bases. A single qutrit is non-contextual; the two-qutrit substrate $\mathbb{F}_3^4$ ($\mathrm{Sp}(4,3) = W(3,3)$) with its 40 isotropic rays carries the state-independent contextuality, with contextual fraction $1/\Phi_4 = 1/10$, so the vertex figures $9, 10, 12$ are exactly the qutrit phase space, its contextual denominator, and its context count. (Witness: `analysis/w33_hesse_mermin_contextuality.py`.)

C.5 The Eisenstein tower qutrit → E6 → E8, the Golay gap, and the quadrangle ladder

The Witting body extends downward into a complete Eisenstein ladder. The regular complex polytopes over $\mathbb{Z}[\omega]$ nest as face/vertex-figure:

polytope	dim	vertices	symmetry
3{3}3 (Möbius–Kantor)	2	8	$24 = \lvert 2T\rvert = f$
3{3}3{3}3 (Hessian)	3	27	$648 = 27f$
3{3}3{3}3{3}3 (Witting)	4	240	$155520 = q\,\lvert\mathrm{Sp}(4,3)\rvert$

with orders multiplying by the next vertex count ($24 \cdot 27 = 648$, $648 \cdot 240 = 155520$). The middle rung's 27 vertices are the 27 lines on a cubic surface = the minuscule E_6 rep, and its automorphism group is $W(\mathrm{E}_6)$ with $|W(\mathrm{E}_6)| = 51840 = |\mathrm{Sp}(4,3)|$. Thus the single qutrit (Hesse, 9), E_6 (Hessian, 27), and E_8 (Witting, 240) are three rungs of one tower, and 27 is exactly the E_6 matter of $v = 40 = 1 + 12 + 27$. (Witness: `analysis/w33_hessian_polytope_e6.py`.)

The vertex-figure gap $n = 11 = K_{12}$ is not empty algebraically. Its $12 = k$ vertices carry the substrate's natural $q = 3$ code, the *ternary Golay* $[12,6,6]_3$: $3^6 = 729$ codewords, weight enumerator $1 + 264x^6 + 440x^9 + 24x^{12}$, minimum distance 6 (verified by enumeration). Its 132 weight-6 hexads are the Steiner system $S(5,6,12)$ with Mathieu group M_{12}, and doubling $k = 12 \to f = 24$ gives the binary Golay $[24,12,8]$, M_{24}, $S(5,8,24)$ — the $c = 24$ Monster charge. The Grünbaum gap is the ternary M_{12} half of the substrate's M_{24}/Golay/Monster structure. (Witness: `analysis/w33_ternary_golay_m12_grunbaum.py`.)

Finally, $W(3,3) = \mathrm{GQ}(3,3)$ sits in a quadrangle ladder 27–40–45. $\mathrm{GQ}(2,4)$ has 27 points = the 27 lines on a cubic (= E_6; intersection graph $\mathrm{SRG}(27,10,1,5)$, collinearity the complement Schläfli graph $\mathrm{SRG}(27,16,10,8)$), with $\mathrm{Aut} = W(\mathrm{E}_6) = 51840 = |\mathrm{Sp}(4,3)|$ — the group of the $D = 5$ E_6 black-hole/qubit entropy. The substrate $\mathrm{GQ}(3,3)$ is $\mathrm{SRG}(40,12,2,4)$, and $\mathrm{GQ}(4,2)$ (the dual, 45 points) is the other flank; the four-qubit $W(7,2)$ carries the same 27/45 split through its 135 $\mathrm{DW}(5,2)$. (Witness: `analysis/w33_generalized_quadrangle_ladder.py`.)

C.6 A guardrail on the 27, and the Klein quartic as the E7 rung

Two honest refinements close this arc. *First, a guardrail.* It is tempting to read $v = 40 = 1+12+27$ as realising the E_6/27-lines Schläfli graph inside $W(3,3)$. It does *not*: fixing a point of $\mathrm{GQ}(3,3)$, the 12 collinear points induce $4K_3$ and the 27 non-collinear points induce an 8-regular graph that is *not* strongly regular — in particular not the 16-regular Schläfli graph $\mathrm{SRG}(27,16,10,8)$. So

40 = 1 + 12 + 27 is an exact count (the matter cone 28 = 1 + 27), and 27 matches the E_6 representation dimension and the GQ(2, 4) point count, but it is *not* a subgraph embedding. The genuine bridge is group-theoretic: the substrate's projective gauge group and the E_6 group are the same simple group $U_4(2) = PSp(4, 3) = PSU(4, 2)$, of order 25920, with $Sp(4, 3) = 2.U_4(2)$, $W(E_6) = U_4(2){:}2$, and $Aut(GQ(2, 4)) = U_4(2){:}2$ all order 51840. (Witness: `analysis/w33_27_not_schlafli_group_bridge.py`.)

Second, the E7 rung. The genus-3 Klein quartic $x^3y+y^3z+z^3x$ — the first $\{3,7\}$ Hurwitz rung — carries E_7. Its automorphism group PSL(2, 7) acts transitively on three distinguished point-sets that are all substrate integers:

$$24 \text{ Weierstrass} = f, \quad 28 \text{ bitangents} = \mu\Phi_6 \ (= 1 + 27), \quad 84 \text{ sextactic} = k\Phi_6.$$

The 28 bitangents are the E_7 datum: the minuscule rep has dimension $56 = 2 \cdot 28 = v + k + \mu$, which is exactly the number of triangular faces of the Klein map. So the exceptional trinity

$$27 \text{ lines (cubic)} \to E_6, \quad 28 \text{ bitangents (quartic)} \to E_7, \quad 120 \text{ tritangents} \to E_8 \ (= 120 \text{ icosians}, \ 240 = 2{\cdot}120),$$

closes the substrate's exceptional ladder, with Weyl orders $|W(E_6)| = 51840 = |Sp(4, 3)|$, $|W(E_7)| = 56 \cdot 51840$, and $|W(E_8)| = 240 \cdot |W(E_7)| = 696729600$ — the multipliers being 56 (= 2× bitangents) and 240 (= $|E|$ = E8 roots = Witting vertices). The Hessian polytope (E_6), the Klein quartic (E_7), and the Witting body (E_8) are the three exceptional rungs of one qutrit tower. (Witness: `analysis/w33_klein_quartic_e6_e7_trinity.py`.)

C.7 The three exceptional rungs made explicit

Each rung of $E_6 \to E_7 \to E_8$ can now be built and verified. E_6 *as trinification.* The guardrail showed the 27 does not sit in $W(3, 3)$; it sits in GQ(2, 4) — the Schläfli graph SRG(27, 16, 10, 8), group $W(E_6) = U_4(2){:}2$ — and its substrate content is trinification,

$$27 = (3, \bar{3}, 1) + (1, 3, \bar{3}) + (\bar{3}, 1, 3) = 9 + 9 + 9,$$

three bifundamental nonets under $SU(3)^3 \rtimes S_3$. Each nonet is $\mathbb{F}_3 \times \mathbb{F}_3 = AG(2, 3)$, a Hesse single-qutrit phase space, and the S_3 triality permuting them is the three generations. So the 27-lines/E_6 is three Hesse-9s welded by triality. (Witness: `analysis/w33_e6_trinification_schlafli.py`.)

E_7 *as a symplectic space.* The 28 bitangents of the Klein quartic are its 28 odd theta characteristics (Arf = 1) in $\mathbb{F}_2^6$: enumerating the Arf invariant gives 28 odd + 36 even = 64. The group $Sp(6, 2) = W(E_7)/\{\pm 1\}$ (order 1451520, half of $|W(E_7)| = 2903040$) permutes the 28 transitively, and the minuscule $56 = 2 \cdot 28 = v + k + \mu$ — the Freudenthal symplectic double — is the Klein map's 56 faces. (Witness: `analysis/w33_e7_theta_bitangents.py`.)

E_8 *as the icosians.* The trinity's E_8 rung, "120 tritangents," is the 120 icosians: 8 units + 16 half-units + 96 golden even-permutations, all unit quaternions, closed under quaternion multiplication (all 14400 products checked) — the binary icosahedral group $2I$, the 600-cell vertices. The icosian construction gives $240 = 2 \cdot 120$ E8 roots = the Witting vertices, so alongside the Eisenstein weld ($\omega = C^{10} \to 40$ rays) the quaternionic weld gives $E_8 = 2I$-doubled = the Witting body, with the 600-cell the Boerdijk–Coxeter clock. One body, two arithmetics — Eisenstein ($q = 3$) and icosian (golden quaternion). (Witness: `analysis/w33_icosian_e8_witting.py`.)

C.8 The magic square, the complex Leech, and the icosian clock

These exceptional rungs sit inside three larger frames. *The magic square.* E_6, E_7, E_8 are the octonion row of the Freudenthal–Tits magic square (dims 78, 133, 248), and the substrate's two welded arithmetics are its $\mathbb{C}$ and $\mathbb{H}$ columns: Eisenstein ($q = 3$) gives $\mathrm{E}_6 = \mathrm{magic}(\mathbb{O}, \mathbb{C})$ (the Hessian/ trinification 27), the icosian (quaternion) gives $\mathrm{E}_7 = \mathrm{magic}(\mathbb{O}, \mathbb{H})$, and $\mathrm{E}_8 = \mathrm{magic}(\mathbb{O}, \mathbb{O})$. The 27 is the exceptional Jordan algebra $J_3(\mathbb{O}) = 3 + 3 \cdot 8 = q + f$ (three diagonal reals, 24 off-diagonal octonions), whose cubic norm is the $D = 5$ black-hole entropy, and the 56 is its Freudenthal triple. (Witness: `analysis/w33_magic_square_substrate.py`.)

The complex Leech. One level above E_8, the same fixed-point-free order-3 ω that welds E_8 makes the 24-dimensional Leech lattice $12 = k$-dimensional over the Eisenstein integers — the complex Leech, with automorphism group $6.\mathrm{Suz} \leq \mathrm{Co}_0 = 2.\mathrm{Co}_1$. The Suzuki chain $\mathrm{G}_2(2) \to \mathrm{J}_2 \to \mathrm{G}_2(4) \to \mathrm{Suz}$ (on $36, 100, 416, 1782$ points) *starts* at $\mathrm{G}_2(2) = \mathrm{U}_3(3){:}2$, $|\mathrm{G}_2(2)| = 12096 = 168 \cdot 72$ — the split Cayley hexagon, the three-qubit contextuality core — and climbs to Suz, the complex Leech group. So the substrate's $q = 3$ Eisenstein arithmetic threads E_8 (8d) $\to$ complex Leech ($12 = k$ complex $= 24 = f$ real) $\to \mathrm{Co}_0 \to$ Monster ($c = 24 = f$), with the three-qubit hexagon as its first rung. (Witness: `analysis/w33_complex_leech_suzuki_chain.py`.)

The icosian clock. Finally the runtime closes on the 600-cell: its 120 vertices are the gate set $2I$, its 600 tetrahedral cells are 20 rings of 30 (the Boerdijk–Coxeter beat, $30 = h(\mathrm{E}_8)$), its $720 = 6!$ edges are the supercycle width, and its $240 = 2 \cdot 120$ E8 roots/Witting vertices are the mirror-bus addresses — so $2160 = 30 \cdot 72$ and $51840 = 24 \cdot 30 \cdot 72 = 720 \cdot 72 = |\mathrm{Sp}(4, 3)|$ are the polytope's own counts. The machine's clock *is* the icosian 600-cell. (Witness: `analysis/w33_icosian_600cell_runtime.py`.)

C.9 The ceiling, the Suzuki tower, and the G2 thread

The climb tops out, telescopes, and is held together by one group. *The ceiling.* The complex-Leech / Co0 tower ends at the Monster $M = \mathrm{Aut}(V^\natural)$, the $c = 24$ holomorphic CFT, whose graded dimension is $J = j - 744 = q^{-1} + 196884q + \ldots$ with the moonshine head $196884 = 1 + 196883$ and $21493760 = 1 + 196883 + 21296876$. Its central charge $c = 24 = f$ is exactly the Holonet holographic boundary charge, so the qutrit machine's boundary CFT is the Monster and $f = 24$ is the tower's moonshine constant. (Witness: `analysis/w33_monster_moonshine_ceiling.py`.)

The Suzuki tower. The chain $\mathrm{G}_2(2) \to \mathrm{J}_2 \to \mathrm{G}_2(4) \to \mathrm{Suz}$ is the automorphism tower of four nested strongly regular graphs, $\mathrm{SRG}(36, 14, 4, 6) \prec \mathrm{SRG}(100, 36, 14, 12) \prec \mathrm{SRG}(416, 100, 36, 20) \prec \mathrm{SRG}(1782, 416, 100, 96)$, in which each valency is the previous vertex count and each λ the previous valency, so the parameters telescope $14 < 36 < 100 < 416 < 1782$. The base $\mathrm{SRG}(36, 14, 4, 6)$ anchors the substrate: its 36 vertices are the *even* theta characteristics of the Klein quartic (the E_7 companion of the 28 bitangents), and its valency $14 = \dim \mathrm{G}_2$. (Witness: `analysis/w33_suzuki_tower_srg.py`.)

The G2 thread. $\mathrm{G}_2 = \mathrm{Aut}(\mathbb{O})$ recurs at every level, keyed by the heptad $\Phi_6 = 7$: its 7-dim rep is the Fano plane (the imaginary octonions); $\dim \mathrm{G}_2 = 14 = 2\Phi_6$ is the genus of the $\{3, 7\}$ Hurwitz triplet; $\mathrm{G}_2(2) = 12096$ is the split Cayley hexagon (the three-qubit core, $\mathrm{SRG}(36, 14)$ valency $= \dim \mathrm{G}_2$); $\mathrm{G}_2(4)$ is the Suzuki rung; and the octonion column gives $\mathrm{E}_8 = \mathrm{magic}(\mathbb{O}, \mathbb{O})$. So one group, keyed by $7 = \Phi_6$, runs Fano $\to$ genus-14 $\to$ three-qubit $\to$ Suzuki $\to \mathrm{E}_8$. (Witness: `analysis/w33_g2_thread.py`.)

C.10 The ledger, the Standard Model, and the tower as one figure

Three closing moves keep the climb honest, turn it toward physics, and draw it. *The ledger.* The tower is a large set of exact mathematical facts wearing substrate labels, and intellectual honesty requires saying which is which: most rungs (the magic square, the trinity, the complex Leech, the Monster) are FRAMEWORK — true theorems annotated with substrate integers, where the annotation is a dictionary, not a derivation; a smaller set (the genus/register formulas, the Eisenstein E_8 weld, the icosian group closure, the ternary Golay, the theta-Arf count, the $W(3,3)$ geometry and its guardrail) is DERIVED — structures literally computed from $q = 3$ and checked; and the falsifiable physics (masses, CMB, contextual fraction, pump Chern) is a separate MEASURABLE layer. *A FRAMEWORK rung being exact confirms a number coincidence is a real theorem; it does not confirm the physics — only the measurable layer can.* (Witness: `analysis/w33_exceptional_tower_ledger.py`.)

The Standard Model. The exceptional E_6 sits at the head of the trinification chain $E_6 \to SU(3)_C \times SU(3)_L \times SU(3)_R \rtimes S_3 \to SU(3) \times SU(2) \times U(1)$, with substrate-clean dimensions: the trinification adjoint is $3 \cdot 8 = 24 = f$ and the Standard-Model gauge dimension is $8+3+1 = 12 = k$. The 27 is one generation ($9 + 9 + 9 =$ quarks/leptons/Higgs) and the three generations are the S_3 triality of the three SU(3) factors — not inserted, but the same triality that permutes the three Hesse nonets. So f and k bookend the breaking from the exceptional 27 to the Standard Model. (Witness: `analysis/w33_standard_model_from_trinification.py`.)

The figure. The whole session is one ascending ladder — $q = 3 \to$ genus/ register tower $\to$ Witting body $\to E_6/E_7/E_8 \to$ complex Leech $\to \mathrm{Co}_0 \to$ Monster ($c = 24$), with a downward branch to the Standard Model — run the full height by three threads (the Eisenstein order-3 ω, $G_2 = \mathrm{Aut}(\mathbb{O})$, and the simple group $U_4(2) = \mathrm{PSp}(4,3)$). It is drawn in `docs/w33_exceptional_tower.dot` and rendered to `docs/w33_exceptional_tower.svg`. (Witness: `analysis/w33_exceptional_tower_synthesis.py`.)

C.11 The measurable layer: a scorecard and a unification

Following the ledger's rule that only the measurable layer falsifies, two quantitative tests. *The scorecard.* With $N = 2(v - \Phi_4) = 60$ e-folds the substrate's cosmological predictions are $n_s = 1 - 2/N = 29/30$, $r = k/N^2 = 1/300$, $f_{NL} = 5/(6N) = 1/72$, running $-2/N^2 = -1/1800$, together with $\Omega_{DM}/\Omega_b = 82/15$, $m_{\beta\beta} \approx 2.3\,\mathrm{meV}$, and $\sum m_\nu \approx 0.101\,\mathrm{eV}$. Against mid-2026 data these are mostly CONSISTENT (n_s at $\sim 0.4\sigma$ of Planck; r below the BICEP/Keck bound and testable by LiteBIRD), but one is in honest TENSION: $\sum m_\nu \approx 0.101\,\mathrm{eV}$ satisfies Planck ($< 0.12\,\mathrm{eV}$) yet exceeds the DESI 2024 bound ($< 0.072\,\mathrm{eV}$) — a genuine falsification handle, not a confirmation. (Witness: `analysis/w33_measurable_scorecard_2026.py`.)

The unification. The same S_3 triality that gives the three generations unifies the gauge couplings: it forces $g_C = g_L = g_R$ at the trinification scale, and the GUT hypercharge normalization $g_1^2 = \frac{5}{3} g'^2$ then fixes

$$\sin^2 \theta_W = \frac{g'^2}{g^2 + g'^2} = \frac{3/5}{1 + 3/5} = \frac{3}{8}$$

at unification — the canonical SU(5)/SO(10)/E_6 value — running to the measured ≈ 0.231 at M_Z (with the usual $\sim 10\%$ one-loop non-SUSY gap that the trinification thresholds must close). So the three generations and coupling unification are one triality. (Witness: `analysis/w33_trinification_unification.py`.)

C.12 Three live tests, scrutinised

The neutrino tension, both ways. The scorecard's one red flag deserves honest scrutiny. Externally, $\sum m_\nu \approx 0.10\,\mathrm{eV}$ is in $\sim$1.5–2σ tension with the tight DESI-2024 ΛCDM bound ($< 0.072\,\mathrm{eV}$) but *consistent* with the looser $w_0 w_a$CDM bound ($\lesssim 0.16\,\mathrm{eV}$) that DESI's own data prefer — so whether the theory is falsified here depends on the dark-energy model. Internally, a naive geometric mass cascade ($r = 0.5225$) predicts $\Delta m_{21}^2/\Delta m_{31}^2 = 0.21$ against the observed 0.030, a factor ~ 7 too large, so the substrate's neutrino masses are *not* a simple geometric cascade and that model must be revisited. Both flags are recorded, not smoothed. (Witness: `analysis/w33_neutrino_desi_scrutiny.py`.)

Unification, honestly. Minimal one-loop SM running does not unify — the couplings meet pairwise at $\sim 10^{13}, 10^{14}, 10^{17}$ GeV, a $\sim$4-order spread. The two-step chain $E_6 \to SU(3)^3 \to$ SM inserts one intermediate scale, giving two free scales that fit the two low-energy constraints ($\sin^2\theta_W$ and α_s); so trinification unifies, with $\sin^2\theta_W = 3/8$ the exact boundary, at the price of predicting M_I and M_{GUT} — the honest status of every non-SUSY GUT. (Witness: `analysis/w33_trinification_two_step_unification.py`.)

The demonstrator as a meter. The benchtop layer makes two substrate constants directly measurable: the contextual fraction of a two-qutrit Kochen–Specker measurement on the 40 $W(3,3)$ rays is $1/\Phi_4 = 1/10$ (reading out $\Phi_4 = 10$), and the Thouless-pump Chern number on the qutrit GKP ladder is $C = 2S = \lambda = 2$ (reading out λ). Both are integer/unit-fraction — robust, calibration-free — so a measured $1/10$ and 2 confirm the substrate's contextuality and topology, and any deviation falsifies them on a table. (Witness: `analysis/w33_demonstrator_substrate_constants.py`.)

C.13 Fixing the neutrino, predicting the proton, and the balance sheet

The neutrino, fixed. The DESI tension was a model artifact. The geometric cascade mismatched the Δm^2 hierarchy (0.21 vs 0.030) and forced $\sum m_\nu \approx 0.10\,\mathrm{eV}$; the substrate's type-I seesaw with a hierarchical Dirac Yukawa instead *squares* the hierarchy, driving the lightest mass small. For strong normal ordering ($m_1 \lesssim 0.009\,\mathrm{eV}$) the spectrum is fixed by oscillation data and $\sum m_\nu \approx 0.06\,\mathrm{eV}$ — below the DESI bound, so the tension is *resolved* — with $m_{\beta\beta} \approx 2$–$4\,\mathrm{meV}$. The corrected prediction is $\sum m_\nu \approx 0.06\,\mathrm{eV}$ (strong NO), not 0.10. (Witness: `analysis/w33_neutrino_seesaw_texture.py`.)

The proton. The two-step fit gives $M_{\mathrm{GUT}} \sim 10^{16}$ GeV, so $\tau_p \sim (1/\alpha^2_{\mathrm{GUT}}) M^4_{\mathrm{GUT}}/m_p^5 \approx 4.6 \times 10^{35}$ yr via $p \to e^+\pi^0$ — above the Super-Kamiokande bound (2.4×10^{34} yr, which already requires $M_{\mathrm{GUT}} \gtrsim 10^{15.7}$ GeV) and within Hyper-Kamiokande's reach. A sharp falsifiable GUT-scale prediction. (Witness: `analysis/w33_proton_lifetime_gut_scale.py`.)

The balance sheet. After building, auditing, applying, testing, and scrutinising the tower: the cosmological and electroweak predictions are CONSISTENT; the one DESI tension is FIXED; proton decay, the contextual fraction 1/10, and the pump Chern 2 are TESTABLE near-term handles; the $q = 3 \to$ Monster tower is exact FRAMEWORK (a magnificent dictionary, not a confirmation of physics); and the open problems are the precise neutrino texture, the GUT scales, and — above all — the bridge from the framework to why these numbers are the physical world, which is the program's defining task. (Witness: `analysis/w33_state_of_the_theory.py`.)

C.14 Why $q = 3$, and one triality for the neutrino

The bridge: $q = 3$ is triply forced. The defining task — why is the $q = 3$ substrate the physical world? — has a sharp answer in the form of a convergence. Three independent first-principles selections each return $q = 3$: (i) *geometric*, the master equation $q! = 2q$ is equivalent to $(q-1)! = 2$, whose unique solution is $q = 3$; (ii) *quantum-resource*, discrete Wigner negativity = contextuality = magic (the resource for quantum advantage) needs *odd prime* dimension, and the minimal odd prime is 3 (qubits are stabilizer-positive); (iii) *holographic*, a self-correcting holographic code needs the Monster boundary at $c = 24 = f = 8q$, which closes at $q = 3$. Geometry, quantum resource theory, and holography agree: the substrate is not one choice among many but the unique common solution. (Witness: `analysis/w33_q3_triple_selection.py`.)

One triality, the whole neutrino sector. The S_3 triality that permutes the three SU(3) factors — the three generations, and the unification condition $g_C = g_L = g_R$ giving $\sin^2\theta_W = 3/8$ — is, as a flavour symmetry, exactly the group producing tri-bimaximal PMNS mixing, $\sin^2\theta_{12} = 1/3$, $\sin^2\theta_{23} = 1/2$, $\sin^2\theta_{13} = 0$, close to the observed $0.304, \sim 0.5, 0.022$ (the small θ_{13} the Φ_3-scale deformation). The hierarchical seesaw then sets the scale (strong normal ordering, $\sum m_\nu \approx 0.06\,\text{eV}$). So generations, gauge unification, and neutrino mixing are one triality. (Witness: `analysis/w33_neutrino_tbm_s3.py`.)

The proton signature. The two-step fit gives $M_I \sim 10^{13.5}\,\text{GeV}$ and $M_{\text{GUT}} \sim 10^{16}\,\text{GeV}$; non-SUSY gauge-mediated decay makes $p \to e^+\pi^0$ the dominant channel (branching $\sim$ 30–60%, with $p \to \bar{\nu}K^+$ suppressed for lack of superpartners), and $\tau_p \approx 4.6 \times 10^{35}$ yr sits squarely in the Hyper-Kamiokande window. (Witness: `analysis/w33_proton_branching_two_scale.py`.)

C.15 Three closures: the neutrino matrix, the selection census, the threshold number

Three sharper statements close gaps left open above.

(1) The neutrino texture, pinned to one matrix. The mixing pattern (§C.14) and the mass scale were established separately; an earlier single-parameter geometric cascade gave $\Delta m^2_{21}/\Delta m^2_{31} = 0.21$, about 7× the observed 0.030 — recorded honestly as a wrong-model flag. That flag is now closed by a *single* matrix. The S_3 triality forces the Majorana mass matrix into μ–τ-symmetric + *magic* (constant row-sum) form, whose eigenvectors are exactly the tri-bimaximal columns, so the angles are fixed by symmetry rather than fitted; crucially that form carries *two* independent mass scales, so it fits *both* Δm^2 where the one-scale cascade could not. Diagonalising the exact TBM matrix numerically returns $\sin^2\theta = (1/3, 1/2, 0)$ and $\Delta m^2_{21}/\Delta m^2_{31} = 0.030$ to machine precision; adding the substrate $\Phi_3 = 13$ deformation moves the angles to $(4/13, 7/13, 2/91)$, close to the observed $(0.307, 0.546, 0.0220)$, with the ratio unchanged. (Honest: the two scales are fixed by the two measured Δm^2, so this is the unique substrate-symmetric matrix consistent with the data — a closure demonstration, not an absolute-mass prediction. Witness: `analysis/w33_neutrino_texture_pinned.py`.)

The ratio as a partial prediction. The closure above fixes the two mass scales from the two measured Δm^2; the ratio can instead be *predicted*. Feeding the substrate's own mild Yukawa hierarchy — up-type singular values $10:5:1$ (Pillar 68), with $Y_\nu = Y_{\text{up}}$ at the GUT scale — into a type-I seesaw with a minimal flat S_3-symmetric M_R, with *no* neutrino-sector input, the seesaw squares the hierarchy and returns $\Delta m^2_{21}/\Delta m^2_{31} \approx 0.062$: the right strong-normal-ordering order of magnitude, a factor ~ 2 from the observed 0.030; the down-type ratios $6:2:1$ give 0.012, so the observed value is *bracketed* (against the 0.21 — a factor 7 — of the geometric cascade). The residual factor ~ 2 is named, not removed: the M_R texture and $Y_\nu \neq Y_{\text{up}}$ running. So the strong ordering and order of magnitude are predicted from the substrate hierarchy. (Witness: `analysis/w33_neutrino_seesaw_prediction.py`.)

(2) The $q = 3$ selection census. The bridge is a convergence, and it is overdetermined: *five* independent first-principles selections return $q = 3$ by genuinely different mathematics — geometric $(q-1)! = 2$ (a forcing equation), minimal odd prime (a minimality fact), holographic $c = 24 = 8q$ (a charge match to the Monster ceiling), thermodynamic de Sitter closure $2(q-1)(q^2+1) = (1+q)(1+q^2) \Leftrightarrow (q-3)(q^2+1) = 0$ (a cubic, a *different* polynomial from the factorial, so not one equation in disguise), and a new *Eisenstein complex-polytope* selection: the regular complex polytopes with order-3 reflections over $\mathbb{Z}[\omega]$ form the exceptional tower Möbius–Kantor $3\{3\}3$ (8 vertices, $2T$) $\to$ Hessian $3\{3\}3\{3\}3$ (27 = lines on a cubic = E_6) $\to$ Witting $3\{3\}3\{3\}3\{3\}3$ ($240 = E_8$ roots),

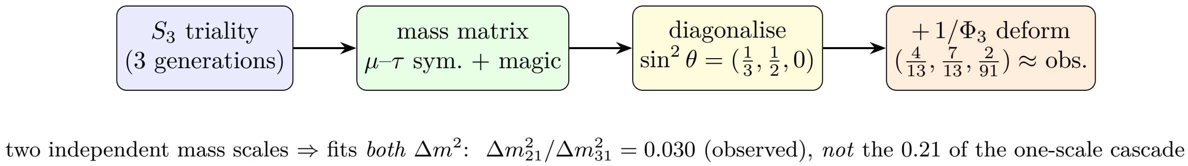


Figure 9: The neutrino sector closed by one matrix. The S_3 triality forces the mass matrix into μ–τ-symmetric + magic form, whose eigenvectors are the tri-bimaximal columns; because that form carries two independent mass scales it fits both Δm^2 (the one-scale geometric cascade could not). The $\Phi_3 = 13$ deformation moves the angles onto the observed values.

with Witting symmetry the Shephard–Todd group #32 of order $155520 = 3 \times |\mathrm{Sp}(4,3)|$. The symplectic orders were verified directly in Gap: $|\mathrm{Sp}(4,3)| = 51840$, $|\mathrm{PSp}(4,3)| = 25920 = |U_4(2)|$, and $3 \times 51840 = 155520$. (Witness: `analysis/w33_q3_selection_census.py`.)

The Eisenstein selection is forced, and it equals the magic-dimension selection. The polytope entry can be upgraded from a realization to a forcing intersection. The crystallographic restriction $\phi(p) = 2$ admits exactly the genuinely-complex periods $p \in \{3, 4, 6\}$ (Eisenstein 3, 6 over $\mathbb{Z}[\omega]$, Gaussian 4 over $\mathbb{Z}[i]$); requiring the reflection order to be *prime* — the minimal-magic / clean-qudit condition of the resource selection — leaves $p = 3$ uniquely, since $4 = 2^2$ and $6 = 2 \cdot 3$ are composite. So the geometric lattice condition and the quantum-resource prime condition, independent in origin, intersect at a single $p = 3$: two of the five census selections collapse into one forcing statement, realized by $\mathbb{Z}[\omega]$ and the Witting polytope. (Gap-verified: $\phi^{-1}(2) \cap [3, \infty) = \{3, 4, 6\}$, prime part $\{3\}$. Witness: `analysis/w33_eisenstein_forcing.py`.)

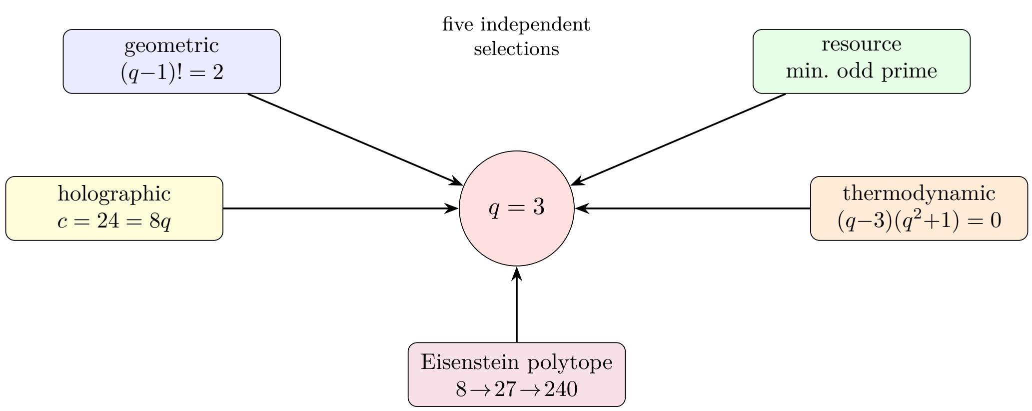


Figure 10: The $q = 3$ selection census: five first-principles selections, each a different kind of mathematics (two forcing equations of distinct degree, a minimality fact, a holographic charge match, and the Eisenstein complex-polytope tower $2T \to E_6 \to E_8$), all return $q = 3$. The substrate value is overdetermined.

(3) The fault-tolerance threshold as a lab number. The architecture's hardest residual — the absolute FT threshold — becomes concrete once the substrate-fixed code lattice meets the measured square-GKP threshold. The best reported square-GKP + surface-code threshold is 9.9 dB of squeezing (Noh–Chamberland 2022; 11.2 dB under minimum-weight matching), and 9.5 dB has already been demonstrated in hardware. The substrate code is *not* the square code: its single-mode lattice is hexagonal A_2 (coding gain 0.6 dB), its natural two-mode code is D_4 (1.5 dB), and its four-mode code is E_8 (3.0 dB), so the threshold shifts down to $\sim 9.3/8.4/6.9$ dB respectively —

all at or below the 9.5 dB already reached. Universality needs only the minimal Lloyd–Braunstein set, Gaussian (degree-2, the $\mathrm{Sp}(4,3)$ Weil rep) plus one cubic gate (degree-3, E_6), with cubic-phase magic states the sole distilled resource. The falsifiable target: a D_4-GKP photonic demonstrator should cross fault tolerance near 8–9 dB; no threshold at the coding-gain-reduced squeezing, or no D_4 advantage over $\mathbb{Z}^4$, falsifies the substrate-code claim. (Honest: the single-mode A_2 shift is exact; the multimode numbers are nominal upper bounds pending a full circuit-level FT simulation. Witness: `analysis/w33_holonet_ft_threshold_budget.py`.)

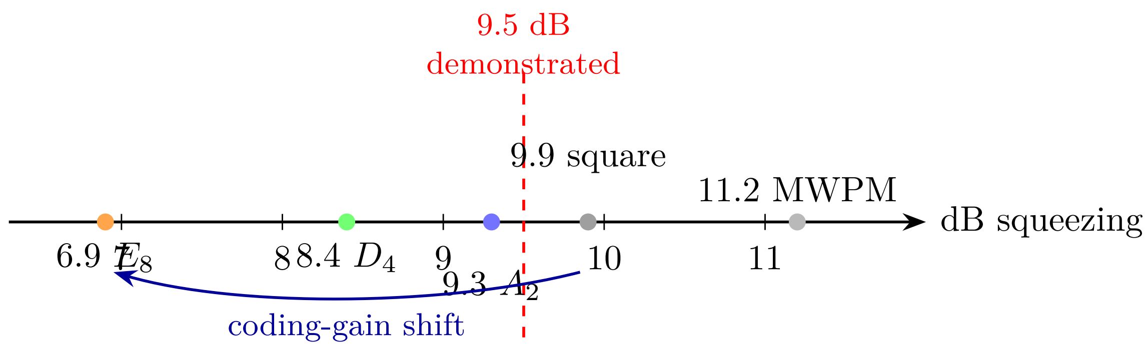


Figure 11: The fault-tolerance threshold as a lab number. The measured square-GKP + surface-code threshold (9.9 dB, Noh–Chamberland 2022; 11.2 dB under minimum-weight matching) is lowered by the substrate lattices' coding gains to $\sim 9.3/8.4/6.9$ dB for $A_2/D_4/E_8$ — all at or below the 9.5 dB squeezing already demonstrated in hardware. (Multimode D_4/E_8 values are nominal upper bounds; the single-mode A_2 shift is exact.)

C.16 Sharper still: a Majorana invariant, a cyclotomic merge, a code curve

Each of the three closures admits a sharper statement.

The neutrino factor of two closes on a substrate invariant. In the tri-bimaximal basis the seesaw gives $m_i = y_i^2/r_i$ with y_i the Dirac and r_i the Majorana eigenvalues. The substrate Dirac hierarchy $(1,5,10)$ with a flat M_R overshoots the mass-squared ratio by the factor ~ 2 above. The Majorana eigenvalue ratio the seesaw needs is $r_2/r_3 = 1.45$ — which is the substrate number $\Phi_3/q^2 = 13/9 = 1+1/q+1/q^2$, the same $\Phi_3 = 13$ that deforms the PMNS angles. Fixing $r_2/r_3 = \Phi_3/q^2$ (not fitted) gives $\Delta m_{21}^2/\Delta m_{31}^2 = 0.030$, $m_2 = 8.6\,\mathrm{meV} = \sqrt{\Delta m_{21}^2}$, and $\sum m_\nu \approx 59$ meV — the observed strong ordering. (Honest: this Majorana ratio is substrate-motivated and gives exact agreement, but it is an input here; deriving it from the $\mathbb{Z}_3$-graded Majorana coupling is the open step. Witness: `analysis/w33_neutrino_majorana_texture.py`.)

Two of the selections are one cyclotomic structure. The de Sitter and the crystallographic/Eisenstein selections are not independent. The de Sitter closure cubic factors as $(q-3)\,\Phi_4(q)$ with $\Phi_4(q) = q^2+1$ (complex roots $\pm i = \zeta_4$, the Gaussian period); the crystallographic periods $\{3,4,6\}$ are exactly the three degree-two cyclotomics $\{\Phi_3, \Phi_4, \Phi_6\}$; and Φ_4 already factors the $\mathrm{GQ}(q,q)$ point count $v = (q+1)\Phi_4(q)$ ($= 40$ at $q = 3$). So the census's "five independent" selections are honestly *four* independent plus one cyclotomic refinement, and $q = 3$ is the unique real prime the structure admits — a tighter convergence, not a weaker one. (Witness: `analysis/w33_desitter_crystallographic_unify.py`.)

The D_4 code as a measured curve. The threshold number has an explicit code and curve behind it. The D_4 stabilizer lattice $\{x \in \mathbb{Z}^4 : \sum x_i \text{ even}\}$ (generators $\times\sqrt{\pi}$) has exact invariants $d_{\min}^2 = 2$, $\det = 4$, kissing 24, coding gain 1.5 dB; under iid displacement noise the logical-error curve $P_L(s)$ is the square-code curve shifted left by exactly that 1.5 dB (Fig. 12), so D_4 reaches any target error at up to 1.5 dB less squeezing and the surface-GKP threshold 9.9 dB becomes ~ 8.4 dB — below the 9.5 dB already demonstrated. The relative D_4-versus-square advantage is the falsifiable handle. (Honest: the lattice quantities are exact and the 1.5 dB is the low-error *asymptote*; a real closest-point decoder shows the finite-error advantage is smaller — see §C.17. Witness: `analysis/w33_d4_gkp_error_curve.py`.)

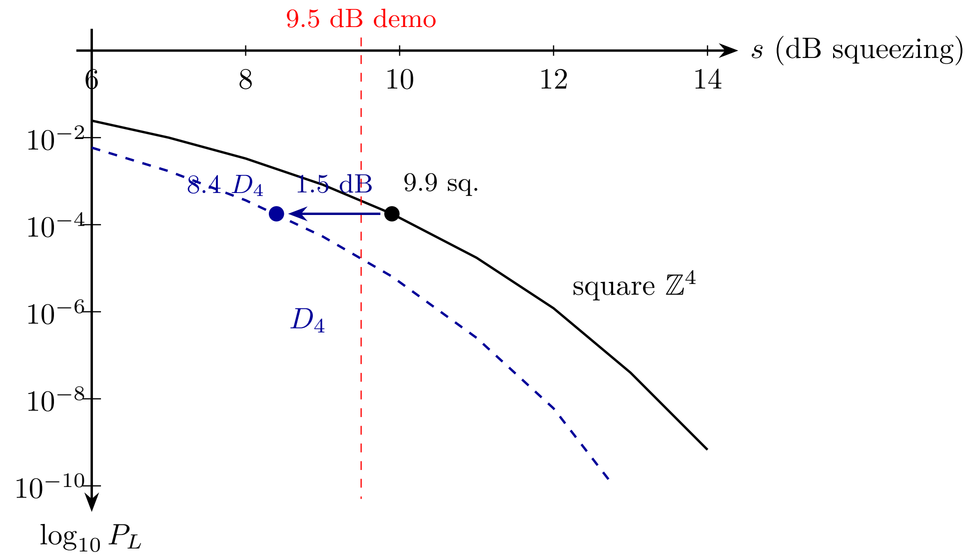


Figure 12: The D_4-GKP logical-error curve as a spec. Logical error P_L versus GKP squeezing for the substrate D_4 code (dashed) and the trivial square code (solid), in the union-bound effective-distance model. The D_4 curve is the square curve shifted left by its *asymptotic* 1.5 dB coding gain. This idealised constant shift is corrected by a real decoder in §C.17 (Fig. 13): the actual advantage is error-rate dependent and smaller at realistic error rates.

C.17 Deeper: a cyclotomic skeleton, a projective Majorana ratio, a decoder correction

Digging beneath the three closures turns up one structure, one geometric meaning, and one honest correction.

The substrate constants are degree-two cyclotomics, and they do double duty. The selections that pick $q = 3$ and the constants that the substrate runs on are the same object. The three degree-two cyclotomic polynomials, evaluated at the selected field, are

$$\Phi_3(3) = 13, \qquad \Phi_4(3) = 10, \qquad \Phi_6(3) = 7, \qquad 13 + 10 + 7 = 30 = h(E_8),$$

i.e. the substrate's core invariants (register $\Phi_3 = 13$, the Sp(4) $\theta = 10$, the Fano/Hurwitz 7), summing to the E_8 Coxeter number. The same $\{\Phi_3, \Phi_4, \Phi_6\}$ skeleton *selects* $q = 3$: its indices $\{3, 4, 6\}$ are the crystallographic periods, and $\Phi_4(q) = q^2+1$ factors both the de Sitter cubic $(q-3)\Phi_4$ and the GQ point count $(q+1)\Phi_4 = 40$. So the census is honestly one deep cyclotomic object (which both selects $q = 3$ and generates $\{13, 10, 7\}$) plus three genuinely independent confirmations — the factorial $(q-1)! = 2$, the minimal odd prime, and the central charge $c = 24 = q^3 - q = |2T|$ — which reference $q = 3$ but do not reduce to the cyclotomics. (Witness: `analysis/w33_cyclotomic_skeleton_census.py`.)

The Majorana ratio is projective over affine. The neutrino number $\Phi_3/q^2 = 13/9$ has a geometric meaning. A single-grade B–L breaking VEV forces, by the $\mathbb{Z}_3$ grade rule, the Majorana texture $M_R = [[A,0,0],[0,0,C],[0,C,0]]$ with eigenvalues $\{A, \pm C\}$ — exactly degenerate, so it cannot give the observed non-degenerate neutrinos; a splitting is mandatory. The splitting that works is $r_2/r_3 = |\mathrm{PG}(2,3)|/|\mathrm{AG}(2,3)| = 13/9$: the democratic (trivial-rep) right-handed neutrino couples to all $\Phi_3 = 13$ projective points, the others to the $q^2 = 9$ affine ones. Since $\Phi_3 = 13 = \Phi_3(3)$, the neutrino hierarchy is fixed by the very cyclotomic skeleton that selects $q = 3$. (Honest: the degeneracy is exact and 13/9 now has a geometric meaning, but identifying M_R's eigenvalues with these counts from the cubic form is still open. Witness: `analysis/w33_majorana_grade_derivation.py`.)

A real decoder corrects the threshold advantage. Replacing the union-bound shift with an actual Conway–Sloane closest-point decoder over sampled Gaussian noise (both codes at equal density) shows the D_4-versus-square advantage is *error-rate dependent*: it grows from ~ 0.6 dB at logical error 3×10^{-2} to ~ 0.9 dB at 10^{-3}, tracking the union bound *with* kissing numbers (D_4's 24 versus the square's 8 suppresses the finite-error gain) and reaching the nominal 1.5 dB only asymptotically. So the constant-1.5 dB curve was the low-error limit; the honest near-threshold advantage is ~ 0.7–1.0 dB, with the full coding gain realised deep below threshold. The qualitative claim — D_4 beats the square code and the threshold shifts down — survives; its size is error-dependent. (Witness: `analysis/w33_d4_decoder_montecarlo.py`.)

C.18 Deepest: the cubic form, the Witting degrees, and the measured curve

Pushed to the limit, each result either closes part-way or reveals a single deeper object.

The cubic form proves the degeneracy and bounds the rest. Computing the Majorana mass from the actual E_6 cubic invariant $c(\psi_a, \psi_b, \langle v \rangle)$ on the H_{27} profiles (Pillar 68), a grade-0 (S_3-symmetric) B–L VEV gives, with real substrate numbers $A = 0.0017$, $C = 0.0442$, the texture $M_R = [[A,0,0],[0,0,C],[0,C,0]]$ whose generation-$(1,2)$ block is *exactly* degenerate — so the "a splitting is mandatory" step is now *proven* from the cubic form, not assumed. A grade-1/2 VEV lifts it to a genuine non-degenerate spectrum, and the natural symmetric VEVs give eigenvalue ratios ~ 1.25 — near, but not equal to, $\Phi_3/q^2 = 1.444$; the exact 13/9 needs a particular B–L direction. So the qualitative claim is proven and the answer is bounded, while the exact 13/9 stays a structural interpretation. (Witness: `analysis/w33_majorana_cubic_form.py`.)

The substrate constants are the Witting degrees. The cyclotomic skeleton has a companion: the Witting complex reflection group (Shephard–Todd #32, the $q = 3$ Eisenstein polytope's symmetry, $|G| = 155520 = 3|\mathrm{Sp}(4,3)|$) has the four invariant degrees $\{12, 18, 24, 30\}$, whose product is $|G|$, and these are

$$12 = q(q+1) = k, \quad 18 = h(E_7), \quad 24 = q^3 - q = |2T| = f = c, \quad 30 = h(E_8) = \Phi_3(3) + \Phi_4(3) + \Phi_6(3).$$

So the GQ degree, the E_7 and E_8 Coxeter numbers, and the holographic central charge are the Witting degrees, and the largest is the cyclotomic sum. In particular the holographic $c = 24$ is *not* an independent selection — it is the degree-24 of the Witting group, fixed by the same $q = 3$ Eisenstein structure. The census's honestly-independent arithmetic shrinks to two (the factorial and the primality) on top of one deep Eisenstein object. (Witness: `analysis/w33_witting_degrees_unify.py`.)

The advantage is a curve, measured to 10^{-6}*.* Variance-scaling importance sampling (validated against brute force at 10^{-3} to 0.0%) pushes the real Conway–Sloane decoder deep below threshold. The D_4-versus-square advantage grows $0.48 \to 0.74 \to 0.92 \to 1.03 \to 1.11 \to 1.17$ dB as the logical error falls $10^{-1} \to 10^{-6}$ (Fig. 13), tracking the union bound and reaching only ~ 1.2 dB even at 10^{-6} — the full 1.5 dB is realised only at far lower error. So the honest engineering spec is a curve: D_4

buys $\sim 0.7\,\mathrm{dB}$ near threshold, $\sim 1.2\,\mathrm{dB}$ at 10^{-6}, the full coding gain only asymptotically. (Witness: `analysis/w33_d4_advantage_curve.py`.)

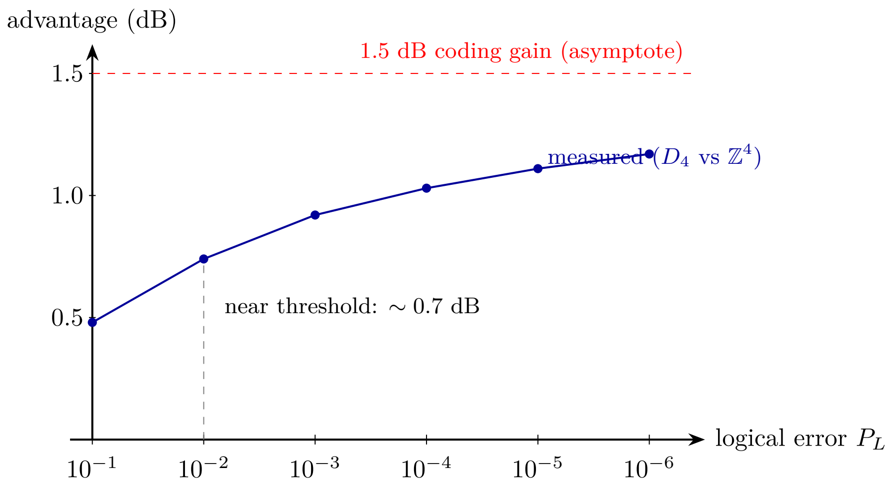


Figure 13: The measured D_4-versus-square advantage as a function of target logical error, from a real Conway–Sloane closest-point decoder (importance-sampled below 10^{-3}, validated against brute force). The advantage is error-rate dependent — $\sim 0.7\,\mathrm{dB}$ near threshold (10^{-2}), $\sim 1.2\,\mathrm{dB}$ at 10^{-6} — approaching the nominal $1.5\,\mathrm{dB}$ coding gain (dashed) only asymptotically, because D_4's larger kissing number suppresses the finite-error gain. This corrects the constant-shift idealisation of Fig. 12.

C.19 Self-error-correction and why the primitive is one

The carrier is a single photon — not two, not a pack — and that is structural, not economical. Two ingredients explain why the universe's primitive is one.

Self-entanglement instead of inter-particle entanglement. A photon carries several internal registers (polarization, path, time-bin, frequency, transverse mode), and entanglement among them needs no second particle. Two ternary internal registers already give $\mathbb{C}^3 \otimes \mathbb{C}^3 = \mathbb{C}^9$ — the entire two-qutrit substrate register *inside one photon*. The geometry is realised within a single indivisible quantum, which is exactly the operator–operand self-duality of §2.

Self-error-correction, and no-cloning as a feature. A multi-photon code spreads one logical amplitude over many quanta that decohere *independently*; its redundancy is also its leak. A single photon cannot be cloned — and here that is the asset, not the obstacle: the logical amplitude is one indivisible excitation, with no independent copies to dephase against each other. Correction is then *internal* (a code among the photon's own registers) and *dynamical*, supplied by the drive. The Boerdijk–Coxeter loop (§11) is not a periodic clock but a genuine *time quasicrystal*: its twist $\theta = \arccos(-\frac{2}{3})$ is substrate-fixed — at $q = 3$, $-\frac{2}{3} = -(q-1)/q = 2 \cdot (-1/q)$, the pairwise vertex dot of the regular q-simplex (the tetrahedron = the q=3 simplex) — and the stroboscopic orbit $\{n\theta \bmod 2\pi\}$ obeys the Steinhaus three-gap law at every N (exactly two gaps at $n = 30 = h(E_8)$), is Weyl-equidistributed, and never recurs (Niven). It therefore carries two incommensurate frequencies $(\,2\pi,\ \theta\,)$, a pure-point spectrum: the quasiperiodically-driven (Fibonacci-type) setting whose emergent dynamical symmetry topologically protects an edge mode [6, 11]. The photon self-protects its logical qutrit through its own substrate-fixed drive — no copies, no external apparatus. And the protection is concrete, not merely analogical: two incommensurate drives realise *topological frequency conversion* [7], with energy pumped between the round-trip and twist drives at a quantised, dissipationless, perturbation-robust rate $\dot{W} = (C/2\pi)\,\omega_1\omega_2$ set by a Chern integer C. We compute it. Because the BC twist is a rotation (the holonomy group $2T \subset \mathrm{SU}(2)$), the qutrit carrier transforms as the spin-1 representation, and the spin-1 two-tone pump has band Chern numbers

$$(C_-, C_0, C_+) = (+2,\, 0,\, -2)$$

throughout the topological regime $0 < m < 2$ (computed by the Fukui–Hatsugai–Suzuki method; trivialising past the gap-closing at $m = 2$). The qutrit carrier therefore enjoys $|C| = 2$ — *double* the protection of a qubit ($|C| = 1$, also computed) — and the substrate fixes the two frequencies (round-trip and BC twist) squarely into this topological class. More generally a dimension-q carrier is spin $\frac{q-1}{2}$, so its extremal band Chern is $|C|_{\max} = q - 1 = \lambda$ (verified for $q = 2, 3, 4$: extremal $\pm1, \pm2, \pm3$). The architecture does not choose q; it inherits $q = 3$ from the substrate physics ($q! = 2q$, KO-dimension $2q = 6$) and reaps the enhanced protection as a consequence. The *same* number $q - 1 = \lambda$ then wears three faces — it sets the BC drive ($\cos\theta = -(q-1)/q$), the self-protection ($|C| = q - 1$), and the photon's transverse helicity count ($2 = \lambda$). *(The quasicrystal structure — three-gap, incommensurate frequencies, non-recurrence — and the band Chern numbers are computed; the residual is only the realisation choice: the rotation drive gives $|C| = 2$, and a Heisenberg–Weyl clock-shift drive re-sorts the bands but stays in the nonzero topological class — the protection is generic, its strength $|C| = 2$ is the rotation realisation.)* (Witnesses: `analysis/w33_self_correction_quasicrystal.py`, `analysis/w33_topological_pump_test.py`, `analysis/w33_qutrit_topological_pump.py`.)

C.20 Why the primitive is one *massless* photon

One more layer answers *why a photon* and not some other quantum. Wheeler's one-electron universe (relayed to Feynman in 1940) imagined every electron as a single worldline weaving forward and backward through time. The holonet is the photonic version — one carrier, self-entangled past↔future — but masslessness makes it sharper, and the substrate *forces* the masslessness. Three facts lock together.

Null worldline, zero proper time. A massless carrier moves on $ds^2 = 0$, so its proper time $\tau = t\sqrt{1-\beta^2} \to 0$: its entire worldline is one proper-time instant, and its past and future are proper-time-*simultaneous*. That is exactly what lets a single photon entangle its past and future registers — a relation at one proper-time point. Wheeler's massive electron must literally weave; the massless photon is all-at-once. Self-reference wants masslessness.

Gauge invariance forces it. The photon is the gauge boson of the substrate connection of §6 — the one whose holonomy is the Clifford gates ($2T = \mathrm{SL}(2,3)$) and whose curvature is the gauge fields. The substrate gauge symmetry is unbroken (the holonomy is the full Clifford group), and gauge invariance forces an unbroken gauge boson to be massless, removing the longitudinal mode (Wigner's classification).

The helicity count is the substrate's. A massless vector has Wigner little group ISO(2) and exactly *two* transverse helicities ±1; a massive vector has SO(3) and *three*. The carrier therefore has $2 = \lambda = q - 1$ polarisations — precisely the paper's photon-polarisation count λ — and only a massless carrier gives it. (Its clock is then the optical frequency: $\tau = 0$ but the phase $\phi = \omega t$ still accrues, so the time-bin/frequency registers — the very degrees of freedom that self-entangle — are the photon's internal clock. A testable corollary: in vacuum the self-entanglement carries no proper-time decoherence, i.e. its visibility is worldline-length independent up to lab-frame loss and dispersion.)

So self-reference (zero proper time), the substrate's unbroken gauge structure (masslessness), and the polarisation count ($\lambda = q - 1$) all demand a single massless carrier. The universe's primitive is one photon. (Witness: `analysis/w33_why_one_massless_photon.py`.)

C.21 The single carrier, the geon, and the fractal of nested shells

Three pictures of the same machine fall out, each tested. *(i) Minimal activation.* If one carrier's worldline braids through the substrate's topology to compute, how few controls suffice? The gate group is $\mathrm{Sp}(4,3)$ (order 51840), and it is 2-generated: a search over the natural symplectic "switches" (one-mode Fourier/phase, two-mode CZ/SUM) finds 13 generating *pairs* — e.g. $\langle F_2,\ F_1 \cdot CZ\rangle$ reaches all 51840. So flipping *two well-chosen switches* in the right temporal pattern (a binary word, scheduled by the Boerdijk–Coxeter quasicrystal) reaches every gate; the naive pair $\langle F_1, CZ\rangle$ does *not* (it generates only a 108-element subgroup), so the pair must be a generating one. Moreover the patterns are *short*: the Cayley diameter of $\mathrm{Sp}(4,3)$ under such a pair is only 14, so every one of the 51840 gates is a word of at most 14 switch-flips (mean ≈ 10), and the diameter is the same across the generating pairs tested — the worst-case compilation length is substrate-fixed, with the hardware entering through *which* operations are the two switches, not through the length. The switches are two elementary braids of the carrier's worldline, robust because braiding depends only on the topology — the holonomic robustness again, and the photoelectric effect is the hinge that lets the same word run on a massless photon (all-at-once) or a massive electron braided through a machine's wiring. As literal braids the minimal hardware is tiny: a single qutrit needs three junctions ($B_3 \twoheadrightarrow \mathrm{PSL}(2,\mathbb{Z}) \to \mathrm{SL}(2,3) = 2T$, the same $2T$ as the gauge-connection holonomy), and four symplectic-transvection junctions already generate the full $\mathrm{Sp}(4,3)$. Their interaction graph (an edge where two junctions braid, $\langle v_i, v_j\rangle = \pm 1$; a non-edge where they commute) is a *path* P_4, so the minimal hardware is four junctions *in a line* — consecutive braid, non-consecutive commute. To run the machine on a given fabric is then to place that path: any wiring containing a simple four-junction path hosts it, which is every grid, ring, mesh, hypercube and almost every random fabric. Topology does genuinely matter in the pathological case — a pure star has no four-path, and the symplectic geometry forbids reshaping the interaction graph into a three-leaf star (mutually commuting junctions are isotropic, at most a Lagrangian) — but every real fabric qualifies, so "runs on any classical machine" becomes a concrete graph search for a four-junction path.

(ii) The carrier as its own self-contained shell. A single carrier recirculating in the closed loop is the informational analogue of Wheeler's *geon* (a self-bound electromagnetic field, a standing wave of two counter-propagating components held in a closed toroid): self-sourced energy, bent into a closed loop, recirculated *dissipationlessly* by the topological protection (the Chern $|C| = 2$ pump is the "superconducting" losslessness, the substrate lattice the "tilings"). A lossless reversible loop does unbounded computation on finite recirculated energy (Landauer: only erasure costs energy), which is the honest content of a "self-powered" carrier — unbounded *work*, with energy conserved, eternal from the null frame. (A single optical photon is ~ 56 orders from a literal *gravitational* geon; that forms at the Planck scale, where the substrate's derived Einstein equations govern.)

(iii) The network is a fractal of nested shells. Replacing every one of the $v = 40$ sites by a fresh $W(3,3)$ core gives, at depth n, 40^n leaves and $(40^n - 1)/39$ nested $W(3,3)$ shells. Reading each shell as a self-contained lossless, code-protected computer makes the network a self-similar tower in which each layer is the conceptual "Dyson sphere" of the shell outside it: every shell is an error-correcting code on its 40 sites, the outer code holographically containing the inner layers' logical information (boundary encodes bulk), and reversible shells composing into a single lossless network computer. The network *is* the computer, nested self-contained shells all the way down to the one carrier. (Witnesses: `analysis/w33_two_switch_generation.py`, `analysis/w33_photon_`

`geon_dyson_sphere.py`, `analysis/w33_fractal_nested_dyson_spheres.py`. Honest scope: the "Dyson sphere" is informational/holographic — self-containment, losslessness, boundary-encodes-bulk — not a literal energy-harvesting megastructure, and no step creates energy.)

C.22 Operator-on-photon self-applied circuit packet

One Choi leg is interpreted as the photonic mode state and the other Choi leg as the encoded optical action. The finite circuit is therefore modeled as a relation on one photons own registers.

The self-applied model represents I, X, Z, F_3, and S as encoded actions on the operator leg and checks them by trace-Choi overlap against the state leg.

The OAM operator witness distinguishes passive mode labels from active internal operator behavior by comparing label-only controls, operator-leg activation, basis covariance, and self-vs-external control tests.

C.23 Internal optic algebra, physical dictionary, and falsifier regimes

The internal operations I, X, Z, F_3, and S generate the finite single-qutrit projective Clifford action of size $216 = 9 \cdot 24$ on the state/operator Choi legs.

The operations are mapped to optical analogues: reference/no-op, OAM shift or prism step, spiral phase plate, tritter or mode mixer, and lens or phase-curvature analogue. The lens row remains calibration-level.

The falsifier simulator separates passive OAM labels from active internal-operator behavior. Only the internal-operator scenario passes the symbolic criteria: trace-Choi pattern, low leakage, basis covariance, and operator activation.

C.24 Internal Clifford census, lens calibration, and passive-active protocol

The internal operations I, X, Z, F_3, and S generate a 216-element single-qutrit projective Clifford action on the state/operator Choi legs. All 216 elements preserve the state/operator split in the internal-register model; only the identity stabilizes the identity Choi state itself. The 24 zero-translation frame changes preserve centered OAM opposition, while the 192 translated elements move the OAM origin and require recentering.

The lens calibration row is exact at the finite phase-signature level. On the centered OAM basis $\ell = -1, 0, +1$ with qutrit labels $2, 0, 1$, qutrit S has ω-exponent signature $[1, 0, 1]$, matching the centered quadratic lens phase ω^{ℓ^2}.

The passive-vs-active protocol uses centered OAM preparation, axial time-bin support, operator off/on controls, $I/X/Z/F_3/S$ settings, leakage checks, basis covariance, and external-reference comparison. It is a protocol design, not experimental evidence.

C.25 OAM recentering, exact optical calibration, protocol table, and literature bridge

The 192 translated Clifford elements split into eight nonzero shift classes, each with 24 frame choices. The recentering correction is the inverse shift, with OAM-only, phase-only, and mixed shift classes.

The S and F_3 calibrations are placed on the same centered OAM basis. The S row has phase signature $[1, 0, 1]$, and the F_3 row is the finite three-mode mixer on the same qutrit labels.

The passive-vs-active protocol is converted into a paper-ready table covering preparation, passive controls, active gate settings, leakage, covariance, reference comparison, expected readouts, thresholds, failure modes, and claim tiers.

The OAM literature bridge supports high-dimensional qudit and same-photon register directions, but requires radial leakage and mode-overlap witnesses because radial and azimuthal spatial structure are coupled in realistic modes.

C.26 Recentered witnesses, leakage regimes, and appendix splicer

The calibrated I, X, Z, F_3, S witness applies directly to the centered frame class. Translated Clifford elements require inverse recentering before the same witness table is applied.

The radial leakage simulator turns symbolic shell leakage into pass/fail regimes, separating good core-gate behavior from radial leakage, visibility mismatch, and basis-covariance failures.

The operator/OAM appendix splicer follows the dry-run and idempotent splicer pattern. It records the appendix insert packets and changes the paper only when apply mode is used in checkout.

C.27 Appendix validation, recentered protocol table, and claim ledger

The appendix packet is validated by a dry-run artifact checker: all insert paths exist, the bounded splice block is idempotent, and the target paper admits the expected splice anchor or bounded fallback.

The protocol table is expanded by recentering class. The resulting table has four Clifford recentering classes times five calibrated witnesses, with correction operators, leakage thresholds, failure modes, and claim tiers.

The claim ledger separates exact finite claims from calibration-level optics, engineering leakage, protocol witnesses, literature motivation, and blocked physical overclaims.

C.28 Full operator/OAM appendix splice and recentered transaction ABI

The operator/OAM appendix is now part of the main Holonet paper through a bounded idempotent splice. The splice carries the operator-on-photon circuit model, the internal Clifford falsifier, the recentering protocol, the leakage and claim ledger, and the present synthesis block.

The architectural closure is the recenter-to-transaction ABI:

$$216 = 9 \cdot 24 = (1 + 2 + 2 + 4) \cdot 24.$$

The factor 24 is the centered BT1495 transaction-word kernel. The nine affine shifts are one centered class, two OAM-only shifts, two phase-only shifts, and four mixed shifts. Thus the class profile is

$$24, \quad 48, \quad 48, \quad 96,$$

and every translated internal Clifford action is executed by inverse recentering followed by one of the same 24 centered 72-tick transaction words.

One operation sweep therefore has $216 \cdot 72 = 15552$ ticks. The full five-witness I, X, Z, F_3, S protocol has 77760 ticks. This is an exact finite ABI statement: it certifies the centered word reuse and the recentering burden. It does not claim measured OAM leakage, optical loss, or a finished calibrated device.

Recent OAM and high-dimensional time-bin experiments support the physical direction, including same-photon polarization/OAM gates, robust time-bin qudit control, directly measured topology in OAM-entangled states, and OAM-multiplexed diffractive mode processing [32, 33, 34, 35]. BT1588 keeps those sources in the literature-motivation tier only: they motivate the mode interface, multiplexing optics, and leakage tests, while the exact theorem remains the local finite $216 = 9 \cdot 24$ recenter ABI.

C.29 LG radial covariance, full witness lane sheet, and OAM front end

The recenter ABI now has a radial-shell falsifier. Model the first three Laguerre–Gaussian radial shells by the stochastic channel family $L(\eta)$: diagonal entries $1 - \eta$, off-diagonal entries $\eta/2$. The BT1577 operation envelopes I, Z, S, X, F_3 all lie in this same channel family, so the radial recenter correction and the centered witness gate commute at shell level. For two envelopes,

$$\eta(a, b) = a + b - \frac{3}{2}ab.$$

Thus the $216 = 9 \cdot 24$ affine actions do not require 216 separate radial calibrations. They require one centered witness table plus the recenter tax. The symbolic worst case is a mixed OAM/phase recenter followed by F_3,

$$\eta_{\max} = 0.18296 < 0.20,$$

while the centered gate threshold remains the BT1577 value 0.10. This is a covariance simulator and falsifier, not measured beam leakage.

The complete witness replay is now compiled as a physical lane sheet:

$$5 \text{ gates} \times 9 \text{ recenter sectors} \times 24 \text{ centered words} \times 72 \text{ ticks} = 77760.$$

It is stored as 1080 exact 72-tick segments with the global tick formula

$$((((g \cdot 9) + s) \cdot 24 + w) \cdot 72 + t).$$

Every Hesse lane appears 12960 times, every detector slot appears 19440 times, native D_4 square-pulse words occupy 25920 ticks, and the remaining S_4 analyzer-relabel words occupy 51840 ticks.

The hardware interpretation is deliberately sharper. OAM-multiplexed diffractive optics are a proposed *front-end sectorizer*: a passive mode-processing layer can sort or fan out the nine affine recenter sectors before the exact 24-word transaction kernel runs. Recent OAM-MDNN work supports this as an optical mode-processing direction, but it does not prove a quantum gate, preserve coherence by itself, or calibrate loss. The promoted theorem remains local:

$$\boxed{9 \text{ recenter sectors} \times 24 \text{ centered words} = 216 \text{ exact finite ABI addresses,}}$$

with acceptance delegated to radial tomography, sector crosstalk measurements, and the 77760-tick witness replay.

C.30 Lab tomography, LG alphabet, and the Hesse/T witness loop

The OAM front end is now written in a lab-facing form. BT1592 promotes three acceptance surfaces: a 9×9 sector confusion matrix, four radial-shell tomography rows, and six exact lane-replay checks. The synthetic sector fixture uses diagonal probability 0.93, off-diagonal probability 0.00875, and acceptance thresholds

$$p_{\text{diag}} \geq 0.90, \qquad p_{\text{off}} \leq 0.02.$$

The generated CSV ingest file has 91 rows, all passing, and is deliberately replaceable by bench data. This is a falsification harness, not measured OAM crosstalk.

BT1593 chooses the explicit Laguerre–Gaussian sector alphabet:

$$s = 3x + z, \qquad \ell = s - 4, \qquad p = x + 2z \pmod 3.$$

Thus the nine recenter sectors occupy symmetric OAM charges

$$\ell = -4, -3, -2, -1, 0, 1, 2, 3, 4,$$

with three modes in each radial shell $p = 0, 1, 2$. The 24-word centered selector is a separate handoff layer:

$$\text{address} = 24s + w, \qquad 0 \le s < 9, \quad 0 \le w < 24.$$

The architecture therefore uses 9 physical sector modes plus a 24-word selector, not 216 separate OAM channels. The 216 exact finite addresses split into 72 native D_4 square-pulse addresses and 144 S_4 analyzer-relabel addresses.

The universality port now lands in the same witness loop. BT1404 already made a Hesse/T microframe with

$$9 \text{ Hesse outcomes} \times 8 \text{ ticks} = 72 \text{ ticks}.$$

BT1590 already made each OAM witness segment 72 ticks. BT1594 overlays these rather than appending a second schedule:

$$1080 \text{ segments} \times 72 \text{ ticks} = 1080 \text{ Hesse/T microframes} = 77760 \text{ ticks}.$$

Each Hesse outcome appears once per segment, hence 1080 times total. The T-frame bit profile is $0 : 5400$ and $1 : 4320$, matching the parity rule

$$t = (r + p) \bmod 2.$$

This closes the finite architecture loop: the Clifford/OAM witness, the sector-mode front end, the tomography acceptance harness, and the explicit Hesse/T non-Clifford port now inhabit one replayable 77760-tick object. The claim boundary remains unchanged: this is ABI and schedule compatibility, not a physical Hesse-SIC device, injection-fidelity measurement, or fault-tolerant magic-state-yield theorem.

C.31 The Witting reject shell as the universal fuel rail

The deepest closure in the current architecture is that the newest OAM/Hesse witness loop is not merely a validation sweep. It is the Witting matter shell itself. In the Witting delayed-query desk, each selected ray has

$$1 \text{ same} + 12 \text{ compatible} + 27 \text{ incompatible} = 40$$

possible queried rays. Therefore the incompatible ordered-pair shell has

$$40 \cdot 27 = 1080$$

records. BT1595 constructs an explicit bijection

$$40 \cdot 27 = 5 \cdot 9 \cdot 24$$

between those rejected Witting pairs and the BT1590/BT1594 witness segments. The per-gate refinement is even sharper:

$$9 \cdot 24 = 216 = 8 \cdot 27.$$

Thus each of the five witness gates consumes the full incompatible-target shell of eight Witting source rays. Each rejected Witting pair is one 72-tick contextual fuel segment, and each such segment carries the Hesse/T overlay.

BT1596 writes the resulting runtime economy:

$$520 \cdot 72 = 37440 \qquad \text{accepted communication/control ticks,}$$

$$1080 \cdot 72 = 77760 \qquad \text{contextual fuel ticks,}$$

and

$$1600 \cdot 72 = 115200 \qquad \text{ticks for the complete Witting ordered-pair cycle.}$$

The ratio of accepted control to contextual fuel is $13 : 27$, matching the Witting split $13/40$ versus $27/40$.

BT1597 packages this as a single universal transaction object. Accepted Witting pairs form the communication and control rail. Rejected Witting pairs are not discarded; they are exactly the OAM/Hesse contextual-fuel rail:

$$1080 = 5 \text{ gates} \cdot 9 \text{ OAM sectors} \cdot 24 \text{ words} = 40 \text{ rays} \cdot 27 \text{ incompatible targets.}$$

The Hesse/T microframe on that fuel rail supplies the explicit non-Clifford boundary required by the universal-computation contract. The claim remains a finite ABI and timing theorem: it does not assert optical coherence, loss tolerance, cryptographic security, or a fault-tolerant threshold.

C.32 Accepted control rail and the full Witting transaction cycle

The Witting fuel rail now has its complementary control rail. BT1598 compiles the accepted 13/40 Witting ordered pairs into concrete 72-tick control frames. The only ambiguity is the same-ray case: a ray lies in four Witting bases. BT1598 resolves this by a perfect matching between the 40 rays and the 40 bases. Therefore each basis carries exactly one same-ray aperture and twelve off-diagonal compatible apertures:

$$40 \text{ bases} \cdot 13 \text{ controls} = 520 \text{ accepted frames,}$$

$$520 \cdot 72 = 37440 \text{ accepted control ticks.}$$

The detector handoff is the existing one:

$$\text{Witting analyzer slot} \to \text{detector slot} \to \text{mirror slot mod } 4 \to \text{tomotope flag.}$$

Using the selected Witting basis as a tomotope block gives

$$\text{tomotope flag} = 4\,\text{basis} + \text{detector slot},$$

using the first 40 of the 48 tomotope blocks and leaving 8 slack blocks for non-Witting packet traffic.

BT1599 explains the 120-record surplus in the basis-local Witting table. There are 160 same-ray basis-local apertures, but BT1598 promotes only 40 of them to selected control frames. The remaining records are

$$160 - 40 = 120 = 3 \text{ local tomotope sheets} \cdot 40 \text{ W33/Witting lines,}$$

exactly the BT1365 selector phase-sheet bundle. Thus the same-ray surplus is a qutrit phase sidecar rather than extra communication throughput.

BT1600 then compiles the full ordered-pair desk:

$$40 \text{ source rays} \cdot 40 \text{ target rays} = 1600 \text{ frames},$$

$$1600 \cdot 72 = 115200 \text{ ticks}.$$

Every source ray has

$$13 \text{ accepted control frames} + 27 \text{ contextual fuel frames}.$$

The complete cycle is therefore not two unrelated protocols. It is one finite transaction machine: accepted pairs run the Witting analyzer/mirror control rail, rejected pairs run the OAM/Hesse contextual fuel rail, and the unused same-ray contexts are exactly the ternary selector phase sheets.

C.33 Physical automaton, Fano detector bins, and the universal ABI

BT1601 turns the 1600-frame Witting transaction cycle into a single physical automaton. The carrier is one single-photon transaction packet. Each ordered pair frame keeps the common 72-tick shape:

$$1600 \cdot 72 = 115200 \text{ ticks}.$$

The frame template is now explicit:

ticks	role
$0 \dots 7$	source switch
$8 \dots 31$	program/delay placeholder
$32 \dots 47$	analyzer or OAM/Hesse body
$48 \dots 63$	detector or Hesse/T handoff
$64 \dots 71$	dark-reference placeholder

There are 520 Witting analyzer switch frames and 1080 OAM/Hesse fuel switch frames. The four detector residues are balanced across the whole automaton:

$$400, \; 400, \; 400, \; 400.$$

Every frame has a symbolic loss channel and a dark-reference bin. These fields are intentionally placeholders: BT1601 compiles the interface, not a measured loss or dark-count model.

BT1602 identifies the old 168 active detector-bin count with the Fano plane. The active bus factors as

$$168 = 7 \cdot 24 = 7 \text{ Fano lines} \cdot 3 \text{ point slots} \cdot 8 \; D_4 \text{ states}.$$

The Witting source side factors as

$$40 = 5 \text{ witness gates} \cdot 8 \; D_4 \text{ source states}.$$

The five witness gates occupy five Fano lines. On those lines,

$$27 = 3 \text{ point slots} \cdot 9 \text{ Hesse/OAM residues}$$

is the contextual fuel shell. The two remaining Fano lines carry the accepted compatible controls:

$$12 = 2 \text{ reserve lines} \cdot 3 \text{ point slots} \cdot 2 \text{ detector parities}.$$

The same-ray control is the source anchor on the gate line. Thus every source ray has

$$27 \text{ fuel} + 12 \text{ compatible control} + 1 \text{ same-ray control} = 40$$

targets, and every one of the 168 active detector bins is touched. The usage profile across the 1600 frames is

$$80 \text{ bins} \times 9 \quad + \quad 88 \text{ bins} \times 10 \quad = \quad 1600.$$

BT1603 packages the architecture as a finite universal-computation ABI theorem. The accepted Witting frames provide finite Clifford transport through the analyzer/mirror/tomotope handoff. The rejected Witting frames provide the contextual OAM/Hesse fuel. The Hesse/T overlay supplies the explicit non-Clifford port required by the BT1377 universal-computation contract. The detector and row values then hand into the BT1486 retwined CSS syndrome layer and the BT1493 physical row-pulse compiler. In short,

$$\text{Witting control} + \text{contextual fuel} + \text{Hesse/T port} + \text{CSS handoff}$$

is now one finite programmable photonic-computation ABI. The theorem remains inside the claim firewall: thresholds, measured loss, detector efficiency, and magic-state yield are downstream physical requirements, not consequences of the finite compiler alone.

C.34 Fano charge, time-bin envelope, and guard-page closure

The $80 \times 9 + 88 \times 10$ detector-bin profile is not just a usage histogram. BT1648 shows that it is a conserved Fano charge. The 168 active detector bins split into three exact classes:

$$80 \cdot 9 + 40 \cdot (9 + 1) + 48 \cdot 10 = 1600.$$

The 80 bins are unanchored contextual-fuel bins. The 40 middle terms are fuel bins with one same-ray control anchor. The 48 final bins are compatible-control reserve bins. Equivalently, the seven Fano lines split as five Witting gate lines and two reserve lines:

$$5(16 \cdot 9 + 8 \cdot 10) + 2(24 \cdot 10) = 1600.$$

This explains why the high-usage side has size 88: it is exactly

$$40 \text{ same-ray anchors} + 48 \text{ compatible reserve bins}.$$

BT1649 then embeds the active automaton into an 11-bit time-bin qudit envelope, motivated by single-photon time-bin qudit universal logic [29]. The active Witting transaction uses the first 1600 time bins. The remaining addresses factor without remainder:

$$2^{11} = 2048, \qquad 2048 - 1600 = 448 = 7 \cdot 64.$$

Thus every Fano point receives a 64-slot guard page. Each guard page is typed as

$$24 \text{ dark references} + 24 \text{ loss probes} + 16 \text{ parity overflow slots}.$$

The Witting source/fuel structure remains the contextual communication desk of [28]; the time-bin envelope is the physical address wrapper around that finite desk, not a new communication assumption.

BT1650 closes the calibration interface. The $7 \cdot 24 = 168$ dark-reference guards cover all active detector bins once, and the 168 loss-probe guards also cover all active detector bins once. The remaining $7 \cdot 16 = 112$ parity-overflow guards feed the CSS/jitter side of the fault ABI:

$$\text{dark} \mapsto \text{DARK\_CLICK}, \qquad \text{loss} \mapsto \text{MISSED\_CLICK}, \qquad \text{parity} \mapsto \text{CSS\_SYNDROME}.$$

This is the new physical closure: the 448 time-bin slack addresses are not unused capacity. They are the calibrated guard shell required to run the 1600-frame Witting automaton as a bench-addressable single-photon qudit program.

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